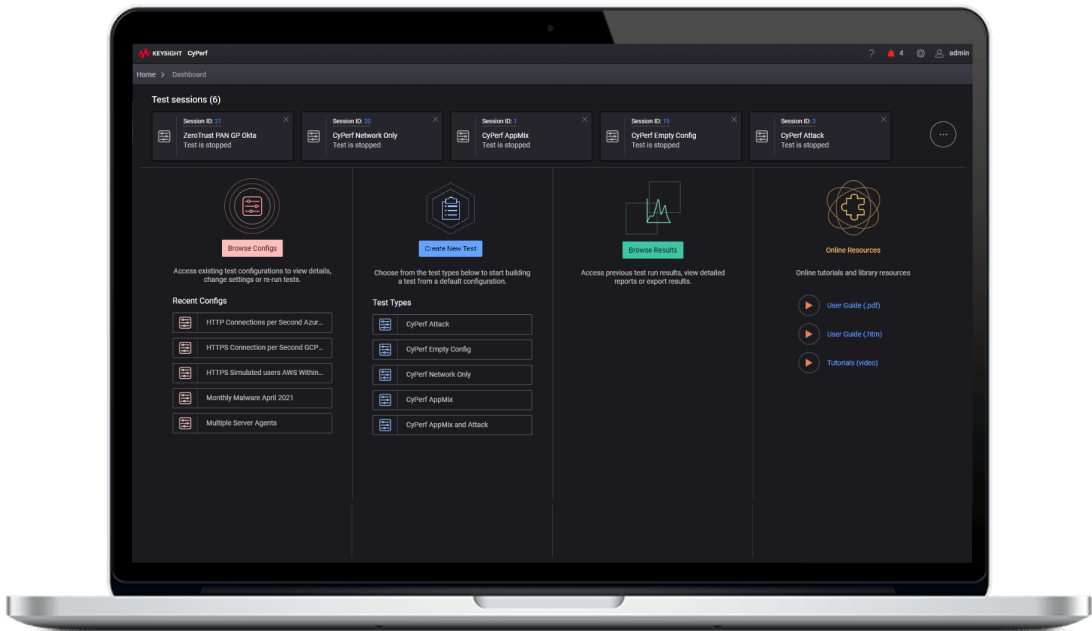


Keysight CyPerf

Release 26.0.0

User Guide



Notices

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Location	Phone number	Local time
<i>Americas</i>		
US, Canada	1-888-829-5558	8h00 – 17h00
Brazil	0800-892-0522	8h00 – 17h00
Mexico	001-888-829-5558	8h00 – 17h00
Other	+1-719-273-6516	8h00 – 17h00
<i>EMEA</i>		
Belgium	0800-18686	8h30 – 17h30
Finland	0800-913-352	8h30 – 17h30
France	0800-917228	8h30 – 17h30
Germany	0800-0824099	8h30 – 17h30
India	1800-18-02552	8h30 – 17h30
Ireland	1800-949245	8h30 – 17h30
Israel	1-809-454975	8h30 – 17h30
Italy	0800-790571	8h30 – 17h30
Luxembourg	0800-25112	8h30 – 17h30
Netherlands	0800-022-9086	8h30 – 17h30

Romania	0213 015 699	8h30 – 17h30
Spain	800-654386	8h30 – 17h30
Sweden	0201-202266	8h30 – 17h30
United Kingdom	0800-0293882	8h30 – 17h30
<i>Asia and Australia</i>		
Australia	1-800-370-558	8h30 – 17h00
China Mainland	800-810-0005	8h30 – 17h30
	400-810-0005	8h30 – 17h30
Hong Kong	800-931-613	9h00 – 18h00
Japan	0120-421-621	9h00 – 17h30
Malaysia	1800-819 092	8h30 – 17h30
South Korea	080-770-0800	8h30 – 17h30
Singapore	800-101-3797	8h30 – 17h30
Taiwan	0800-699-880	9h00 – 18h00
Other	+65 6215 7600	8h30 – 17h30 (Singapore)

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Secure Software Design Framework (SSDF)

This release is in compliance with Secure Software Development Framework (SSDF).

Keysight complies with multinational regulations for the cybersecurity of its own products and is committed to providing information to assist you in protecting your products and solutions from external cyber threats.

This section guides you on how to:

- [Keep your system secure](#)
- [Report vulnerabilities and receive fixes](#)
- [Review Keysight security management](#)
- [Securely decommission your system](#)

To sign up for Support portal access, go to <https://support.keysight.com/> and select **Register**.

Security advisories

To review security advisories and keep your system secure:

1. Access the Support portal at <https://support.keysight.com/>.
2. Select **Network Tests and Visibility Support (NAS)**.
3. Select the tile labeled **Product & Solution Cyber Security**.
The Product & Solution Cyber Security web page appears.
4. Select the tile labeled **Security Advisories**.
The Security Advisories web page appears.
5. Review and follow the instructions in the relevant security advisories.
6. If desired, download advisories.

Report a vulnerability or security issue

To report a vulnerability or security issue:

1. Access the Support portal at <https://support.keysight.com/>.
2. Select **Network Tests and Visibility Support (NAS)**.
3. Select the tile labeled **Product & Solution Cyber Security**.
The Product & Solution Cyber Security web page appears.
4. Below the images, select the link at the end of the sentence in red text that says:
"To report a cyber security issue on a Keysight product please go [here](#)".
The Product and Solution Cyber Security web page appears.

5. Select either the red text hyperlink labeled "**Report an Issue**", or scroll down and select the red **Report Issue** button:



Report Issue

After you report the issue, Support works with you to resolve it. If you need a software fix, Support provides it.

Keysight security management

To review Keysight security management:

1. Access the Support portal at <https://support.keysight.com/>.
2. Select **Network Tests and Visibility Support (NAS)**.
3. Select the tile labeled **Product & Solution Cyber Security**.
The Product & Solution Cyber Security web page appears.
4. Below the images, select the link at the end of the sentence in red text that says:
"**To report a cyber security issue on a Keysight product please go here**".
The Product and Solution Cyber Security web page appears.
5. Scroll to the bottom of the web page, and under **Additional Information**, select the red text hyperlink labeled "**Keysight Security**".

This opens a PDF on Keysight global security policy for all products. It includes information on:

- Security management
- Product and solution security
- Information security
- Government security
- Supply chain security
- Physical and site security
- Data privacy policy
- Enterprise risk management
- Additional resources

Decommission a system

Because each product line is different, contact Support for product-specific instructions on how to decommission a system.

CHAPTER 1

CyPerf Overview

Keysight CyPerf is a scalable, zero trust test solution for distributed hybrid cloud networks. It is a subscription-based software solution that re-creates every aspect of a realistic workload across a variety of physical and cloud environments, offering insights into end-user experience, security posture, and performance bottlenecks.

Keysight CyPerf simultaneously generates both legitimate traffic mixes and malicious activities across a complex network of proxies, software-defined wide area networks, TLS inspection, elastic load balancers, and web application firewalls. Combined with the ability to interleave applications and attacks to model user behavior and security breaches, it enables a holistic approach in replicating distributed customer deployment environments. It uses elastically scaling traffic agents that you can set up and tear down dynamically during a test to validate auto-scale policies. This process enables you to fine-tune the balance between user experience and security.

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Supported Tests

Keysight CyPerf currently supports a two-arm test type, where Keysight CyPerf emulates clients and servers. Client-only or server-only emulation is currently not possible. Keysight CyPerf can simulate both legitimate and attack traffic running as plaintext or TLS encrypted from traffic agents deployed in a distributed, hybrid manner.

The following section describes a series of common test configurations.

CyPerf AppMix

CyPerf AppMix is a default test template which contains an Application Profile and the emulated network segments: one Client network segment and one Server network segment, both with automatic IP assignment. Starting from this test, users would need to add the desired applications in the Application Profile, as well as configure the DUT element if the emulated network segments are interacting with the real Device Under Test (as in the case of Reverse Proxy, ALB, ELB).

① **NOTE:** You can use Auto IP when the test interface has an IP already assigned to it.

CyPerf Attack

CyPerf Attack is a default test template which contains an Attack Profile and the emulated network segments: one Client network segment and one Server network segment, both with automatic IP assignment. Starting from this test, users would need to add the desired attack list in the Attack

Profile, as well as configure the DUT element if the emulated network segments are interacting with the real Device Under Test (as in the case of Reverse Proxy, ALB, ELB).

CyPerf AppMix and Attack

CyPerf AppMix and Attack is a default test template which contains an Application Profile, an Attack Profile, and the emulated network segments: one Client network segment and one Server network segment, both with automatic IP assignment. Starting from this test, users would need to add the desired applications in the Application Profile, attack list in the Attack Profile, as well as configure the DUT element if the emulated network segments are interacting with the real Device Under Test (as in the case of Reverse Proxy, ALB, ELB).

CyPerf Network Only

CyPerf Network Only is a default test template which contains only the emulated network segments: one Client network segment and one Server network segment, both with automatic IP assignment. Starting from this test, users would need to add traffic elements (Application Profile and/or Attack Profile) as well as configure the DUT element if the emulated network segments are interacting with the real Device Under Test (as in the case of Reverse Proxy, ALB, ELB).

CyPerf Empty Config

CyPerf Empty Config is a default blank template. Starting from this test, users would need to add the desired applications in the Application Profile as well as configure the DUT element if the emulated network segments are interacting with the real Device Under Test (as in the case of Reverse Proxy, ALB, ELB).

Prerequisites

The following components are required to run a test from Keysight CyPerf's web UI:

Component	Description
Licenses	See Manage Licenses section.
Web browsers	• Chrome - v.80 (or later) • Firefox - v.75 (or later) • Edge Chromium - v.82 (or later) • Safari - v14.0.3

CHAPTER 2

How To

All the procedures presented in this section assume that you have successfully logged in to Keysight CyPerf. For more details, refer to [Access the CyPerf Controller through the Web UI](#).

You can perform the following actions from Keysight CyPerf:

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Access the CyPerf Controller through the Web UI

To start using Keysight CyPerf's web UI, the following information is required after deploying the Keysight CyPerf controller instance:

- Keysight CyPerf's IP address or host name (for example, `https://{host name}` or `https://{IP address}`). You can retrieve the IP address or host name while deploying the Keysight CyPerf controller instance virtual machine.

For more information on this, see the *Keysight CyPerf Deployment Guide*.

- A local Keysight CyPerf user account with its associated credentials. For more information, see [Create a User Account](#).

Create a User Account

After deploying the Keysight CyPerf controller instance, you need a Keysight CyPerf user account to access the web UI. CyPerf provides a default administrator account, which you will use on your initial login and for subsequent administrative tasks.

CyPerf user accounts can be of one of the following types:

- *Administrative user*: Can access the User Management functions and perform various administrative tasks, including the creation and management of other user accounts.
- *Regular user*: Can access the application and use all of the resources involved in test creation, execution, and analysis.



NOTE: All users are recommended to close non-active test sessions after tests complete in order to avoid resource depletion for the rest of the users.

The CyPerf administrator can create regular user accounts.

To add a new user account:

1. Log in with your default administrator account credentials:

Username: **admin**

Password: **CyPerf&Keysight#1**



NOTE:

It is recommended that you change your administrator password when you first log in the CyPerf web UI.

To change your administrator password:

- a. From the **Settings** menu, select **User Management**.
- b. In the top right corner of the web UI, select **Default Admin > Manage Account**.
- c. On the page that opens, select the **Password** tab.
- d. Then, enter the old and new passwords in the required fields, and select

2. Under the **Settings** menu () , select **User Management**. The **Keycloak Admin Console** opens in a new browser tab.
3. On the page that opens, select **Users** from the list of **Manage** functions (in the navigation pane).
4. Select the **Add User** button. The **Add User** page opens.
5. Enter the required information in the **Add user** form.

The following options and fields are available:

- ID
- Created at
- Username (which must be unique within the realm)
- Email
- First Name
- Last Name
- User Enabled (set to **ON**)
- Email Verified (automatically set to **ON** if the user email is verified)
- Groups
- Required User Actions (sets the action that the user is required to complete when they log in)

 **NOTE:** To enable user registration, go to **Settings > User Management > Realm Settings > Login**, and enable **User Registration**.

6. Select **Save**. The new user account is created, and the new user account page is displayed.
7. To allow the new user their first login, you need to set an initial password for their account:
 - a. On the newly created user account page, select the **Credentials** tab.
 - b. Under **Set Password**, enter and confirm a password for the new account.
 - c. Enable the **Temporary** option if the user will be required to change the password upon initial login.
 - d. Select **Set Password**. You are prompted to confirm the action. Select **Set password** again for confirmation.

 **NOTE:** Passwords must be at least 8 characters in length and contain at least one special character, one upper case character, and one lower case character.

8. Then, select the **Role Mappings** tab and make sure that under **Assigned Roles** both the **default-roles-keycloak** and one of **cyperf-user** or **cyperf-admin** are listed.
The new user account is now available for use, and the user will be able to log in with the credentials you set as an administrator.

For more detailed information on the **User Management** options, see the [Keycloak](#) documentation.

Reset Password for Regular User

Administrative users can reset a regular user's password.

NOTE: Passwords must contain at least one uppercase letter, one special character, one numerical character and have a length of at least 8 characters.

1. Select the **Settings** menu, and then select **User Management**.
2. Select **Users** from the list of **Manage** functions, and select the **Edit** button.
3. Select the **Credentials** tab.
4. Enter the new password.
5. Re-enter the new password in the Password Confirmation field.
6. Set the **Temporary** option to **ON** if the user will be required to change the password upon initial login.
7. Select the **Reset Password** button. A confirmation dialog opens.
8. Select the **Reset Password** button to confirm the action.

Log in

To log in to the Keysight CyPerf browser-based Web UI:

1. Open a supported web browser. For more details, see [Prerequisites](#).
2. Type `https://<IP address>` into the browser's URL bar. This is the IP address of the deployed Keysight CyPerf controller machine. The Keysight CyPerf login page appears.

NOTE: If you are logging in for the first time, you are required to register and create a new user.

3. Type the username and password of your Keysight CyPerf user account.
4. Select **Log In**. Upon successful login, the Keysight CyPerf **Dashboard** page opens.

NOTE: If you are logging in for the first time, you are required to accept the **Keysight Software End User License Agreement** and to update your password before you can log in.

Log out

To log out of your Keysight CyPerf user account:

- Select the user name in the top right corner
- Select **Log Out**

The Keysight Web Application Platform login page opens. You can use this page to log in again with a different user account or to register a new account.

Manage Licenses

CyPerf is a licensed product. You can manage licenses using either the integrated CyPerf [License Manager](#) or a centralized [License Server](#) that is managed by your organization.

Licensing Requirements

The license server is shipped as a separate `.ova` file available for download as the **License and User Manager** file on the CyPerf Downloads & Updates Support page.

After deploying the `.ova`, you will have access to a web interface for the license server by entering the server's IP address in a web browser address field (for example: `https://10.38.156.169`).

The **License and User Manager** web UI allows you to:

- [Activate licenses](#)
- [Sync licenses](#)
- [Manage licenses offline](#)
- [View license statistics](#)
- [Deactivate licenses](#)

After activation, the licenses and features will be available in the CyPerf web UI.

CyPerf Licenses

The following types of CyPerf licenses are available:

- Keysight, CyPerf 10 Agents and single 10 Gbps Performance Unit (1-year subscription)
- Keysight, CyPerf 10 Agents and single 10 Gbps Performance Unit (perpetual license)
- Keysight, CyPerf 10 Agents and single 100 Gbps Performance Unit (1-year subscription)
- Keysight, CyPerf 10 Agents and single 100 Gbps Performance Unit (perpetual license)
- Keysight, CyPerf 40 Agents and single 400 Gbps Performance Unit (1-year subscription)
- Keysight, CyPerf 40 Agents and single 400 Gbps Performance Unit (perpetual license)
- IXIA, CyPerf Single Agent Floating License (1-year subscription)
- IXIA, CyPerf 10G Performance Floating License (1-year subscription)
- IXIA, CyPerf Single Agent Floating License (30-day subscription)
- IXIA, CyPerf 10G Performance Floating License (30-day subscription)
- IXIA, CyPerf Single Agent Floating License (perpetual license)
- IXIA, CyPerf 10G Performance Floating License (perpetual license)
- IXIA, CyPerf 4 Agents and 1 Gbps of Performance Bundle (1-year subscription)
- IXIA, CyPerf 2 Agents and 10 Gbps of Performance Bundle (1-year subscription)
- IXIA, CyPerf 10 Agents and 10Gbps of Performance Bundle (1-year subscription)
- IXIA, CyPerf 2 Agents and 1 Gbps Bundle (1-year subscription)

- IXIA, Security & Performance Bundle with 2 x 10 Gbps BreakingPoint VE, CyPerf 10 Agents, and 10Gbps CyPerf performance bundle (perpetual license)
- IXIA, CyPerf Perpetual 2 Agents, and 10Gbps CyPerf performance bundle (perpetual license)
- IXIA, CyPerf Perpetual for 10 Agents and 10Gbps, CyPerf performance bundle (perpetual license)
- IXIA, CyPerf 10 Agents and 10Gbps of Performance Bundle (30-day subscription)
- IXIA, CyPerf 10 Agents and 10Gbps of Performance Bundle (90-day subscription)
- IXIA, CyPerf Single Agent Floating License (90-day subscription)
- IXIA, CyPerf 10G Performance Floating License (90-day subscription)
- IXIA, L47 Promotional Cloud University Bundle for 15 seats x CyPerf 1Gbps floating 1-year subscription license
- IXIA, CyPerf Starter Bundle with 2 Agents and up to 10 Gbps (30-days subscription)
- IXIA, CyPerf Starter Bundle with 2 Agents and up to 2 Gbps (30-days subscription)

License Manager

The first time you use CyPerf, you need to active at least one license. You can activate and manage your licenses using the **License and User Manager** functions, which are accessed from the license server's web UI **Settings** menu (⚙️).

Access the License Manager

To access the License Manager:

1. Access the License Server's web UI by entering the server's IP address in a web browser address field (for example: <https://10.38.156.169>).
2. On the **Title** bar, under the **Settings** menu, select **License Manager**.

Activate a license

To activate one or more CyPerf licenses:

1. From the License Server's web UI, select **License Manager** under the **Settings** menu.
2. Select **Activate licenses**. The **Activate Licenses** dialog opens.
3. Enter your license data in the dialog box. You can use either activation codes or entitlement codes.
4. Select **Load Data**, indicate the number of licenses you want to activate, then click **Activate**. Your new licenses, which should now be listed on the License Manager page, are now available for running tests.

Deactivate a license

To deactivate one or more CyPerf licenses:

1. From the License Server's web UI, select **License Manager** under the **Settings** menu.
2. Select **Deactivate licenses**, then and indicate a new quantity for each of the existing licenses.
3. Select **Perform the Activation** to complete the task.

Sync licenses

After any licensing operation, you may need to synchronize licenses to make sure the latest licensing changes you made are reflected in your environment. To sync licenses:

1. From the License Server's web UI, select **License Manager** under the **Settings** menu.
2. Select **Sync licenses**. The latest changes should now be reflected on the **License Manager** page.

Reserve a license

To reserve licenses:

1. From the License Server's web UI, select **License Manager** under the **Settings** menu.
2. Select the **Manage Reservation** icon. The **Manage License Reservation** dialog opens.
3. Select the license you want to reserve.
4. Enter the number of desired licenses in the **New Reserved Count** field.
5. Enter the duration of the reservation (in hours) in the **Duration to Reserve** field.
6. Click **Reserve**. The initial reserved count and duration is overwritten when a new reservation is performed.



TIP: The **License Statistics** window shows all reserved features, ordered by count and reserved time.

Get license statistics

To view license statistics such as the number of consumed or reserved licenses, or the remaining reservation duration:

1. From the License Server's web UI, select **License Manager** under the **Settings** menu.
2. Select **License statistics**. The **License Statistics** dialog opens listing all the available licenses.
3. Expand the license for which you want to view statistics. Details about the license are listed on the dialog under **Consumed Count**, **Reserved Count**, or **Remaining Reservation Duration**.

Manage licenses offline

Offline license management is required for cases in which your test network is operating in an isolated environment with no internet access, or for registering a license on a server running over an *IPv6-only* infrastructure that does not support 6to4 routing.

To perform offline license operations:

1. From the License Server's web UI, select **License Manager** under the **Settings** menu.
2. Select **Offline operations**. The **Keysight Licensing Offline Operations** dialog opens.
3. Click **Generate request**.

4. Using a system that has internet connectivity, access the **KSM Offline Operations Page**, and follow the steps provided for the desired operation.
5. From your offline system, return to the **Keysight Licensing Offline Operations** dialog, then click **Import license**.
6. Click **Finish** to complete the task.

License Server

Rather than using the internal CyPerf [License Manager](#), you can use a centralized License Server that is managed by your organization, and then add or remove the installed licenses from the CyPerf web UI.

Add a License Server

To register a new license server from the CyPerf web UI:

1. Log in the CyPerf web UI.
2. Under the **Settings** menu, select **Licensing > License Servers**. The **License Servers** dialog opens.
3. Enter the license server IPv4 or IPv6 address, or hostname in the empty input field.



NOTE:

To register a license on a server running over an *IPv6-only* infrastructure, use the Keysight Licensing Offline Operations (see [Manage licenses offline](#)). However, if the IPv6-only environment supports 6to4 routing, you can use the online license activation process.

4. Make sure the **Interactive Authentication** option is enabled. This option is enabled by default.
5. Then, select the **Add**  button next to the server IP field. The **Fingerprint validation** dialog opens.
6. After validating the license server's key fingerprint, type the user name and password for the license server.

The default credentials are:

- username: `admin`
- password: `CyPerf&Keysight#1`

7. Select **Connect** to confirm your action. The connection to the server initiates. If validation is successful, the **License Servers** dialog displays an **Established** label showing that the connection was successful.



TIP: To see the list of installed licenses, you need to access the license server in a web browser by entering the server's IP address in the browser address bar: `https://<license server IP>`

Remove a License Server

To remove a license server that was previously added in the CyPerf web UI:

1. Log in the CyPerf web UI.
2. Under the **Settings** menu, select **Licensing > License Servers**. The **License Servers** dialog opens listing the previously set license servers.
3. Select the **Delete** button next to the license server that you want to remove.
4. Select **Close** to confirm your action and close the **License Servers** dialog.

Work with the Resource Library

CyPerf allows you to view, import and export various resources such as payloads, TLS certificates, TLS keys, TLS DHS, playlists, media files, captures and custom applications. You can access the Resource Library from the **Settings** (⚙️) menu > **Library > Resource Library**.

See the following sections for details on how to:

- [Import Captures](#)
- [Create Custom Applications from Imported Captures](#)
- [Edit Custom Applications](#)
- [Export Custom Applications](#)
- [Import Custom Applications](#)
- [Export Other Resources](#)

Import Captures

You can import and view available captures from the **Settings** (⚙️) menu > **Library > Resource Library**. You can use imported captures to create custom applications from the CyPerf UI. The *PCAP* or *PCAPNG* file formats can be used for importing operations.

The following limitations apply when importing captures:

- Capture files must include at least one TCP flow or one UDP flow.
- The maximum capture size is 1 GB.
- The maximum number of flows per capture is 10,000 (with one exchange per flow).
- The total number of exchanges per capture is 10,000.

To import a packet capture:

1. Log in to the CyPerf UI.
2. Under the **Settings** menu, select **Library > Resource Library**. The Resource Library page opens.
3. Under **Browse resources**, select **Captures**.
If no captures have been imported yet, the page will be blank. If capture files have previously been imported, they will be listed on the page with a capture ID and a file name. To view more details about each capture, expand each entry by clicking the **Expand** (➤) button. You can view each flow ID, source and destination addresses, source and destination ports, transport

protocol, available exchanges for each flow, including the first 1024 bytes of payload for each direction of the exchange.

4. Click **Import** in the left bottom corner of the screen.

The **Import capture** dialog opens allowing you to upload a capture file and *optionally*, an SSL key log file.



NOTE: In addition to HTTP captures, CyPerf supports importing and decrypting HTTPS captures. If the capture contains encrypted HTTPS traffic, you can *optionally* provide an SSL key log file (SSLKEYLOGFILE) to enable decryption of HTTPS flows.

5. Browse to the capture file you want to import and select it. Then, click **Open**.
6. (Optional) Click **Upload SSLKEYLOGFILE** and select the SSL key log file that will be used to decrypt the HTTPS traffic in the capture. If no SSLKEYLOGFILE is provided, the capture is still imported successfully, but HTTPS payloads remain encrypted.

After the capture file is uploaded, it will be displayed on the list of available capture files. The time it takes to upload the capture depends on its complexity. For example, a higher number of exchanges or smaller average payload sizes will increase the upload time.

You can also delete an existing capture by selecting the capture on the available list and clicking **Delete**.

See also [Create Custom Applications](#).

Create Custom Applications from Imported Captures

After you import a capture, you can create a new custom application that you can use in your CyPerf tests.

An application can have multiple actions and one action can be created from:

- one full capture,
- a subset of flows from one or multiple captures, or
- a subset of exchanges from one or more flows and from one or multiple captures.

After the application is created it will be available in the list of applications just as the pre-canned applications.

The following limitations apply for creating custom applications from imported captures:

- Custom applications on ports 80 or 443 cannot be mixed with pre-canned web applications.
- Importing invalid packet captures will return an error message.
- Importing packet captures with unsupported IP protocols will return an error message.
- The UI displays only the first 1024 bytes of each client-to-server (C2S) and server-to-client (S2C) payload; however, the full payload size is visible.
- The total number of flows (connections) in a single application cannot be bigger than 64.
- The total number of parameters cannot be bigger than 10,000. The total number of parameters

is equal to the sum of endpoints per action multiplied by the number of parameters per endpoint for the same action.

To create a custom application:

1. Log in to the CyPerf UI.
2. Under the **Settings** menu, select **Library > Resource Library**. The Resource Library page opens.
3. Under **Browse resources**, select **Captures**.
The list of available captures opens. If no captures are displayed on the page, first [import a capture file](#).
4. Click **Create Apps** at the bottom of the screen. The Create App dialog opens.
5. Under each column on the Create App dialog, select the capture(s), flow(s) and exchanges(s) you want to use for the new application.
6. Then click the **Add**  button to add actions from the selected entries.
7. Click the **Create App** button in the bottom right corner of the dialog.
The new application is created based on your selections and displayed under **Browse resources > Custom Applications** along with other pre-canned applications.

Edit Custom Applications

After you create a new custom application, you can further customize it to fit the tests you want to run.

To edit a custom application:

1. Log in to the CyPerf UI.
2. Under the **Settings** menu, select **Library > Resource Library**. The Resource Library page opens.
3. Under **Browse resources**, select **Custom Applications**.
The list of available custom applications opens. If no applications are displayed on the page, first [create a new custom application](#).
4. Click the **Expand** icon next to the application you want to edit. The available application actions are displayed.
5. To navigate between actions, flows, or exchanges, click through each action/ flow/ exchange fly-out panel.
6. To view the Client to Server and Server to Client payload, click the **C2S Payload** button or the **S2C Payload** button respectively.
7. To add a new action/ flow/ exchange:
 - a. Click the **Add**  button next to the component you want to add. The **Edit App** dialog opens, listing the available captures, the application name, and the already available actions/ flows/ exchanges for the current application.
 - b. Under **Select captures**, select the capture you want to use as the basis for your action/ flow/ exchange.

- c. Then, under **Edit App**, click the **Add**  button under the action/ flow/ exchange list to add the selected capture as a new action/ flow/ exchange. The new component is added to the list under a default name which you can update by clicking the **Rename AI** button and typing a new custom name for the action/ flow/ exchange.
 - d. You can also click the **Edit**  icon to update the selected action/ flow/ exchange. For example, when clicking **Edit**, you can see the list of flows and exchanges for an action and add new flows or exchanges by clicking **Add** .
 - e. To confirm all the changes and close the Edit App dialog, click the **Edit App** button in the lower right corner of the dialog. The new action/ flow/ exchange is displayed under the custom application.
8. To delete any action/ flow/ exchange, click the **Delete**  button next to the component you want to remove.
 9. To change the order of actions/ exchanges, click and hold the **Reorder**  button. Then, drag and drop the item in the correct position.
 10. To save all the changes you made for the selected custom application, click **Save App**.
The changes you made to the application are saved on the server. However, if the application was included in any test configurations before being edited, the changes will not be automatically applied, and the configurations may be left in an inconsistent state. Please remove the old version of the application and add the current one to make sure everything works as expected.

Export Custom Applications

After you import a capture and create a new custom application, you can export it from the CyPerf UI.

To export a custom application:

1. Log in to the CyPerf UI.
2. Under the **Settings** menu, select **Library > Resource Library**. The Resource Library page opens.
3. Under **Browse resources**, select **Custom applications**.
The list of available custom applications opens. If no custom applications are displayed on the page, first [create a custom application](#).
4. Select the application you want to export.
5. Click **Export** at the bottom of the applications list. The selected application is downloaded on your machine as a .zip file. You can use this downloaded archive to import the same application on other CyPerf setups for example.
You can also export all the custom applications by clicking the **Export all** button under the **Browse resources** pane.

Import Custom Applications

To import a custom application:

1. Log in to the CyPerf UI.
2. Under the **Settings** menu, select **Library > Resource Library**. The Resource Library page opens.
3. Under **Browse resources**, select **Custom applications**.
The list of available applications opens.
4. Click **Import** under the **Browse resources** pane.
5. Browse to the application you want to import and select the .zip file. Then click **Open**.
The custom application is loaded on the server and displayed under the list of **Custom Applications**.

Export Other Resources

You can export other types of resources besides custom applications from the CyPerf UI, both system resources and custom resources (such as payloads, TLS certificates, TLS keys, TLS DHS, playlists, media files, captures).

To export resources:

1. Log in to the CyPerf UI.
2. Under the **Settings** menu, select **Library > Resource Library**. The Resource Library page opens.
3. Under **Browse resources**, select the tab corresponding to the resources you want to export (for example, **Payloads**, **TLS Certificates**, etc.).
The list of available resources opens.
4. Select the resources you want to export by clicking the check box next to each item, or select all the resources on the page by clicking the check box next to the **Name** column.
5. Click **Export** at the bottom of the resources list. The selected resources are downloaded on your machine as a .zip file. You can use this downloaded archive to import the same resources on other CyPerf setups for example.
You can also export all the available resources by clicking the **Export all** button under the **Browse resources** pane.

Add a Controller Proxy

A controller proxy consists of a message broker and other communication management components.

A CyPerf controller proxy is required in some hybrid deployment scenarios, where each of the distributed agents cannot directly access the CyPerf controller. For example, when the CyPerf controller is deployed on premise and some CyPerf agents are in the cloud, they can still communicate through a CyPerf controller proxy. In this case, agents register to the controller proxy, and the public IP address of the controller proxy is configured in the CyPerf controller.

To register a controller proxy from the CyPerf web UI:

1. Log in the CyPerf web UI.
2. Under the **Settings** menu, select **Agent Management**. The **Agent Management** page opens.
3. Click the **Controller Proxies** button. The **Controller Proxies** dialog opens. Enter the controller proxy IP address or hostname in the empty input field.

 **NOTE:** Both IPv4 and IPv6 addresses are supported.

4. Make sure the **Interactive Authentication** option is enabled. This option is enabled by default.
5. Then, select the **Add**  button next to the controller proxy IP field. The **Fingerprint validation** dialog opens.
6. After validating the controller proxy's key fingerprint, type the following credentials for the controller proxy:

The default credentials are:

- username: `admin`
- password: `CyPerf&Keysight#1`

During deployment, you will be prompted to change the default password. If you have changed the default password, for the **Password** field, copy and paste the controller proxy secret that can be found on the proxy in the `./cyperf/local_secret` file

To get the controller proxy secret:

- a. Establish an SSH connection to the controller proxy, and run the following command:

```
cat ./cyperf/local_secret
```

The command will output the content of the `local_secret` file to the terminal.

- b. Copy the controller proxy secret from the terminal, and paste it in the **Password** field on the **Controller Proxy** dialog in step 6 above.

7. On the **Fingerprint validation** dialog, select **Connect** to confirm your action. The connection to the controller proxy initiates.

If validation is successful, the **Controller Proxy** dialog displays an **Established** label showing that the connection was successful.

See also [Manage Server Fingerprints](#).

Migrate Controller Data

CyPerf allows you to migrate controller data from one setup to another. You can export authentication data and other data (such as configurations, external license servers, controller proxies, and results) from a source controller, which you can then import to a new target controller. When the import procedure is complete, the resources that you imported from the source controller will have replaced all the existing data on the target controller.

 **NOTE:** Controller data migration is supported starting with CyPerf 2.1 release.

Install Controller Data Migration Patch

 **IMPORTANT:** Follow the steps in this procedure *only* if you want to migrate controller data from an older 1.7 or 2.0 CyPerf setup to a 2.1 or newer setup.

To migrate controller data from a 1.7 or a 2.0 CyPerf controller, you first need to download and install a CyPerf migration patch:

1. Download the migration patch (for example, `cyperf-controller-migration-patch-2-20-316.tar`) available on the Keysight CyPerf Support page.
2. Log in to the CyPerf UI.
3. Under the **Settings** menu, select **Software Updates**. The **Software Updates** page opens.
4. On the **Software Updates** page, click the **Select packages for upload** button, choose the migration patch you downloaded in step 1, and then click **Open**.

The migration package starts uploading. When the upload is complete, the migration patch will be listed on the page.

5. Click the **Start update** button.

Installation of the migration patch begins, and the progress is displayed on the screen. After the patch installation is complete, do a manual refresh of the UI. After refreshing the page, the **Settings** menu will display a new **Data Migration > Export Controller** option, which allows you to collect and download the export package to be migrated to a 2.1 or newer setup.

See [Export Controller Data](#) for the export procedure.

Before starting the patch installation, you are also given the option to remove the migration patch upload by clicking the **Reset** button. However, after installation begins, the **Reset** button is no longer available.

Export Controller Data



IMPORTANT: The export procedure cannot run when the available controller disk space is lower than 50%. To free up disk space by removing diagnostics, logs, or migration archives, use the **System cleanup** option under the **Settings** menu > **System Monitor**. See [System Monitor](#).

To export data from the source controller:

1. From the **Settings** menu, select **Data Migration > Export Controller**. The **Export Controller Data** window opens informing you of the components to be exported.
2. Click the **Start** button. A notification at the bottom of the screen informs you that the migration package is being queued in preparation for export, and the **Export Controller Data** will automatically close.

When all the export resources are collected and available on your local system, a notification informs you that the export package was created.

The export package is downloaded on your local system in the form of a compressed .zip file, under your usual downloads location (for example, *C:\Users\<user name>\Downloads\migrate_package1676028828*).

Import Controller Data

To import data on the target controller:

1. From the **Settings** menu, select **Data Migration > Import Controller**. The **Import Controller Data** window opens.
2. Click **Select migrate package for upload**, and select the migration package that you exported from the source controller, and click **Open**.
A warning is displayed informing you that the import procedure cannot be canceled or reverted after it is initiated.
3. Confirm acknowledgment of data loss by selecting the **I understand this will wipe existing data** check box.
4. Click the **Start** button.
The import procedure initiates. The application screen is grayed out during import and will become active again when the import procedure is complete.
5. After the import procedure is finished, close the **Import Controller Data** window.
6. Log out of the web UI, and log in again to see the newly imported controller data.

Manage Controller Certificates

From the **Certificate Management** page in the CyPerf web UI, you can:

- [View](#) the current controller certificate information
- [Manage certificate](#)

View certificate

To access the **Certificate Management** page and view the current controller certificate information:

1. Log in to the CyPerf UI.
2. Under the **Settings** menu, select **Controller Security** > **Certificate Management**. The **Certificate Management** page opens, listing the following information about the controller certificate: issuer, DNS names, IP addresses, generation method, and validity period.

Manage Certificate

You can change the certificate used by the CyPerf controller in three different ways:

- [Automatic](#): CyPerf detects changes in your network configuration and updates the certificate to reflect the current network information. The previous certificate will be overwritten, and the **Certificate Management** page will reload with the new certificate.
- [Custom](#): Generates a self-signed certificate using the information you provide. This option is useful in complex network topologies, such as those involving proxies.
- [Upload](#): Obtain a valid certificate through an external method and upload it via the **Change Certificate** dialog for the CyPerf controller to use.

Auto-generate a controller certificate

 **NOTE:** The controller's certificate is auto-generated *by default* unless you manually overwrite the current certificate using the [Custom](#) option, or the [Upload](#) option.

Use the **Automatic** method to auto-generate a new controller certificate:

1. Under the **Settings** menu, select **Controller Security** > **Certificate Management**. The **Certificate Management** page opens.
2. Click the  **Manage Certificate** button. The **Change Certificate** dialog opens.
3. Select **Automatic**.
4. Click the **Change certificate** button. The previous certificate is overwritten, and the **Certificate Management** page reloads listing the information for the newly-generated certificate.

Generate a custom certificate

Use the **Custom** method to generate a certificate with information you provide, such as DNS name, IP address, and validity period:

1. Under the **Settings** menu, select **Controller Security > Certificate Management**. The **Certificate Management** page opens.

2. Click the  **Manage Certificate** button. The **Change Certificate** dialog opens.

3. Select **Custom**.

4. Enter a new DNS name.

Optionally, click **Add**  to include more DNS names. To remove a DNS name from the list, click **Delete** .

5. Enter a new IP address.

Optionally, click **Add**  to include more IP addresses. To remove an IP address from the list, click **Delete** .

6. In the **Valid for** field, enter the validity period for the new certificate.

7. Click **Change Certificate**. A new certificate is generated based on the provided network information, which will replace the old certificate.

Upload a new controller certificate

Use the **Upload** method to upload a valid certificate you can get through any other external method:

1. Under the **Settings** menu, select **Controller Security > Certificate Management**. The **Certificate Management** page opens.

2. Click the  **Manage Certificate** button. The **Change Certificate** dialog opens.

3. Select **Upload**.

4. Click **Upload certificate**.

5. Browse to the certificate file you want to upload, select it, and click **Open**. The selected certificate file is displayed on the dialog.

6. Then, click **Upload private key**.

7. Browse to the certificate key you want to upload, select it, and click **Open**. The selected key file is displayed on the dialog.

8. Click **Change certificate**. The new certificate-key pair is uploaded replacing the old certificate.

Manage Server Fingerprints

You can copy or revoke a server fingerprint from the **Key Management** dialog in the CyPerf web UI.

To copy a server fingerprint:

1. Log in the CyPerf web UI.
2. Under the **Settings** menu, select **Controller Security > Key Management**. The **Key Management** dialog opens.
3. Copy the key by clicking the **Copy** button (📋) and use it for authentication when connecting the agent to the controller or controller proxy.
See the *CyPerf Deployment Guide* for more details on how to connect agents to a controller or controller proxy.

To revoke a server fingerprint, click the **Revoke server fingerprint** button, which generates a new key pair.

ⓘ **IMPORTANT:** All active connections are terminated when activating the **Revoke server fingerprint** button.

Update Software

You can update all the required CyPerf and ATI software components, including the CyPerf agents connected to the current controller, from the **Software Updates** page that is accessible from the **Settings** menu of the CyPerf web UI. You can also update CyPerf agents from the **Agent Management** page as described in [Update Agents from the Controller UI](#).

ⓘ **NOTE:**

- An automatic agent update during the CyPerf controller update is available only for **4.0** and newer CyPerf versions, up to CyPerf **6.0**.
- Upgrade from CyPerf **6.0** to CyPerf **7.0** is *not supported* because of the migration to Debian base images.

To update the required components to a newer version:

1. From the [Keysight CyPerf Support](#) page, download the CyPerf controller upgrade package or the ATI upgrade package, and save it locally.
User registration is required to access the [Keysight CyPerf Support](#) page.
2. Access the CyPerf web UI and log in.
3. Under the **Settings** menu, select **Software Updates**. The **Software Updates** page opens.
4. Click **Select packages for upload**, and choose the upgrade package you downloaded in step 1.
5. Then, click **Open**. The **Upload file** dialog opens informing you of the estimated time it takes to upload the selected package, and warning you against refreshing the page during the upload.

6. Click **Upload** to confirm the action.



IMPORTANT: Upload progress will be lost upon page refresh.

The upgrade package starts uploading and progress is shown on the screen. When the upload is complete, a notification is displayed under **Events** informing you that the upload finished successfully. The changes that will be applied are highlighted for each corresponding component.

If an error occurs during the upload process, an **Upload report** will be displayed on the **Software Updates** page.

7. Click **Start update** to initiate the software update process.

Starting from release 4.0, CyPerf agents that are identified not to be running the recommended version will be automatically updated during the CyPerf controller update. When this happens, a notification appears in the lower right corner of the screen, informing you of the number of agents that require an update and the version to which they will be updated.

For agents that don't support the update feature from the controller UI (specifically, those running on versions older than 3.0), a notification prompts for a manual update. However, the CyPerf controller update continues even if an agent update fails for any reason.



NOTE: These notifications will not appear immediately after initiating the controller update. Only when the deployment service installs the package that checks which agents support the update operation, will these notifications be displayed. Even if you miss these notifications during the update process, you can access them later from the **Events** page.

When the update process successfully completes, the newer software version for each upgraded package will be listed on the **Software Updates** page.

Software Updates Page

The following UI elements are available from the **Software Updates** page:

UI Element	Description
<i>Installed versions</i>	
CyPerf ATI Update	Shows the currently installed ATI version.
CyPerf Controller	Shows the currently installed CyPerf controller version.
CyPerf Agent (recommended)	Shows the recommended agent version for the currently installed CyPerf controller.
Select packages for upload button	Click this button to choose and start the upload process for the upgrade package you want to install. See also Update Software .
Upload report	Shows any errors triggered during the package upload process. See Update Software .
CyPerf ATI Update	Shows the current ATI version, and the version to which the package will be upgraded.
CyPerf Controller	Shows the current CyPerf controller version, and the version to which the package will be upgraded.
CyPerf Agent (recommended)	Shows the current CyPerf agent version, and the version to which the package will be upgraded.
Start Update button	Click this button to start the CyPerf controller and agent update or the ATI package update. See also Update Software .
Reset button	Click this button if you want to remove the upgrade packages you uploaded during the Select packages for upload process.

Update Agents from the Controller UI

Starting from release 3.0, the CyPerf controller package includes an agent installation package that is compatible with the current version of the controller. The recommended agent version for use with the current controller version is displayed under the **Installed Software** section on the **Software Updates** page. If the agent version you are currently using differs from the recommended version, you can update multiple traffic agents directly from the **Agent Management** page of the controller's user interface, as described below. Starting from release 4.0, you can also automatically update agents during the controller update as described in the [Update Software](#) section.



NOTE:

- To use the agent update feature from the controller's UI, agents need to be registered and connected to the current controller.
- Agents must support the update feature from the controller's user interface (UI). When an agent registers with the controller, it sends a message containing a list of features, including the update feature. Based on this registration message, the controller can distinguish between agents that support or do not support the update feature. If you attempt to update an agent that does not support the update feature from the controller's UI, a warning notification will be displayed on the screen. This notification informs you that the update operation cannot be performed for the selected agent(s) through the controller's UI, and a manual update is required.

To update the CyPerf agent from the Agent Management page:

1. Access the CyPerf web UI and log in.
2. Under the **Settings** menu, select **Agent Management**. The **Agent Management** page opens listing the available agents and their current versions under the **Traffic Agents Info** column. If an agent has a different version than the recommended version, it will be flagged with a yellow warning icon which, when hovered over, will list the recommended agent version to run.
3. Select the agent(s) you want to update.
4. Click the **Update** button.

During the update operation, the `.deb` file for the agent recommended version is uploaded and installed on each selected agent. If the operation is completed successfully, a pop-up notification is displayed for each updated agent in the lower right corner of the screen.

If you click the **Update** button for agents that are already running the recommended version, the controller skips the operation for these agents, and a notification is displayed informing you that no update is required. For agents that do not support the update feature from the controller UI (i.e., agents running on versions older than the CyPerf 3.0 release), you can do the update manually as in pre-3.0 CyPerf versions.

See also [Software Updates](#).

Manage Test Sessions

When you create a new test, CyPerf establishes a *test session* which remains available until you decide to delete it. This way, you can access existing test configurations to change the settings and to view details, or to re-run a test.

Keysight CyPerf allows you to:

- [Create a View a New Test Session](#)
- [Access Recent Test Sessions](#)
- [Delete a Test Session](#)
- [Navigate between Test Sessions](#)

Create and View a New Test Session

After you access the web UI, the **Dashboard** page is displayed, which allows you to create, delete, view, or navigate between test sessions.

You can create a test session either directly from the **Dashboard** page, or from the **Browse Configs** page.

To create a new test session from the Dashboard page:

1. Access the web UI from a supported browser and log in. The **Dashboard** page opens.
2. Under the **Create New Test** panel, click one of the tests listed under **Test Types**, or click the **Create New Test** button.

The **Test Overview** page opens displaying different areas available for configuration, including a graphical representation of the test topology. CyPerf assigns a session number and a name to the test, and displays that information in the **Navigation** bar. You can now start configuring the test.

To create a new test session from the Browse Configs page:

1. Access the web UI from a supported browser and log in. The **Dashboard** page opens.
2. From the **Dashboard** page, click **Browse Configs**. The **Browse Configs** page opens.
This page displays the recently used configurations as well as all the predefined test configurations shipped with CyPerf (grouped under different tabs, such as CyPerf Features Configs, CyPerf Performance Configs, and other config types).
3. Create a new test session by clicking any of the configurations listed on the **Browse Configs** page.

If a session using the same configuration is already available, you are prompted to choose which action you want to do next: either create a new session or open the existing one. Click **Create** to add a new session based on the same configuration. (Alternatively, click the existing session to open it.)

The **Test Overview** page opens displaying different areas available for configuration, including a graphical representation of the test topology. CyPerf assigns a session number and a name to

the test, and displays that information in the **Navigation** bar. You can now start configuring the test.

NOTE: It is recommended that you decide on an IP addressing scheme before you start configuring a test. Alternatively, you can determine the IP addressing as you configure the test settings. Although you may choose to completely configure each node one at a time (including IP addresses), it is recommended that you start by configuring the IP addresses for the entire test topology.

To view the newly created session, follow the steps below:

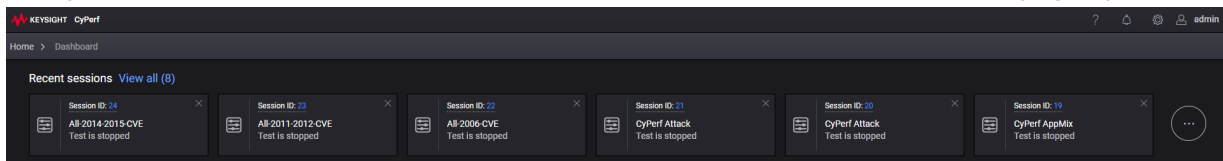
1. After creating the new test session, click **Home** on the **Navigation** bar to return to the **Dashboard** page.
The newly created test session is displayed on the **Dashboard** page under the **Test sessions** panel, along with other already created test sessions.
2. To see the full list of sessions available in your environment, click **View All** next to the **Recent Sessions** label, or the **View More**  button in the right hand corner of the **Test sessions** panel. The **Browse Sessions** page opens listing all the available sessions.

For more details, see also the [Dashboard Page](#) chapter.

Access Recent Test Sessions

To enter an already created test session:

1. Log in the web UI. The **Dashboard** page opens.
2. From the **Recent sessions** panel under the **Home > Dashboard** page, select the test session that you want to access. The selected session is loaded, and the test **Overview** page opens.



Navigate Between Test Sessions

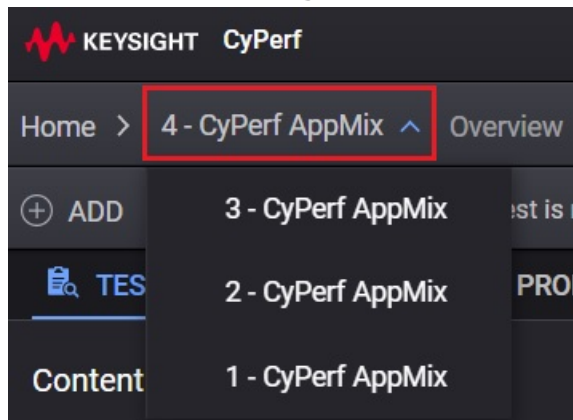
After you access the web UI, the **Dashboard** page is displayed, which allows you to create, delete, view, or navigate between test sessions.

NOTE: The procedure below assumes that at least two test sessions have already been created for the currently logged on user. For more details on how to create a session, see [Create a Test Session](#) section.

To navigate from one test session to another:

1. Access the web UI from a supported browser.
The **Home > Dashboard** page opens listing any previously created test sessions under the **Test sessions** panel.
2. From the **Recent sessions** panel, select any test session tile. The **Overview** page for the selected test is displayed.
3. From the **Overview** page, select the test drop down list on the upper left side of the **Navigation** bar, and select another test session that you want to enter. The selected test's **Overview** page is displayed.

Optionally you can navigate back to the **Dashboard** page by selecting **Home** in the upper left side corner of the **Navigation** bar and enter another test session as shown in step 2 above.



Reuse a Test Session

Keysight CyPerf allows you to reuse existing test sessions by saving them as test configurations. The **Browse Configs** section offers access to existing test configurations so that you can view details, change settings, or re-run tests.

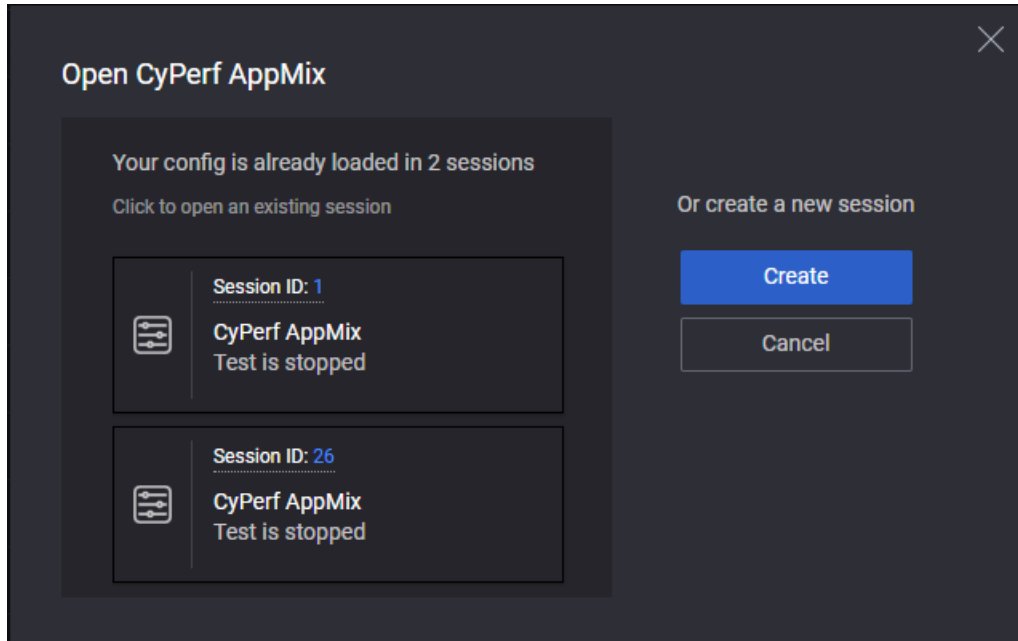
To use pre-canned configurations:

1. Select **Browse Configs** from the **Dashboard** page.
2. Select any of the configuration tabs under **Config Categories**.

To create a new test configuration:

1. Open the test session you want to reuse.
2. Select the **Save** button  in the upper right side corner of the page.
3. Select **Save config as**.
4. Enter a name for your test configuration, and then select **Save**. The newly created test configuration is now available in the **Browse configs** section.

To open a saved test configuration, select it from the **Browse Configs** page or from the **Browse Sessions** page. If your test configuration is already loaded in a test session, you are prompted to either open the existing session, or create a new one.



See also [Browse Sessions Page](#).

Delete a Test Session

After you access the web UI the **Dashboard** page is displayed, which allows you to create, delete, view, or navigate between test sessions.

To delete a previously created test session:

1. Access the web UI from a supported browser, the **Dashboard** page opens.
2. From the **Test sessions** panel on the **Dashboard** page, select the **Close** button corresponding to the test tile you want to remove.
3. Select **DELETE** when you are prompted to confirm the delete action. The test session is removed from the server and is no longer displayed on the **Test sessions** panel.

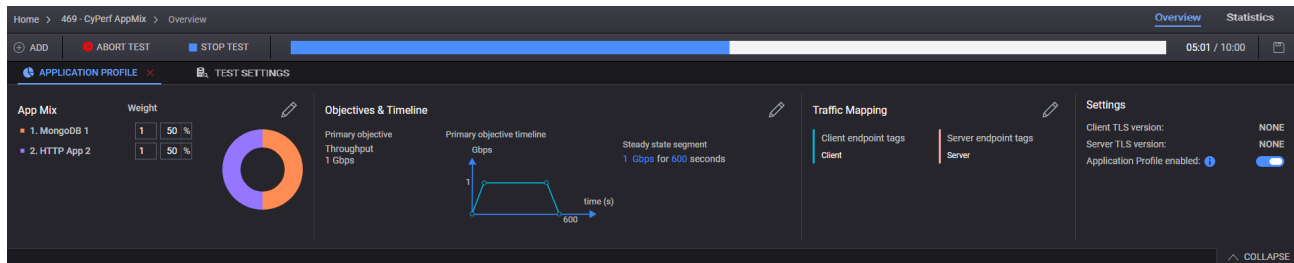


NOTE: Deleting locked or pinned sessions is not currently supported from the web UI.

Start and Stop Tests

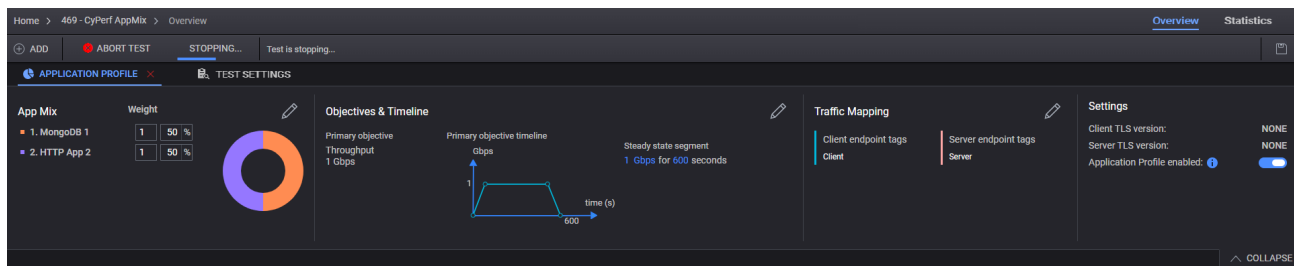
After you set up all the required test configuration parameters, you can start the test by selecting the **START TEST** button on the test progress bar. The progress bar shows the stages that the test goes through, for example the test is initializing, progressing, and then stopping. The test status is also displayed on the test session tile on the **Dashboard** page.

After the test starts, the **STOP TEST** button becomes available on the progress bar. The estimated total time that it takes the test to complete and the time passed from the test start are also displayed on the progress bar. Once the test is running, the **STATISTICS** page is displayed allowing you to view the real-time and test specific statistics.

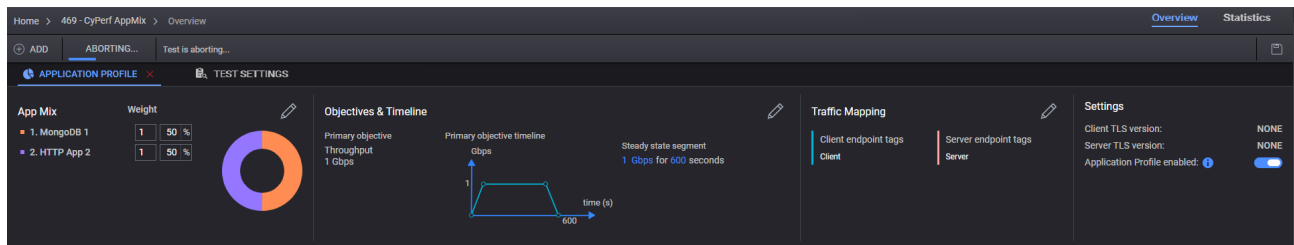


If for any reason you want to stop or abort the test before it completes, select the **STOP TEST** or the **ABORT TEST** button respectively.

The **STOP TEST** button initiates a *Graceful stop*, which can end successfully, or end in a *Soft abort*, as described below.



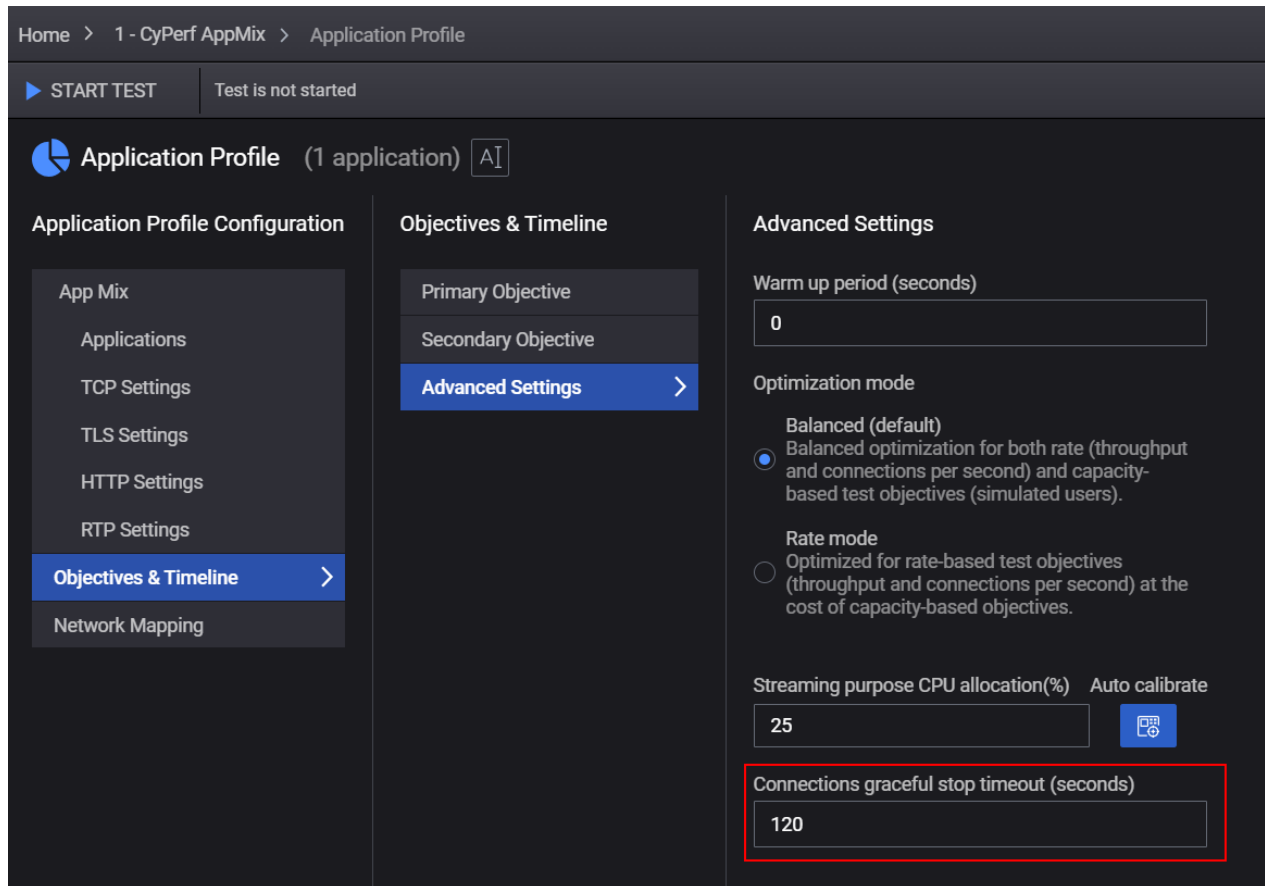
The **ABORT TEST** button initiates a *Hard abort*. The **ABORT TEST** button is available while the test is running, and also after a *Graceful stop* is launched. However, when a *Hard abort* is initiated, the *Graceful stop* operation is no longer an option.



There are three ways a test can end:

Graceful stop

A graceful stop is initiated when the test timeline is completed, or by clicking the **STOP TEST** button. The agent gracefully closes open connections after it waits for flows either to be completed or to time out based on the value you configured for the **Connections Graceful Stop Timeout** parameter. This value can be configured for both Application profiles (under **Objectives and Timeline** > **Advanced Settings** > [Connections Graceful Stop Timeout](#)) and Attack profiles (under **Timeline** > **Timeline** > [Connections Graceful Stop Timeout](#)).



Soft abort

A soft abort is initiated when the **Connections Graceful Stop Timeout** is reached, and the agent immediately closes open connections without waiting for flows to finish. The timeout for this operation has a default value of 5 seconds, which is not configurable.

Hard abort

A hard abort is initiated by clicking the **ABORT TEST** button. This operation can be done while the test is running or after clicking the **STOP TEST** button. The ABORT operation kills the traffic agent process, and it may influence the statistics accuracy at the end of the test. If the ABORT operation is not successful, the agent's logs and capture files will not be accessible. However, the agents will be released and available for other tests.

Manage Network Topology

Each test's **Overview** page displays the test topology used in the current configuration. From the test topology section, you can:

Configure a DUT	41
Configure Multiple DUTs	42
Configure a ZeroTrust DUT	43
Map a Network Segment to a DUT	45
Configure Network Segments	46
Configure VLAN Range Properties	47
Assign Traffic Agents	50
Manage Application Network Mapping	51
Manage Attack Network Mapping	52



NOTE:

If the DUT is enabled in the UI, you can allocate more than one agent for a network segment, and you can use more than one IP for the server network segment IP range. For more information on the UI elements visible on the test topology section, see [Test Topology](#).

Configure a DUT

You can configure the DUT(s) you want to use in your tests from each DUT's **Configuration** page. The DUT's **Configuration** page allows you to set up both simple configuration options (HTTP, HTTPS, TCP Health Check) and advanced configuration options (Forward Proxy, Zero Trust PEP, Reverse Proxy).

See also [DUT Properties](#).

To configure the device under test:

1. Enable the DUT configuration using the toggle on the **DUT Network** tile in the test topology on the test **Overview** page.
2. Select **Click to configure** on the **DUT Network** tile. The DUT's **Configuration** page opens, which allows you to set up the host, and configure the DUT's HTTP, HTTPS, and TCP Health Check options, as well as advanced configuration options such as Forward Proxy, [Zero Trust PEP](#), Reverse Proxy.

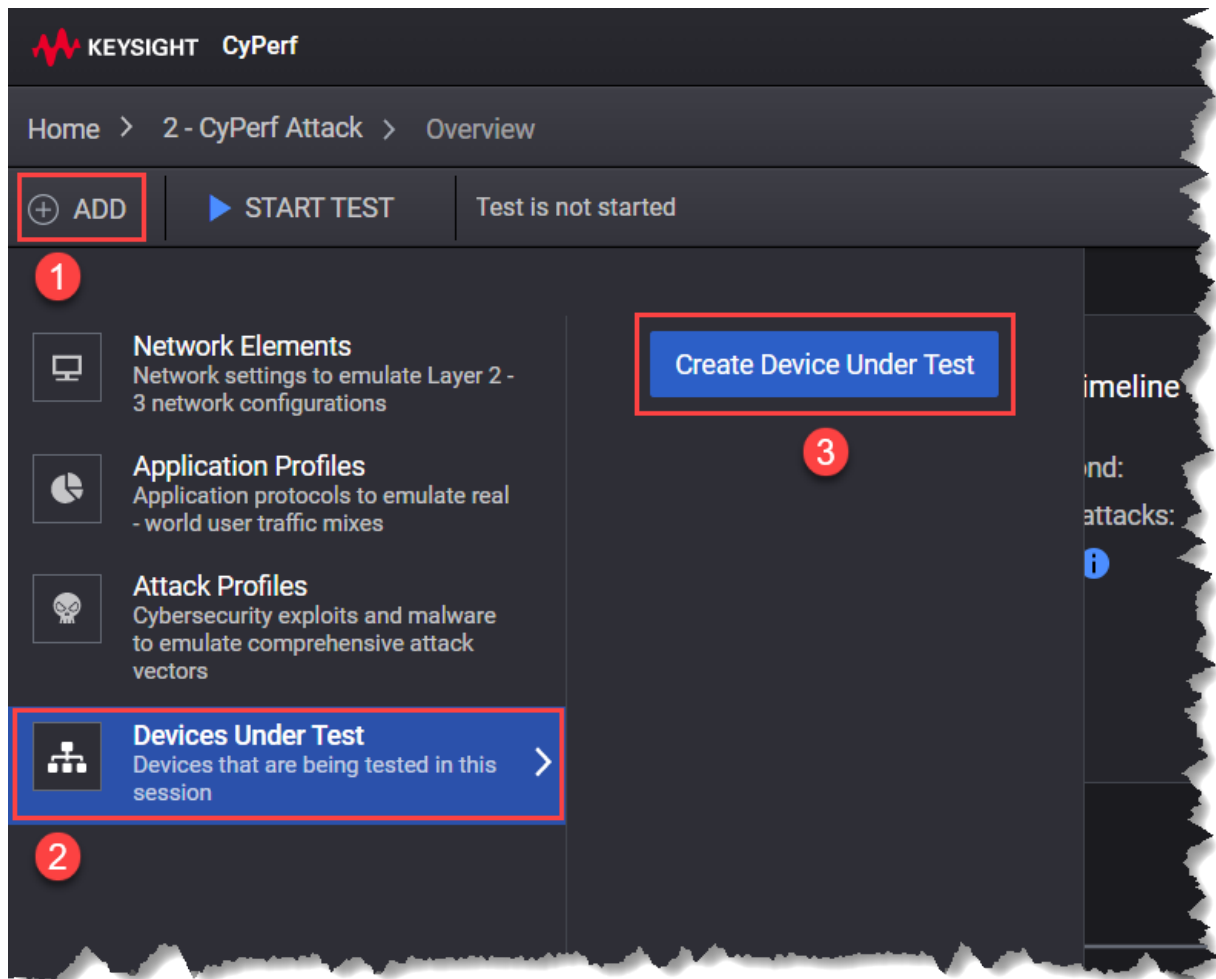
To rename an existing DUT:

1. Select the **Rename** button next to the current DUT Network name.
2. Enter the new name for the DUT. Click anywhere within the page to save your modification.

Configure Multiple DUTs

To add more than one DUT to a test:

1. Enter an existing test session.
2. Select **Add > Devices Under Test > Create Device Under Test**.



The new DUT is added to the Topology section of the test **Overview** page.

You can now enable the new DUT's configuration and also add new Network Segments for the new DUT if required.

NOTE: If you add Network Segments before adding multiple DUTs, these Network Segments are automatically mapped to the default DUT. For more information, see [Configure Network Segments](#).

Configure a ZeroTrust DUT

With **ZeroTrust PEP** configuration enabled, you can run CyPerf HTTP applications through a PEP¹ device that requires authentication at the Application layer.

When a ZeroTrust PEP device is added to a test, all the applications mapped to that device will perform authentication if requested by the device, as described below:

1. Applications will follow HTTP Redirect responses (3xx).
2. Two scenarios are possible:
 - a. If they are redirected to the configured real IdP² (for example, the third party Okta service), they will provide the configured credentials to that IdP.
 - b. If using a Simulated IdP instead, the client agent will generate and sign the appropriate SAML response internally, and provide those to the PEP directly.
3. If authentication with the IdP is successful, they will follow the IdP responses back to the PEP and provide any cookies or SAML responses in the required format.
4. Authentication will be automatically done whenever the PEP device requests it.
5. Cookies will be automatically deleted after a configurable number of Application executions to force a new re-authentication.

CyPerf supports various PEP devices and real IdPs based on **Authentication Profiles**. These profiles contain instructions on how to process PEP and IdP responses at scale. The currently supported profiles are:

- For **Forward Proxy**:
 - **Juniper Zerotrust Device**: Juniper Zerotrust Device PEP, with SAML authentication through the Okta IdP
 - **PAN Explicit Proxy** PEP, with SAML authentication through the Okta IdP, or the Simulated IdP
- For **Reverse Proxy**:
 - **Cisco Firepower**: Cisco Firepower PEP with SAML authentication through the Simulated IdP
 - **Cisco Umbrella**: Cisco Umbrella PEP with SAML authentication through the Okta IdP, or the Simulated IdP
 - **F5 BIG-IP Identity Aware Proxy** PEP, with SAML authentication through the Okta IdP, or the Simulated IdP
 - **PAN GlobalProtect Clientless VPN** PEP, with SAML authentication through the Okta IdP, or the Simulated IdP
 - **Generic PEP**, with SAML authentication through the Simulated IdP

¹. Policy Enforcement Point (PEP): a device placed between clients and a secured network, which verifies the authentication status of every connection and applies per-user access rules.

². Identity Provider (IdP): a service or device that performs authentication (typically user/password based, potentially including Multifactor Authentication)

To configure a ZeroTrust DUT:

1. Enable the DUT configuration using the toggle on the **DUT Network** tile under the test topology on the test **Overview** page.
2. Select **Click to configure** on the **DUT Network** tile. The DUT's **Configuration** page opens.
3. Under **DUT Configuration > Config settings**, select the **Advanced configuration** option.
4. On the **Forward Proxy** or the **Reverse Proxy** tabs, use the toggle to enable Forward Proxy or Reverse Proxy configuration. Then, enter the hostname, port number, and other required settings. Optionally, you can also configure a domain suffix for Forward Proxy.
See also [Advanced Configuration](#) for descriptions of the ZeroTrust PEP settings for Forward and Reverse Proxy.
5. On the **ZeroTrust PEP** tab, use the toggle to enable configuration of the ZeroTrust PEP for Forward or Reverse Proxy, and select an available authentication profile for the real or simulated IdP.
6. Configure the parameters that become available for the authentication profile you selected.

To configure Simulated IdP:

If choosing **Simulated IdP** as the IdP type in the previous section, an additional configuration group will appear for the Simulated IdP settings.

1. Configure these parameters with the desired values.
Additional information for each Simulated IdP parameter is available in [Advanced Configuration](#) for both Forward Proxy and Reverse Proxy.
2. After the parameters are configured, click the **Download Simulated IdP metadata** button.
You should see a file download starting.
3. Use the file generated at the previous step to register the Simulated IdP as an authentication option in the PEP device that is being tested.

Map a Network Segment to a DUT

To map a Network Segment to a specific DUT:

1. Select a Network Segment from the **Topology** section of the test **Overview** page.
2. Under **Network Configuration**, select **DUT Connections**.
3. Select a specific DUT Network Segment, or choose **Select All** if you need a client Network Segment to go through globally.

 **NOTE:** Repeat the steps above for all your Network Segments.

To remove or modify the Network Topology DUT for an Attack or Application Profile:

1. Select the **Edit** button on the **Attack List** or **Application Mix** tile.
Depending on your selection, either the **Attack Profile Configuration** page or **Application Profile Configuration** page opens.
2. Add the attack or application you want to use (if not already added), and navigate to **Network Mapping > Network Endpoints > Network Topology**.
3. Under **Select DUTs**, choose the DUT(s) you want the application or attack to go through.

 **NOTE:** You can see any changes you made to the traffic under the Topology section of the test **Overview** page.

Configure Network Segments

Add new Network Segments

Network Elements are network settings used to emulate Layer 2-3 network configurations.

To add a new Network Element:

1. Select **Add > Network Elements > Create IP Network**.
Your new network topology is visible on the **Overview** page.

How to configure Network Segments

To change the configuration of an existing Network Segment:

1. Select the Network Segment on the **Overview** page.
2. The **Network Configuration** page opens. Here you can configure **Network Stacks** with **IP Stacks**, **TLS VPN Stacks**, **IPSec Stacks**, and **MAC DTLS Stacks**, properties for **Emulated Router**, **Ethernet Range**, **DNS Resolver**, **DNS Server**, and properties for the connected DUTs in **DUT Connections**.

You can also:

- Leave the IP Range settings on **AUTOMATIC**; in this case, the test uses the IP Addresses configured on the agent's test interface through CLI.
Setting Multiple Automatic IPs is also supported on local setups or in Cloud. For detailed configuration steps on Cloud setups, see the specific Cloud documentation.
- Set the **Automatic IP type** which supports IPv4 only, IPv6 only or IPv4 and IPv6.
- Set a **Static IP** to be used in the test if the **AUTOMATIC** setting is disabled on an IP Range.



NOTE:

- When you run an **Application Profile** test using multiple IP Ranges, ensure that the **Simulated Users** variable is set to values that accommodate all of the IP's set by the IP Ranges. Similarly, when you run an **Attack Profile** test using multiple IP Ranges, set the **Max Concurrent Attacks** variable to values large enough to accommodate all of the IP's set by the IP Ranges.
- When the defined IPs are larger than the **Simulated Users** or **Max Concurrent Attacks** variables, only a limited number of IPs from the defined IP Ranges are used.

For more information on the **Ethernet Range** settings, see [Network Segments Properties](#).

Rename Network Segments

To rename a Network Segment:

1. Select the **Rename**  button next to the current IP Network Segment name.
2. Enter the new name. Select anywhere within the page to save your modification.

Configure VLAN Range Properties

To access the VLAN range properties:

1. Select an IP Network.
2. Select **Network Configuration > Network Stacks > IP Stacks**.
3. Select the IP Range and enable VLAN under **VLAN Range Properties** using the toggle button. Configure the available VLAN options.

The following fields are available under **VLAN Range Properties**:

- **VLAN ID**, the first VLAN ID to be used for VLAN. The minimum value is 1 and the maximum value is 4094
- **Increment by**, the value to be added to the VLAN ID for each new assignment. The maximum value is 4094 and the default value is 1
- **Count**, the number of unique VLAN IDs that will be created. The default value is 1 and the maximum value is 4094.
- **Count per agent**, the number of VLAN IDs that are assigned to each agent in round-robin order.



NOTE: After the total count is exhausted, the next agent re-starts from the first VLAN ID.

- **Tag Protocol Identifier:**
 - the Inner VLAN range needs to be 0x8100 (non-editable)
- **802.1Q Priority**, the VLAN priority where the minimum value is zero and the maximum value is 7.

VLAN configuration examples

The below examples represent two VLAN and IP count cases.

1. When the VLAN count is less than or equal to the IP count:

- The IP Range properties:

IP Start	10.0.0.10
IP increment	0.0.0.1
Count	4

- The VLAN Range properties:

VLAN ID	100
Increment by	1
Count	2
Tag Protocol Identifier	0x8100
802.1Q Priority	1

This configuration results in the below combinations:

<i>IP VLAN</i>	
10.0.0.10	100
10.0.0.11	101
10.0.0.12	100
10.0.0.13	101

2. When the VLAN count is greater than the IP count:

- The IP Range properties:

IP Start	10.0.0.10
IP increment	0.0.0.1
Count	2

- The VLAN Range properties:

VLAN ID	100
Increment by	1
Count	4
Tag Protocol Identifier	0x8100
802.1Q Priority	1

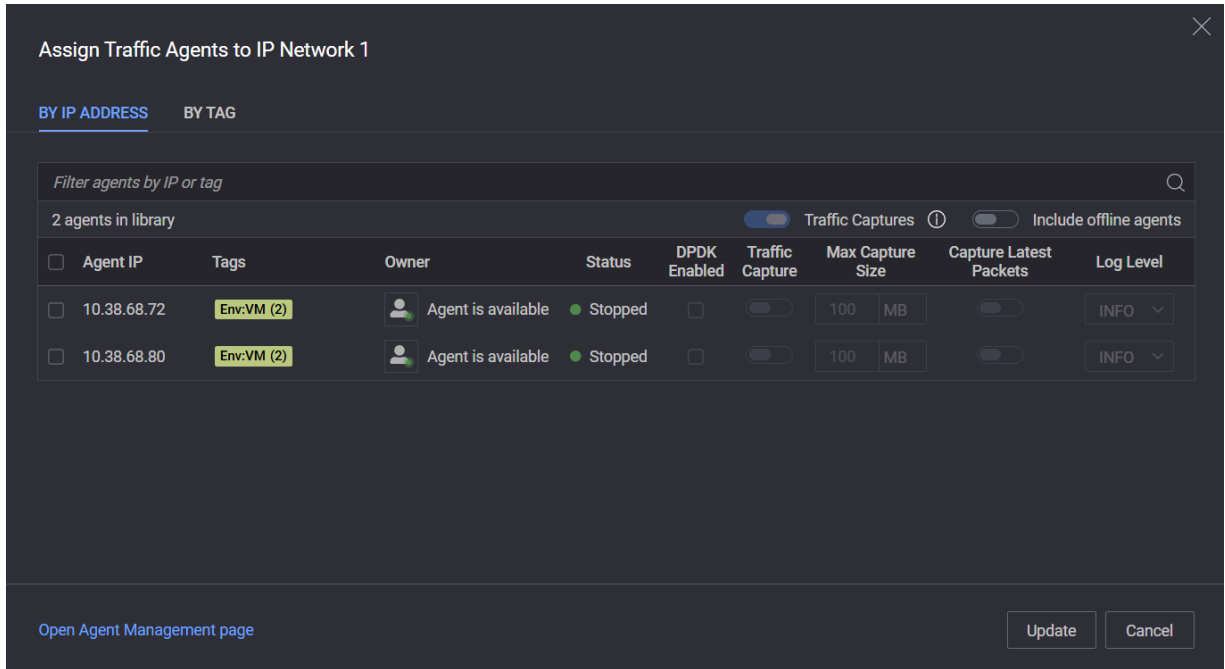
This configuration results in the below combinations:

<i>IP VLAN</i>	
10.0.0.10	100
10.0.0.11	101
10.0.0.10	102
10.0.0.11	103

Assign Traffic Agents

To assign traffic agents to an IP network segment:

1. Select the yellow icon  on the IP network segment you want to configure. A dialog box opens, displaying the list of available traffic agents.



Assign Traffic Agents to IP Network 1

BY IP ADDRESS BY TAG

Filter agents by IP or tag

2 agents in library

Traffic Captures  Include offline agents 

☐	Agent IP	Tags	Owner	Status	DPDK Enabled	Traffic Capture	Max Capture Size	Capture Latest Packets	Log Level
☐	10.38.68.72	Env:VM (2)	 Agent is available	 Stopped	☐		100 MB		INFO 
☐	10.38.68.80	Env:VM (2)	 Agent is available	 Stopped	☐		100 MB		INFO 

Open Agent Management page

Update Cancel

2. Select a traffic agent by clicking the checkbox icon () on the **By IP Address** tab. Alternatively, you can use the **By Tag** tab to select agents based on their assigned tags.



NOTE:

If a traffic agent is grayed out, it is currently offline.

3. Use the toggle button to turn **Traffic Capture** on or off.
4. Configure the maximum capture size.
5. Use the toggle button to indicate whether you want to capture the latest packets.
6. (Optional) For debugging purposes only, you can change the **Log Level** for the selected agent. By default, the **INFO** log level is enabled. See also [Decrypt TLS Packets](#).
7. Click **Update** to assign the selected traffic agent to your IP network segment.

Manage Application Network Mapping

To configure the traffic flow for an application, do the following:

1. On the **Overview** page of an existing test session, under the **Application Profile** tab, select the **Edit**  button on the **Traffic Mapping** tile. The **Application Profile** page opens with the **Network Mapping** tab selected.
2. Under **Network Endpoints**, select the application you are interested in.
Under **Network Topology**, select the network segment you are interested in. You can choose from:
 - **Clients**
 - **Servers**

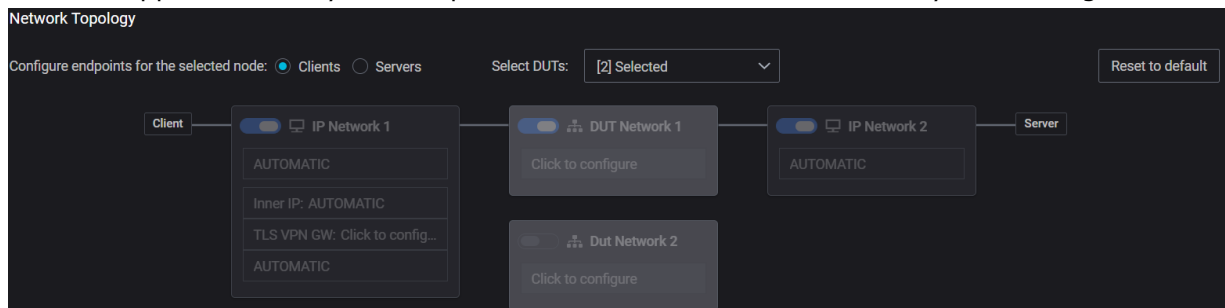
The network topology reflects your selection by briefly highlighting the targeted network segment.



NOTE:

You can also map traffic and applications by IP Ranges. You can add tags to IP ranges by navigating to **Overview > Automatic/IP Range > Add tags** or from **Network Segment > IP Ranges > Network tags**.

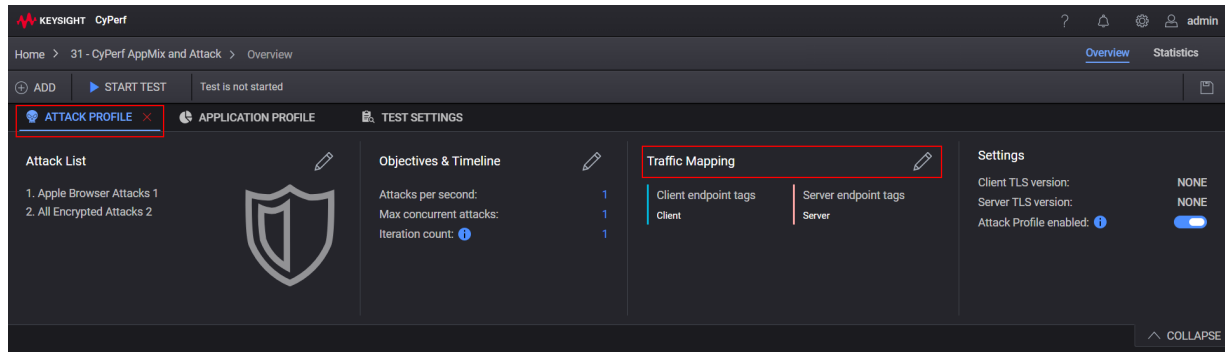
3. Select the tags assigned to the network segment you are interested in. A yellow dot is shown next to the application name under the **Network Endpoints** section. This indicates that the item is mapped differently than its parent. You can also select other newly created tags.



Manage Attack Network Mapping

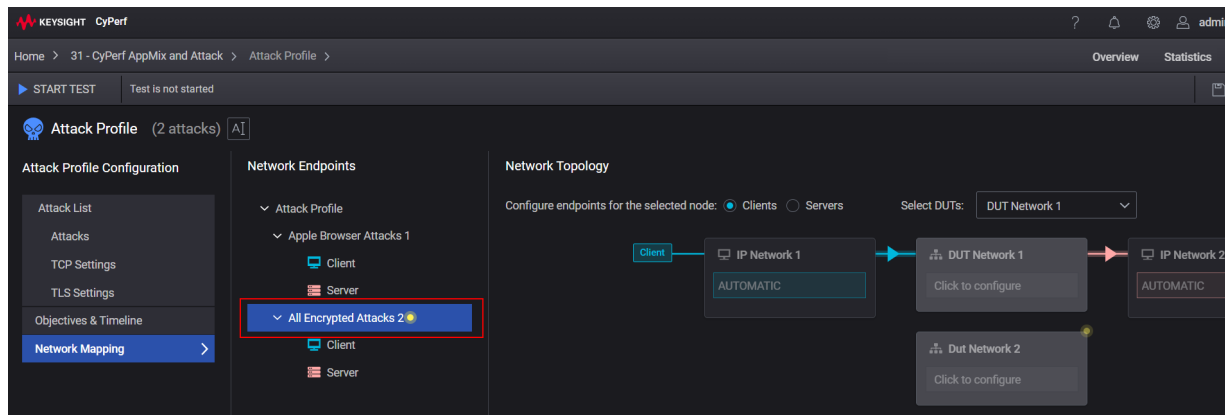
To configure the traffic flow for an attack, do the following:

1. In the **Attack Profile** section, select the **Edit** icon in the **Traffic Mapping** sub-section. The **Network Endpoints** sub-section appears.



2. In the **Network Endpoints** sub-section, select the attack you are interested in.
3. Configure the endpoints for the selected node by selecting the DUT and the tags assigned to the network segment you are interested in.

If a yellow dot is shown next to the attack name under the **Network Endpoints** section, this indicates the item is mapped differently than its parent.



Customize Application and Attack Profiles

Keysight CyPerf allows you to create custom traffic and attack profiles to employ in your tests.

You can perform the following actions from CyPerf:

Manage Applications and Application Actions	53
Manage Attack Actions and Strikes	56

Manage Applications and Application Actions

You can manage applications from the **Application Profile Configuration** page which allows you to add new applications, configure applications and application actions, and delete applications.

Add Applications

1. In an existing test session, under the **Application Profile** tab, select the **Edit**  button on the **Traffic Mapping** tile.
The **Application Profile Configuration** page opens. Under **Application Profile Configuration > Applications**, all the already added applications are listed. If no applications are listed, you can add the applications you want to use.
2. Click the **Add**  button. The **Add Application(s)** window opens.
3. Search for the application you want to add, and select the **Add**  button next to the application (s) you want to add.
4. Then, click the **Add** button to confirm your action and close the **Add Application(s)** window.
The newly added applications are now listed on the **Application Profile** page under the **Applications** list. You can now further configure the added applications by adding or deleting application actions, or configuring TCP, TLS, or HTTP settings at the profile level or the per-application level.

Add Application Actions

1. In an existing test session, under the **Application Profile** tab, select the **Edit**  button on the **Traffic Mapping** tile. The **Application Profile Configuration** page opens. Under **Application Profile Configuration > Applications**, all the already added applications are listed.
2. Select an already created application from the **Applications** list. The **Actions** section becomes available.
3. Select the **Add Actions**  button. The **Add Actions** window opens.
4. Select one or more actions from the existing library, and click the **Insert Actions**  button.
5. Then, click the **Add** button to confirm your action and close the **Add Actions** window. The **Actions** list is now updated with the newly added action(s).

**IMPORTANT:**

When you add two applications on the same server with the default settings and try to start a test, an error similar to the following may be triggered:

Failed to initialize test: Operation finished with error: FTP 1 and FTP 2 are running on the same agent and port. Make sure the actions are in a different order or have different parameters values for one of the actions preceding LIST: CWD

This indicates that the two applications (named FTP 1 and FTP 2, in this example) are not unique enough for the server to know which request it is serving. The error indicates two of the possible solutions you can implement to avoid the error the next time you start the test:

- The order of actions can be changed for one of the applications.
For example, you can change the order of the **SYST** and **MKD** actions in FTP 2.
- The parameters of the **CWD** action can be changed for one of the applications.
For example, you can change the **Directory path** parameter in FTP 2 to `/cyperf2` instead of the default `/cyperf`.

Delete Application(s)

You can delete applications one by one, or delete multiple applications simultaneously.

To delete one application:

1. From the **Applications** list, click the **Delete**  button next to the application you want to remove.

To batch delete multiple applications:

1. From the **Applications** list on the **Application Profile Configuration** page, select the **Edit** button.

Home > 1 - CyPerf AppMix > Application Profile

▶ START TEST Test is not started

Application Profile (3 applications) AI

Application Profile Configuration

- App Mix
- Applications** >
- TCP Settings
- TLS Settings
- HTTP Settings
- Objectives & Timeline
- Network Mapping

Applications Edit +

Q Search applications by name...

Active	Name	TLS	Weight	
☒	Adobe Reader ...		1 33.33%	AI ⚙️ 🗑️ ⇅️
☒	AWS Console ...		1 33.33%	AI ⚙️ 🗑️ ⇅️
☒	eBanking Chro...		1 33.33%	AI ⚙️ 🗑️ ⇅️

The list of applications is now unlocked, allowing you to select the applications you want to remove.

Home > 1 - CyPerf AppMix > Application Profile

▶ START TEST Test is not started

Application Profile (3 applications) AI

Application Profile Configuration

- App Mix
- Applications** >
- TCP Settings
- TLS Settings
- HTTP Settings
- Objectives & Timeline
- Network Mapping

Applications Cancel 🗑️

Q Search applications by name...

- ☐ **Name**
- ☐ Adobe Reader Updates Firefox
- ☐ AWS Console Microsoft Edge
- ☐ eBanking Chrome to Apache

2. Select each application you want to delete. You can also delete all the applications on the list (use the check box next to the **Name** label).
3. Then, click the **Delete** 🗑️ button. The applications are now removed from the list.

For examples of the available application actions and action parameters, see [Application Actions](#).

Manage Attack Actions and Strikes

You can customize your attacks by adding or removing attack actions and strikes to simulate various vulnerabilities.

Add Attack Actions and Strikes to a New Attack

1. Go to **Attack Profile > Attack Profile Configuration > Attacks**, and select the **Add**  button. The **Add Attack(s)** dialog opens with the **ATTACK LIBRARY** tab active.
2. Filter the attacks based on category name or value, for example **Browser**, **Date**, **Direction**, **Intent**, and others. Then, from the filtered attack list, click the **Add**  button next to the attack to add it to your list.
3. To customize the attack, switch from the **ATTACK LIBRARY** tab to the **CUSTOMIZE ATTACK** tab. A four-pane window opens.
4. Optionally, select a target application either from an existing **Application Profile** or the **Application Library**.
To deselect the application, click the same highlighted target application.
5. Filter strikes based on category name or value, for example **Browser**, **Date**, **Direction**, **Intent**, and others.
6. From the filtered strikes list, select one or more strikes, and add them using the **Add**  button. If you skipped **Step 3**, simply insert strikes in the order you see fit. A standalone strike is also a valid attack.
7. Click the **Add** button in the lower right corner of the dialog to create your custom attack and close the **Add Attack(s)** dialog.

Add Attack Actions and Strikes to an Existing Attack

1. Go to **Attack Profile > Attack Profile Configuration > Attacks** and select an existing attack. The **Actions** panel is now visible.
2. Select the **Add Actions**  button. The **Add Actions and Strikes** window opens.
3. Under the **Add Actions** tab, select one or more attack actions and add them using the **Insert Action** button.



NOTE: The actions can be added only to the pre-canned attacks, not to individual strikes.

4. Select the **Add Strikes** tab.
5. Select one or more strikes from the list and insert them using the **Insert Strike** button.
6. Select **Add** to complete the configuration.

Delete Attack(s)

There are two ways to delete attacks:

1. Delete individual elements from the attacks list by selecting the delete icon.

Or

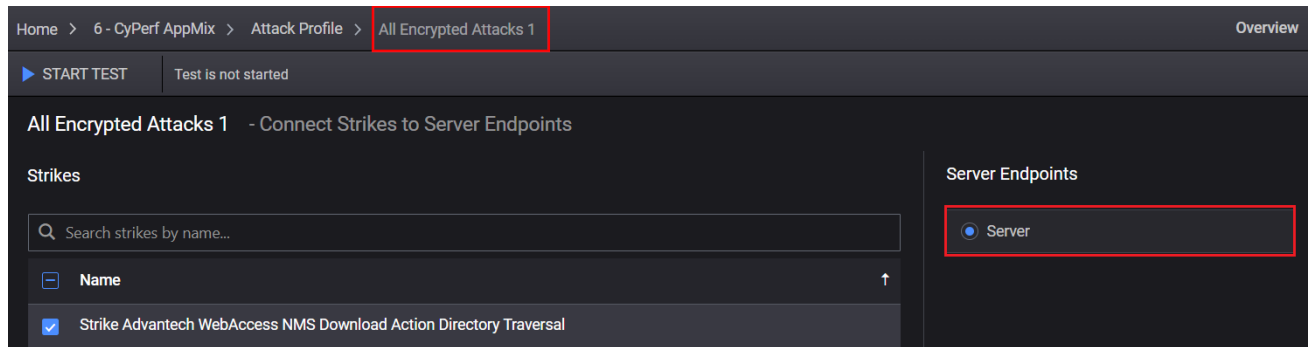
1. Go to **Attack Profile > Attack Profile Configuration > Attacks**.
2. Select the **Edit** button. The list of attacks is now unlocked.
3. Select the names of the attacks you want to batch delete.
4. Select the **Delete** button.

For more information about attack actions, see [Attack Actions](#).

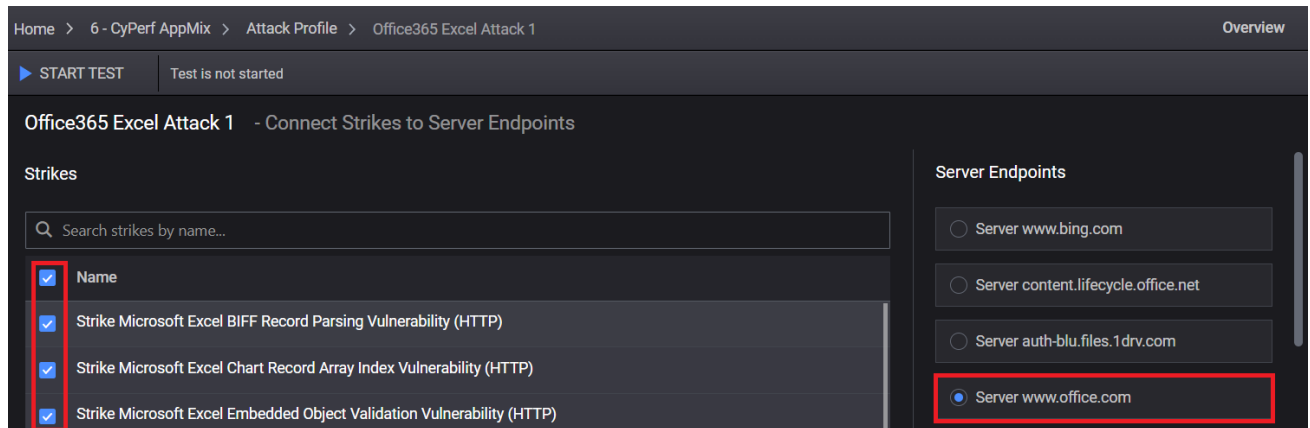
Connect Strikes to Server Endpoints

1. Go to **Attack Profile > Attack Profile Configuration > Attacks**, and select the attack you want to configure. The **Actions** section is now visible.
2. Select the strike you want to connect to the server endpoint. Then, click the **Connect Strikes to Server Endpoints**  button.

There is only one server available for any pre-canned attack or strike as a standalone attack, which is selected by default:



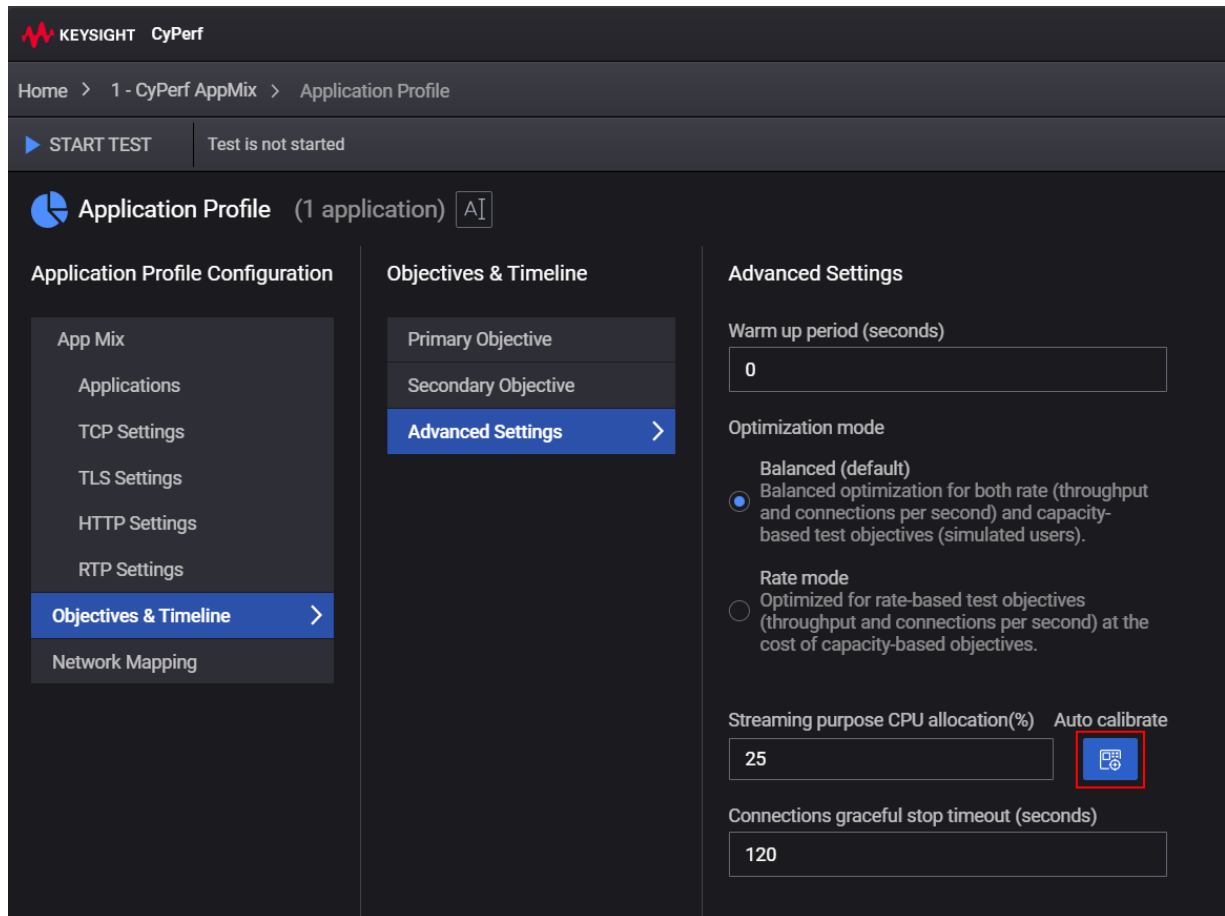
There are multiple servers available for selection for all Outlook Custom Attacks. You can change the default selection for all available strikes:



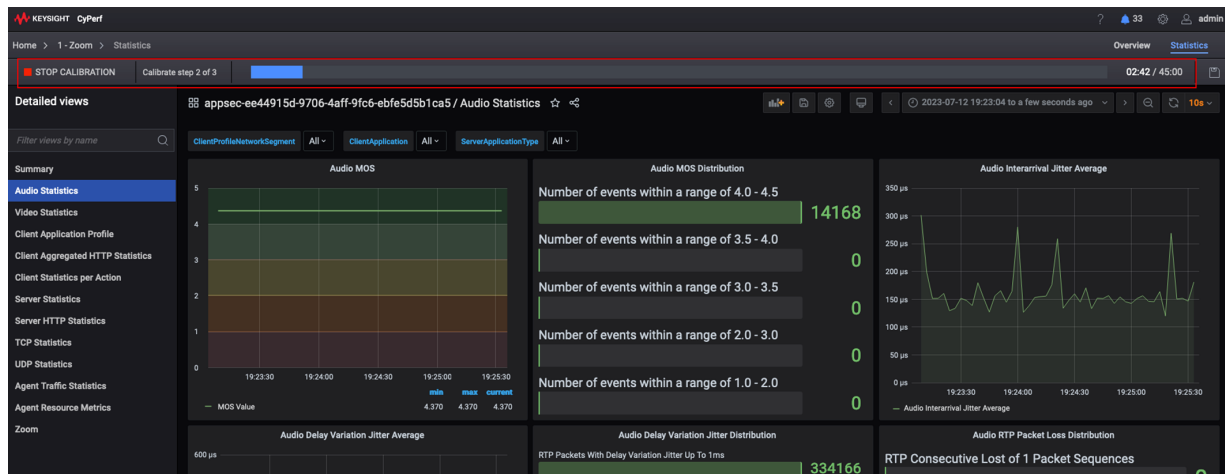
Auto Calibrate Streaming Purpose CPU Allocation

To automatically find and configure the streaming purpose CPU allocation value:

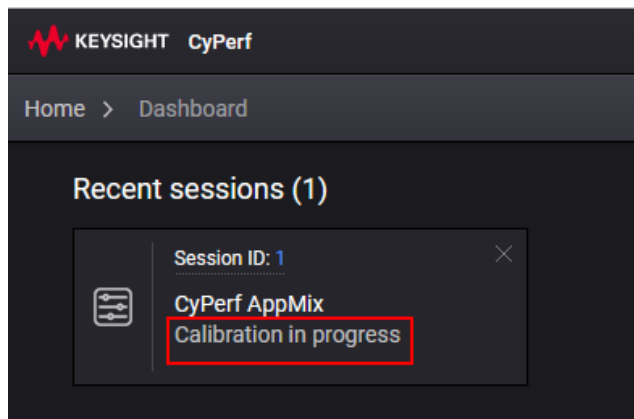
1. In an existing test session, open the **Application Profile** page by clicking the **Edit** button on the **Application Profile** tab.
2. Under **Application Profile Configuration** > **Objectives & Timeline** > **Advanced Settings**, click the **Auto calibrate** button.



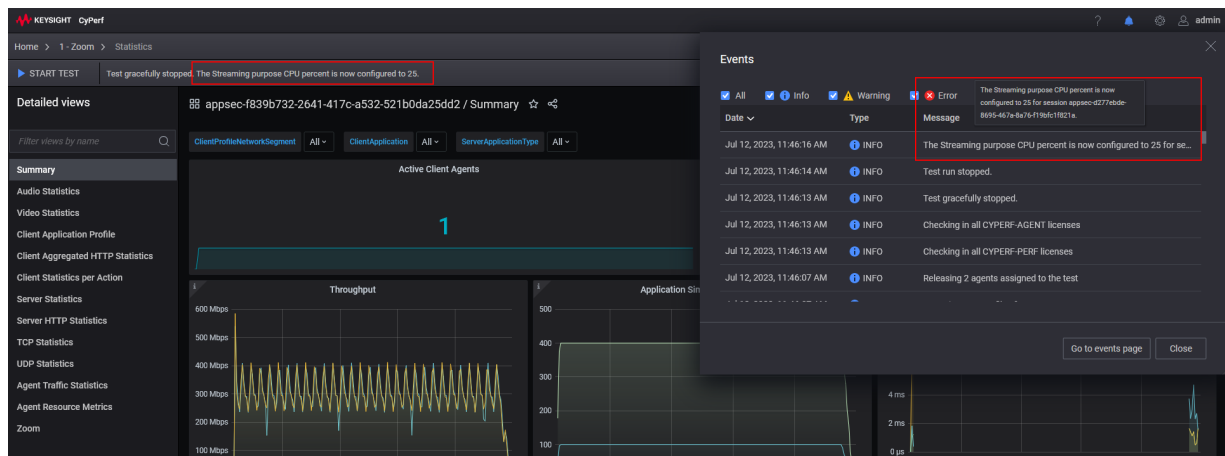
The calibration process starts, and CyPerf uses an automatic algorithm to calculate the optimal streaming purpose CPU allocation value for the given test and deployment environment. Information about the calibration steps, their progress and the maximum test duration is displayed on the test status bar.



The corresponding test session tile also displays the **Calibration in progress** status.



When calibration ends, the status bar displays the value that was automatically configured for the **Streaming Purpose CPU allocation (%)** parameter. The **Events** window also displays the automatically configured value.



Benchmark an LLM Inference Deployment

The LLM benchmarking feature allows you to use CyPerf as an inference client to test and benchmark deployed large language model (LLM) inference servers, such as vLLM or Ollama. CyPerf generates controlled test traffic toward supported inference deployments (for example, servers compatible with the OpenAI Chat Completions API) and collects performance and token-related metrics from the responses. You can configure benchmark scenarios by selecting predefined or user-supplied prompts and by controlling traffic parameters such as prompt rate and concurrency. This enables evaluation of inference behavior under different load conditions and prompt characteristics.

The LLM Benchmarking feature operates in one-arm mode, i.e., CyPerf functions as the client communicating directly with the inference server under test, and does not simulate or include any server-side components.

During an LLM benchmark run, CyPerf measures and displays latency and token statistics, including time to first token (TTFT), time to last token (TTLT), average inter-token latency (ITL) within a single response, request latency, prompt rate, and other statistics. These metrics help you assess responsiveness and throughput of your inference server.

Supported Inference Servers and Models

The following inference servers and models are supported for benchmarking:

- Inference servers which support the *OpenAI Chat Completions API* over *HTTP/1.x* and which provide *token usage statistics* for both *streamed* and *non-streamed* responses. For streamed responses, the server API must support the following inputs:

```
"stream": true,
"stream_options": {"include_usage":
true}
```

- All *text generation models* which can be served through the OpenAI Chat Completions API.

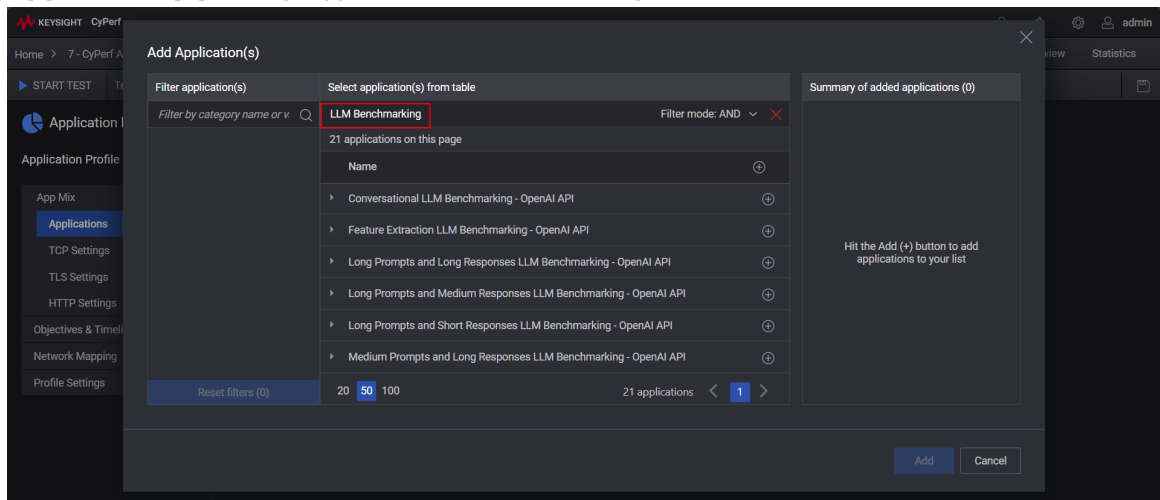
Configure a Test for LLM Benchmarking

1. From the CyPerf **Dashboard** page, create a new test by selecting the **CyPerf AppMix** test type under the **Create New Test** panel. The new test **Overview** page opens displaying the default test topology containing two network elements.
2. Delete the server network element by clicking the **Delete** button (✗) on the **IP Network 2** tile.
3. Configure the DUT network:
 - a. Enable the **DUT Network 1** segment.
 - b. Open the **DUT Configuration** page by clicking the **DUT Network 1** tile.
 - c. On the **DUT Configuration** page, select **Advanced configuration**.
 - d. Click **Reverse Proxy** to open the Reverse Proxy configuration panel, and enter the required settings:

- **Hostname:** Either the IP address or the FQDN of the inference server.
 - **Port:** TCP port on which the inference server API can be reached.
4. Configure the client network (i.e., **IP Network 1**) as desired.

NOTE: If you configured the DUT network segment with FQDN in the Reverse Proxy step, then provide a DNS resolver configuration in the client network.

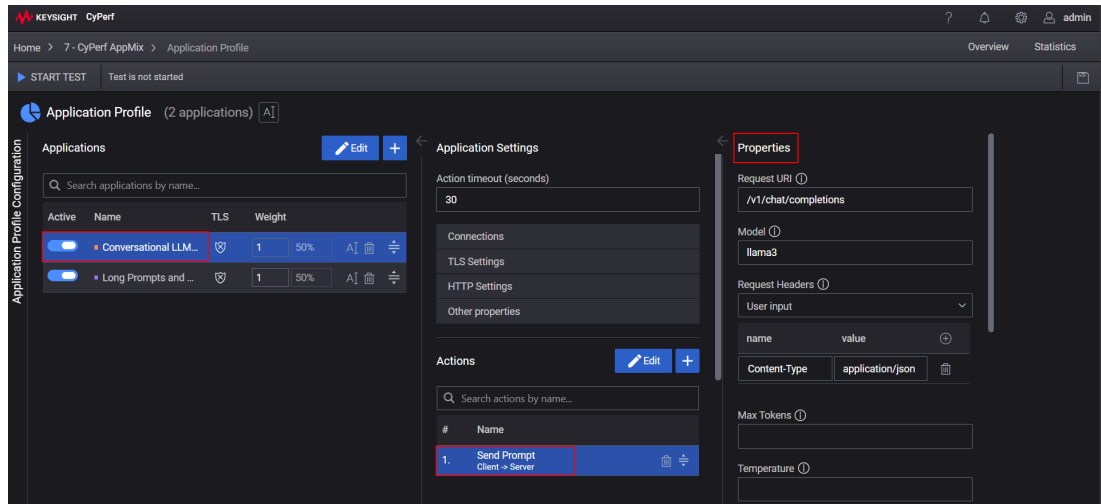
5. Add the LLM benchmarking specific apps in the AppMix:
- Edit the AppMix by clicking the **Edit** button (✎). The Application Profile page opens.
 - On the Application Profile page, click the **Add** button (+) to open the **Add Application(s)** dialog.
 - Under the **Filter applications by name, description** search box on the **Add Application(s)** dialog, type **LLM Benchmarking**.



All the available LLM Benchmarking applications are filtered and displayed under this list. The LLM Benchmarking applications can be split in two categories, each offering multiple pre-canned applications:

- Simple LLM Benchmarking: single turn, single prompt tests that evaluates baseline inference latency (e.g. *Short Prompts and Short Responses LLM Benchmarking - OpenAI API*, *Summarization LLM Benchmarking - OpenAI API*, and others)
 - Multi-turn LLM Benchmarking: long multi-turn conversations, ideal for stressing Key Value (KV) cache and measuring latency slowdowns as context grows (e.g. *MultiTurn LLM Benchmarking - OpenAI API*, *Long Prompts and Long Responses LLM Benchmarking - OpenAI API*, *Very Long Prompts and Long Responses LLM Benchmarking - OpenAI API*, and others)
- d. Choose the applications you want to use from the filtered list and add them to the AppMix by clicking the **Add** button (+) corresponding to each application. You can choose applications based on desired traffic or prompt characteristics, such as type or length of prompts.

- e. After selecting all the applications you need, click **Add** to finalize adding the applications to the AppMix and to close the Add Application(s) dialog.
- f. On the Application Profile page, select each application in turn from the application list. For each application:
 - i. Under **Actions** for the selected application, click the **Send Prompt** action to configure it.
 - ii. On the **Send Prompt > Properties** panel, inspect the available parameters and edit them as needed for your use case.



The following parameters are available:

- **Model:** Specify the exact LLM model identifier. This value is sent to the inference server in each request, so it must match the model name as it is defined on the server.
 - **Max tokens:** Sets the maximum number of tokens the model can generate for a request, as defined by the OpenAI API. This value must be a positive, non-zero integer. You may leave this field empty if you do not want to specify a limit.
 - **Temperature:** Controls the randomness of the generated response, as defined by the OpenAI API. This field accepts a floating-point value between 0 and 2. You may leave this field empty if you do not want to set a temperature.
 - **Stream:** Enables or disables streamed responses from the inference server. When set to **False**, the application cannot provide meaningful Time To First Token (TTFT), Time To Last Token (TTLT), Inter-Token Latency (ITL), and Events Per Second statistics.
 - **System Prompt:** A system prompt can be provided, or it can be left empty.
 - **Prompt:** The prompt sent to the inference server during testing. You can edit the prompt as needed. You can configure prompts as a single text string, or you can upload a properly formatted CSV file to use as a playlist for the test.
 - **API Body Template:** This advanced parameter allows modification of the default OpenAI API request body to support specialized use cases. This feature is experimental and is not recommended for user modification at this time.
6. Under the **Application Profile Configuration** panel, select the **Objectives & Timeline** tab to configure the load control parameters:

- **Primary Objective:** Must be set to **Prompts per second**
- **Secondary objective:** Can be set to **Simulated Users** if desired. This can be used to control concurrency.

7. Start the test.

When the test starts, the Statistics page opens displaying the relevant statistics under the **LLM Benchmarking Token Statistics** section. See [LLM Benchmarking Token Statistics](#) for more details.

LLM Benchmarking Token Statistics

The following statistics are available under the **LLM Benchmarking Token Statistics** section of the Statistics page:

Statistics	Definition
Aggregated Token Latencies	Displays the minimum, maximum, and average Time To First Token (TTFT), Time To Last Token (TTLT), and Inter-Token Latency (ITL ¹) values for the entire test, visualized as a time-series graph. The data can be filtered by application or by network segment (when multiple client network segments are configured).
Request Latency	Displays the minimum, maximum, and average Request Latencies for the entire test as a time-series graph.
Events Per Second	Displays the current rate of response events (server-sent events from the inference server) as received by the client.
Instantaneous Prompt Rate	Shows the current number of prompts sent per second and prompts completed per second, visualized as a time-series graph.
Test Average Token Rates	Shows the cumulative average input and output tokens per second, visualized as a time-series graph. NOTE: This graph does <i>not</i> represent instantaneous rates. It reflects how the cumulative averages evolve throughout the duration of the test.
Benchmark Test Level Average Rates	Shows cumulative <i>averages</i> for Queries Started per second, Queries Succeeded per second, Output Tokens per second, and Input Tokens per second.
Benchmark Totals	Shows total counts for: Requests Started, Requests Succeeded, Requests Failed, Output Tokens Received, and Input Tokens Sent.
Agent Token Latency Percentiles (Cumulative)	Shows cumulative percentile values of TTFT, TTLT, and ITL for each agent in the test. Clicking an agent entry opens a detailed view showing:

1. Average inter-token latency within a single response

Statistics	Definition
	- Instantaneous percentile values of TTFT, TTLT, and ITL (for the last measurement window) as time-series graphs. - Application Token Latency Percentiles (Cumulative): Shows cumulative TTFT, TTLT, and ITL percentiles for each application configured in the test. - Application Token Metric Percentiles (Cumulative): Shows cumulative Input Sequence Length (ISL¹) and Output Sequence Length (OSL²) percentiles for each application configured in the test.
Agent Token Metric Percentiles (Cumulative)	Shows cumulative ISL and OSL percentile values for each agent. Clicking an agent opens the same detailed view as described for Agent Token Latency Percentiles (Cumulative).

Validate LLM API Intercept Guardrail Solutions

The LLM API intercept feature allows you to use CyPerf to test the effectiveness of third-party LLM API intercept guardrail solutions, such as Google Model Armor and Palo Alto Networks Prisma AI Runtime Security. CyPerf generates controlled application and attack traffic toward a supported API intercept guardrail provider and collects performance metrics to assess how well it identifies, filters, and blocks unsafe or policy-violating requests and responses. You can configure test scenarios by selecting from provider-specific applications and attacks to evaluate guardrail behavior under different benign and malicious traffic conditions.

The LLM API intercept feature operates in one-arm mode, i.e., CyPerf functions as the client communicating directly with the API intercept guardrail service, and does not simulate or include any server-side components.

During a test run, CyPerf measures and displays results on the Statistics page, including the number of initiated connections and how the API intercept guardrail handled traffic. These metrics help you assess the effectiveness of the API intercept solution in identifying and blocking specific attack patterns, validate guardrail performance, and compare multiple guardrail implementations for suitability to your use case.

Supported API Intercept Providers

The following API intercept providers are supported:

- **Palo Alto Networks Prisma AI Runtime Security:** 1 application and 39 attacks
- **Google Model Armor:** 1 application and 31 attacks

Configure LLM API Intercept Tests

1. From the CyPerf **Dashboard** page, create a new test by selecting one of the test types corresponding to the requirements of your use case (i.e. AppMix, Appmix and Attack, or Attack). The new test **Overview** page opens displaying the default test topology containing two

1. Number of tokens in a prompt
2. Number of tokens in the LLM response

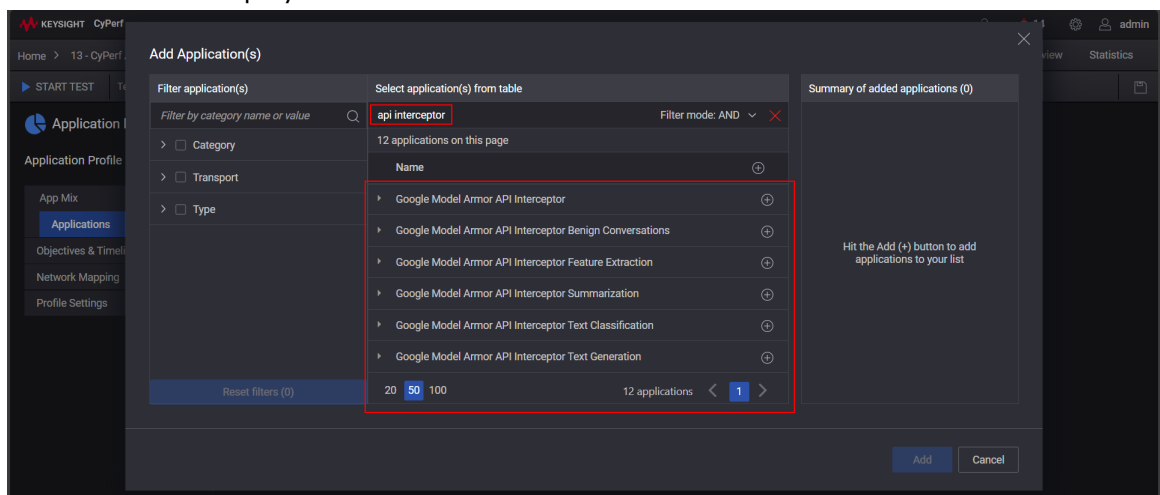
network elements.

2. Delete the server network element by clicking the **Delete** button (✗) on the **IP Network 2** tile.
3. Configure the DUT network:
 - a. Enable the **DUT Network 1** segment.
 - b. Open the **DUT Configuration** page by clicking the **DUT Network 1** tile.
 - c. On the **DUT Configuration** page, configure the DUT in one of these scenarios:
 - i. If the API intercept service uses a standard TCP port (80 or 443):
 - Select the **Simple configuration** option.
 - Enter the **Hostname** using either the IP address or the FQDN where the service is reachable.
 - ii. If the API intercept service uses a non-standard TCP port:
 - Select the **Advanced configuration** option.
 - Click **Reverse Proxy** to open the Reverse Proxy configuration panel, and enter the required settings:
 - **Hostname**: Either the IP address or the FQDN of the inference server.
 - **Port**: TCP port on which the inference server API can be reached.
4. Configure the client network (i.e., **IP Network 1**) as desired.

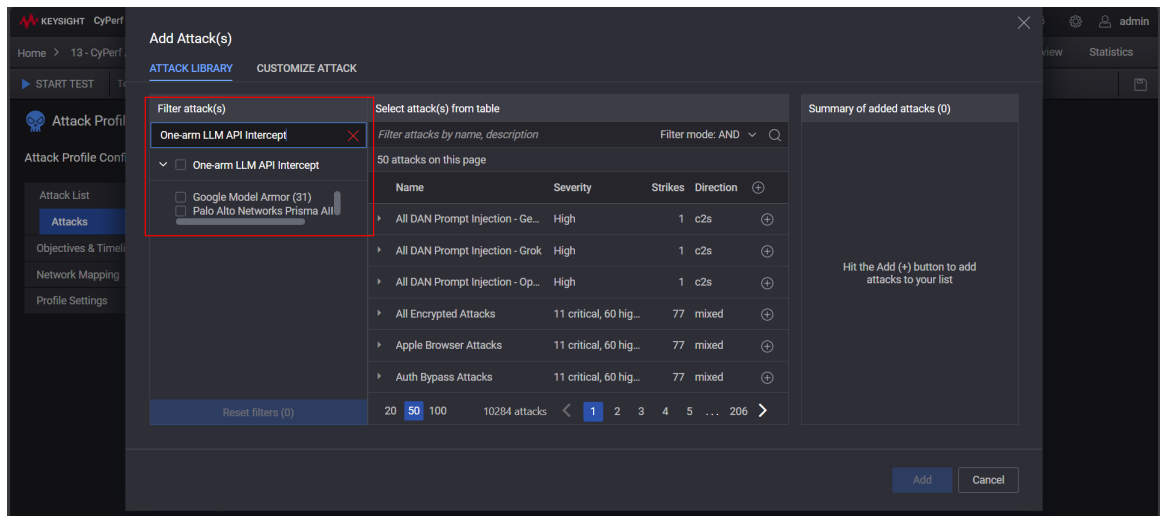


NOTE: If you configured the DUT network segment with FQDN in the Reverse Proxy step, then provide a DNS resolver configuration in the client network.

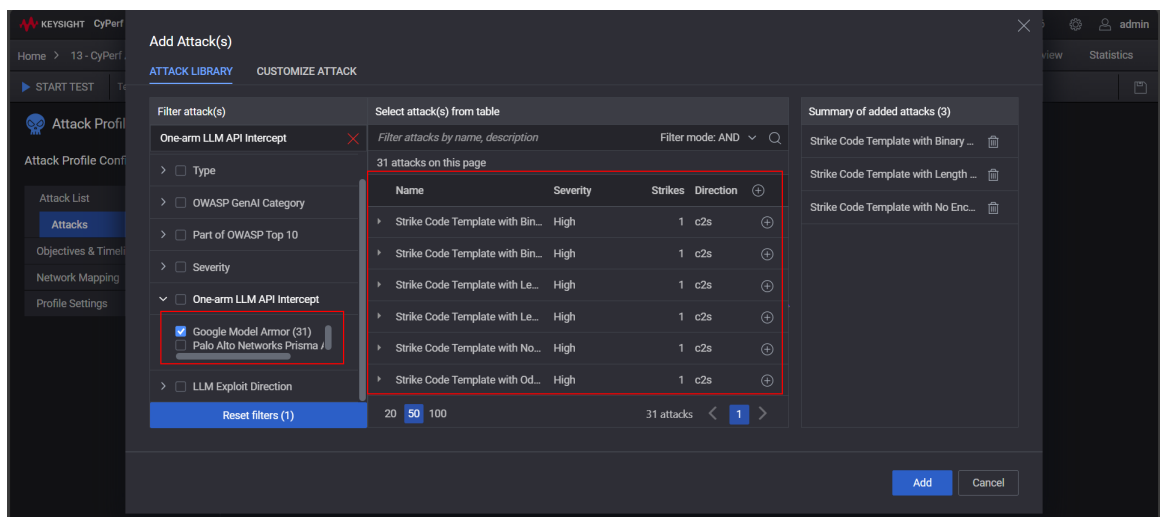
5. (Optional) Configure the Application Profile:
 - a. Edit the AppMix by clicking the **Edit** button (✎). The Application Profile page opens.
 - b. On the Application Profile page, click the **Add** button (+) to open the **Add Application(s)** dialog.
 - c. Under the **Filter applications by name, description** search box on the **Add Application(s)** dialog, type **API Interceptor**. All the available API intercept applications are filtered and displayed under this list.



- d. Choose the applications you want to use from the filtered list and add them to the AppMix by clicking the **Add** button () for each application. Choose the application(s) corresponding to your API intercept service provider; each application name includes the provider identifier (i.e., **Google Model Armor** or **PRISMA AIRS**).
- e. After selecting all the applications you need, click **Add** to finalize adding the applications to the AppMix, close the Add Application(s) dialog, and return to the Application Profile page.
- f. (Optional) From the Application Profile page, enable client TLS for the Application Profile:
 - i. Under **Application Profile Configuration > App Mix > TLS Settings > Client TLS**, use the toggle to enable TLS traffic.
 - ii. Configure the other client TLS parameters as required.
- g. Then, configure the **Send Benign Prompt** and **Send Benign Response** actions for each application you added to the App Mix.
Under **Actions** for the selected application, configure the parameters according to the provider in use.
 - i. For **Google Model Armor**:
 - **Project ID**: GCP Project ID associated with the Model Armor deployment.
 - **Template ID**: Template ID for the Model Armor deployment.
 - **Location**: GCP location where the Model Armor template is deployed (for example, `us-central1`, `asia-south1`).
 - **Access Token**: Google Cloud access token. Generate using gcloud CLI: `gcloud auth print-access-token`
 - **Prompt / LLM Response**: The prompt or response string sent to the API intercept service for validation. If needed, the prompts or responses can be customized and configured as a single string value, or as a CSV file uploaded as a playlist for use during the test.
 - ii. For **Palo Alto Networks Prisma AI Runtime Security**:
 - **PAN Auth Token**: Authentication token for the Prisma AIRS security profile.
 - **AIRS Profile Name**: Configured AIRS profile name.
 - **AIRS Profile Id**: Configured AIRS profile ID. Either **AIRS Profile Name** or **AIRS Profile Id** must be populated.
 - **Prompt / LLM Response**: The prompt or response string sent to the API intercept service for validation. If needed, the prompts or responses can be customized and configured as either a single string value, or as a CSV file uploaded as a playlist for use during the test.
6. (Optional) Configure the Attack Profile.
 - a. Edit the Attack List by clicking the **Edit** button () . The Attack Profile page opens.
 - b. On the Attack Profile page, click the **Add** button () to open the **Add Attack(s)** dialog.
 - c. Under **Attack Library**, in the **Filter by category name or value** search box, type **One-arm LLM API Intercept**. All the available API intercept attack categories are filtered and displayed under this list.



- d. Select the category corresponding to your provider, either Google Model Armor, or Palo Alto Networks PRISMA AIRS.
- e. From the filtered list, click the **Add** button (+) next to each desired attack to select it.
- f. When finished, click **Add** to include the selected attacks in the Attack List.



- g. (Optional) From the Attack Profile page, enable client TLS for the Attack Profile:
 - i. Under **Attack Profile Configuration > Attack List > TLS Settings > Client TLS**, use the toggle to enable TLS traffic.
 - ii. Configure the other client TLS parameters as required.
- h. Then, configure the strike parameters under **Strikes and Actions** for each attack you added to the Attack List. Configure these parameters according to the provider in use.
 - i. For **Google Model Armor**:
 - **Project ID**: GCP Project ID associated with the Model Armor deployment.
 - **Template ID**: Template ID for the Model Armor deployment.

- **Location:** GCP location where the Model Armor template is deployed (for example, `us-central1`, `asia-south1`).
 - **Access Token:** Google Cloud access token. Generate using gcloud CLI: `gcloud auth print-access-token`
- ii. For **Palo Alto Networks Prisma AI Runtime Security**:
- **PAN Auth Token:** Authentication token for the Prisma AIRS security profile.
 - **AIRS Profile Name:** Configured AIRS profile name.
 - **AIRS Profile Id:** Configured AIRS profile ID. Either **AIRS Profile Name** or **AIRS Profile Id** must be populated.
7. Configure the **Objectives & Timeline** for all the profiles you created for the test as usual based on your use case.
8. Start the test.

When the test starts, the Statistics page opens displaying relevant metrics to help you assess how the API intercept service responded to the configured traffic:

- **Allowed:** The API intercept service evaluated the prompt or response as safe and permitted it.
- **Blocked:** The API intercept service evaluated the prompt or response as unsafe and prevented it.
- **Errored:** The strike or action failed due to an error (for example, a connection failure, an HTTP 4xx/5xx response, or another operational error).

Debug

Keysight CyPerf offers support for debugging capabilities so that you can:

View Notifications and Test Events	69
View Statistics	69
Manage Granular Statistics	70
Manage Test Results	74
Decrypt IPSec Packets	75
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View Notifications and Test Events

The **Navigation** bar displays a notifications icon  and a counter showing the total number of triggered notifications since the counter was last reset for the current CyPerf controller instance. The icon and the counter are visible from all the pages of the Keysight CyPerf web UI. When a notification is triggered, a color-coded banner is also displayed on the lower right corner of the screen: green for informational messages, orange for messages informing you of actions you are not allowed to perform, and red for error messages.



NOTE:

While on the **Dashboard** page, the notifications icon displays notifications strictly at cluster or system level.

While inside a **test session**, the notifications icon displays notifications for the current test as well as notifications at the cluster and system level.

To view more details on the events triggered, select the notifications icon. The **Events** page opens listing the date time stamp, type of event, and a description of the event. For example: Jun 6, 2020, 10:55:48 AM Initializing 2 newly reserved test agents.

You can also filter events by:

- Typing a keyword and pressing enter
- Selecting a to and from date
- Selecting the notification type to be viewed

View Statistics

After you start a test, you can view the real-time statistics on the **STATISTICS** page.

To view real-time test statistics:

1. In an active test session, ensure the test is started.
2. Select the **Statistics** tab on the navigation bar.
3. Filter the available statistics by selecting an appropriate option from the **Detailed view** section. For example, you can choose to display statistics about the client attack profile, traffic agents, server, as well as individual applications and attacks.

See also [Statistics Page](#).

Manage Granular Statistics

Granular statistics can be enabled and customized for any test where the configured applications or attacks use a **Playlist** file in their actions. The **Granular Statistics** option is disabled by default. When granular statistics are enabled, specific statistics for each entry from a playlist will be displayed under the **Summary** panel on the **Statistics** page, and they will be positioned last in the statistics list.

The screenshot shows the KeySight CyPerf interface. The top navigation bar includes 'Overview' and 'Statistics'. The left sidebar has a 'Detailed views' section with a search filter 'appsec-19865862-9c57-4...'. The main content area displays two tables of statistics for 'eBanking Chrome to Apache 1'.

Table 1: eBanking Chrome to Apache 1 - Action 1 - Top 3 Failed

Profile - Network Segment	Application & Attack	Action	Playlist Variant	Succeeded	Failed	Started
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	1	ixia1:Appsec@2019	44568	0	44568
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	1	ixia2:Appsec@2019	44564	0	44564
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	1	viadtest:appsec123#	44559	0	44559

Table 2: eBanking Chrome to Apache 1 - Action 2 - Top 3 Failed

Profile - Network Segment	Application & Attack	Action	Playlist Variant	Succeeded	Failed	Started
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	2	ixia1:Appsec@2019	44568	0	44568
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	2	ixia2:Appsec@2019	44564	0	44564
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	2	viadtest:appsec123#	44559	0	44559
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	2	cyperf1:appsec2019	44551	0	44551
App Client Application Profile-IP ...	eBanking Chrome to Apache 1	2	cyperf2:appsec2019	44546	0	44546

The left sidebar also shows a list of statistics categories, with 'Application Profile Granular Stats' and 'Attack Profile Granular Stats' highlighted in red.

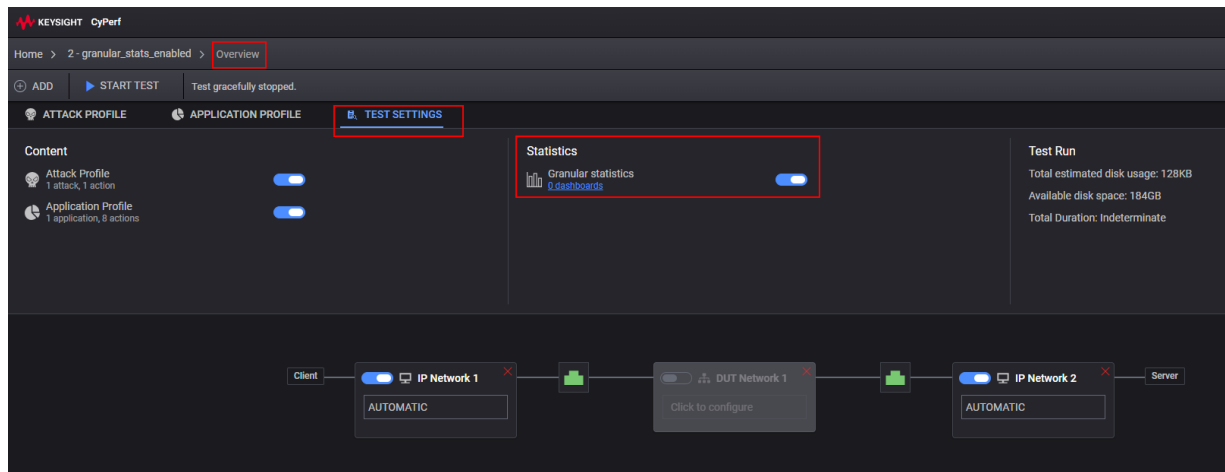
Granular statistics are also saved in the CSV report that is available for download at the end of a test run.

To enable and customize granular statistics dashboards:

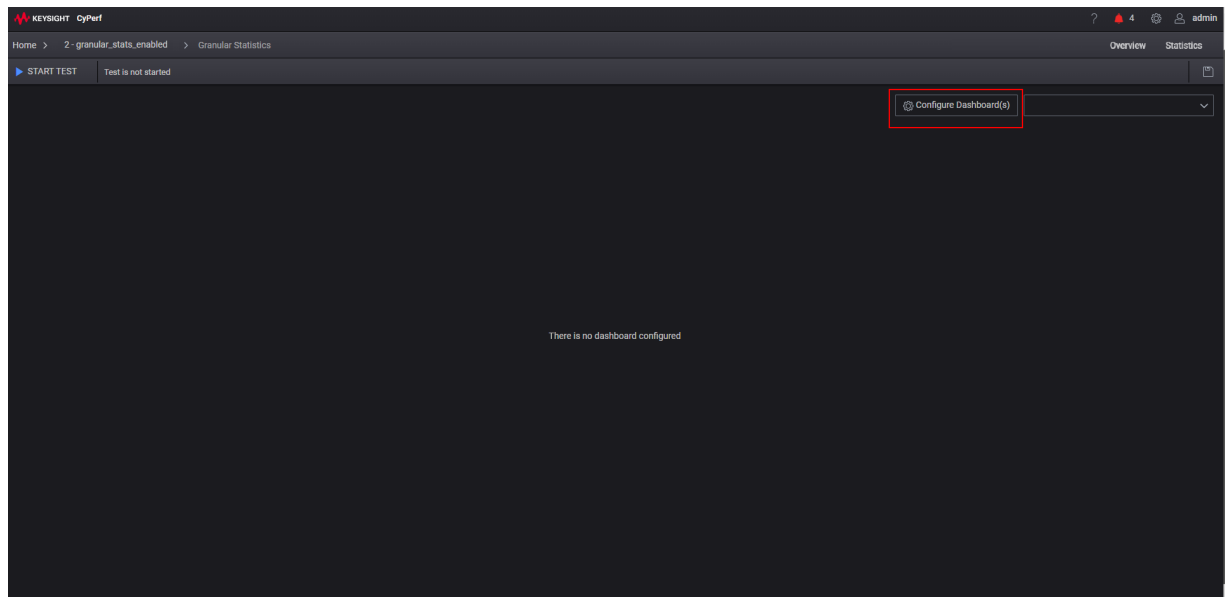
1. From an existing test session, select the **Test Settings** tab on the test **Overview** page.
2. Under the **Statistics** panel, use the toggle to enable the **Granular Statistics** option.



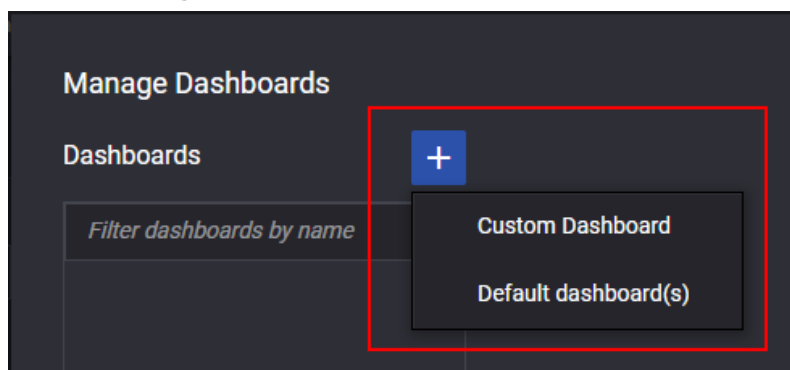
NOTE: Enabling granular statistics generates a large volume of statistics which require significant memory resources to store and process.



- Under the **Granular Statistics** option, click the active link displaying the number of active dashboards. The **Granular Statistics** page opens.
- From the **Granular Statistics** page, add a new dashboard by clicking **Configure Dashboard(s)**. The **Manage Dashboards** window opens.

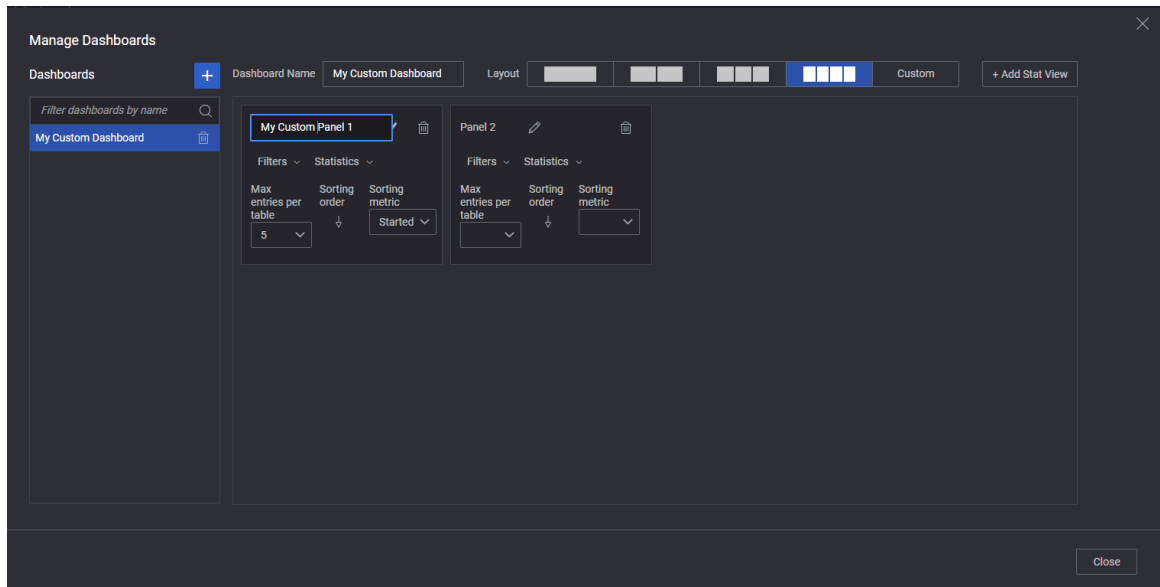


- On the **Manage Dashboards** window, click .

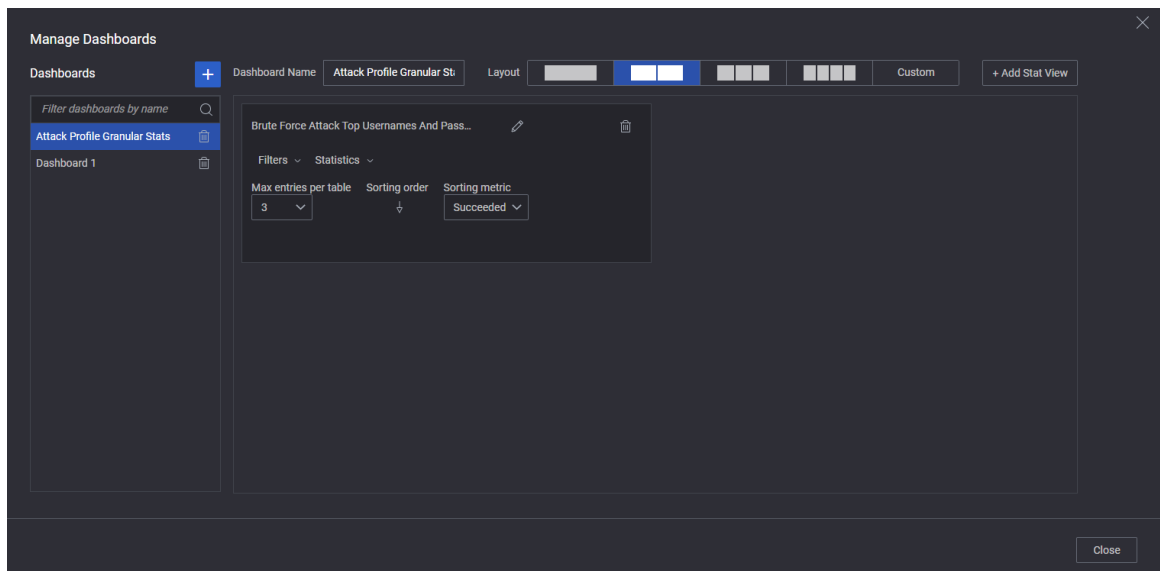


Then select one of the available options:

- **Custom Dashboard:** adds a custom dashboard for which you can create custom statistics panels (by clicking **Add Stat View**) for each action or attack that uses a **Playlist** file.



- **Default Dashboard(s):** adds all the available dashboards for the applications or attacks that use a **Playlist** file. The newly created statistics panels on each dashboard have default names relative to the application or attack using the **Playlist** file.



6. For custom dashboards, set the name for the dashboard by typing the desired name in the **Dashboard Name** field.
7. Add new statistics panel(s) by clicking **Add Stat View**, and enter custom names for each panel by clicking the **Edit** icon next to each panel name.

8. When working with multiple panels, choose the **Layout** to be used for displaying the statistics panels.
9. From **Filters** on each statistics panel, choose the action(s) for which to display granular statistics.
10. From **Statistics** on each statistics panel, choose the statistics you want to display.

The available options are:

- Started
- Succeeded
- Failed
- Skipped
- Payload Bytes Sent
- Payload Bytes Received
- Responses Received (1xx)
- Responses Received (2xx)
- Responses Received (3xx)
- Responses Received (4xx)
- Responses Received (5xx)
- Responses Received (400)
- Responses Received (401)
- Responses Received (403)
- Responses Received (404)
- Responses Received (502)
- Responses Received (503)
- Responses Received (504)
- Responses Received (other)

11. From **Max Entry per table** on each statistics panel, choose the maximum number of entries to be displayed for each table.

The available options are:

- 1
- 3
- 5
- 10
- 50
- 100
- 250
- 500
- 1000
- 5000

12. From **Sorting order** on each statistics panel, sort the values in ascending or descending order.
13. From **Sorting metric** on each statistics panel, choose the primary filter for the granular statistics.

Manage Test Results

The **Browse Results** page allows you to view, download, load, or delete test results, such as CSV files, PDF reports, capture files, or configuration files.

To download test results:

1. Go to the **Browse Results** section on the **Dashboard** page. The **Browse Results** page opens listing the results of previously run tests.
2. Select the results you are interested in. Hold the **Ctrl** key for multiple selection.
3. Select the **Download** button, and choose the files you want to download: **CSV**, **Report**, **Captures**, or **Config**.

To load test results:

1. Go to the **Browse Results** section on the **Dashboard** page. The **Browse Results** page opens listing the results of previously run tests.
2. Select the results you are interested in. Hold the **Ctrl** key for multiple selection.
3. Select **Load** and view the test session and its statistics.

To delete test results:

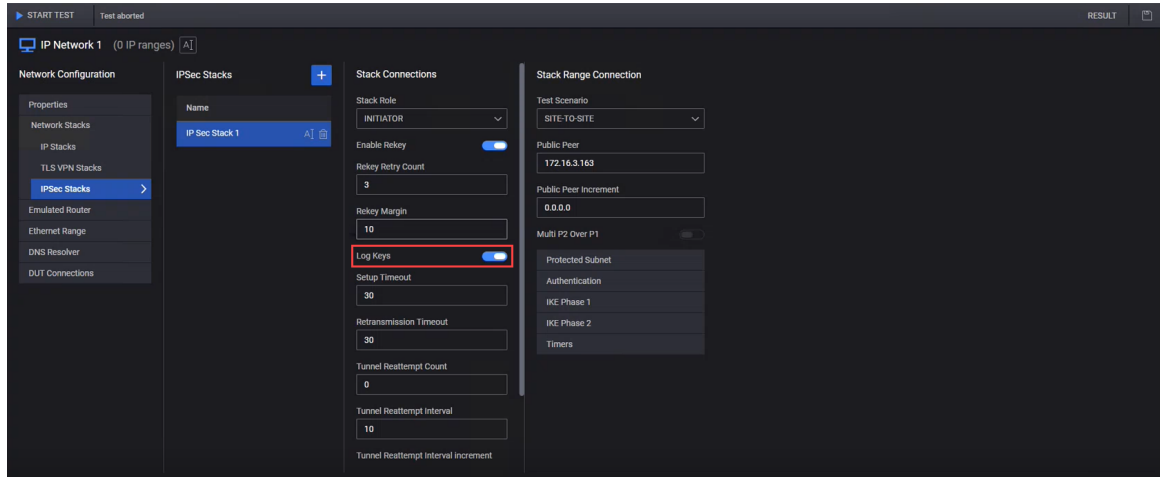
1. Go to the **Browse Results** section on the **Dashboard** page. The **Browse Results** page opens listing the results of previously run tests.
2. Select the results you are interested in. Hold the **Ctrl** key for multiple selection.
3. Select **Delete**. The selected results are deleted from the database.

Decrypt IPsec Packets

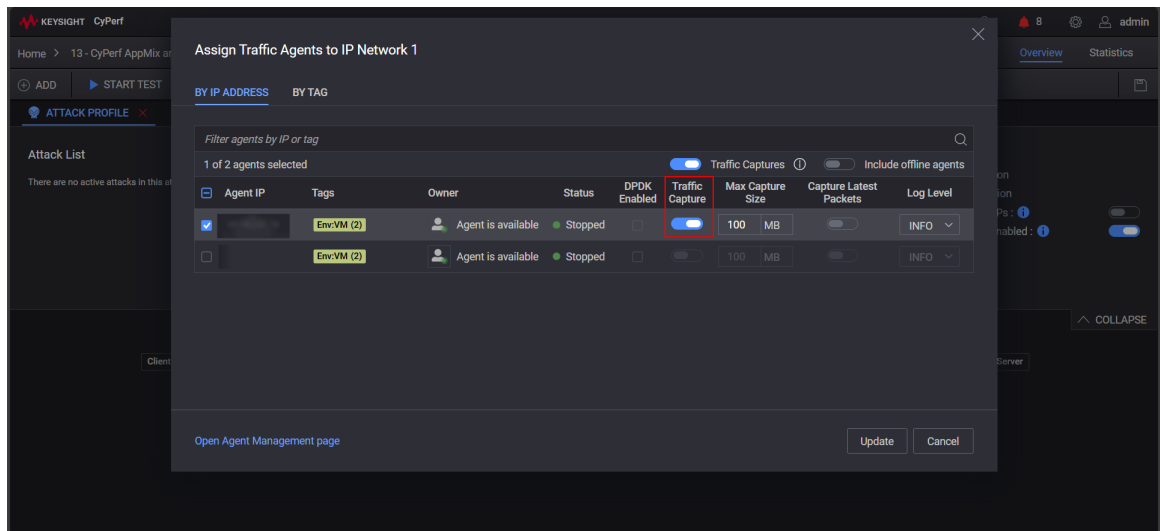
CyPerf allows you to download traffic captures and decrypt IPsec packets in order to do packet level analysis using Wireshark.

To decrypt and analyze CyPerf IPsec packets using Wireshark:

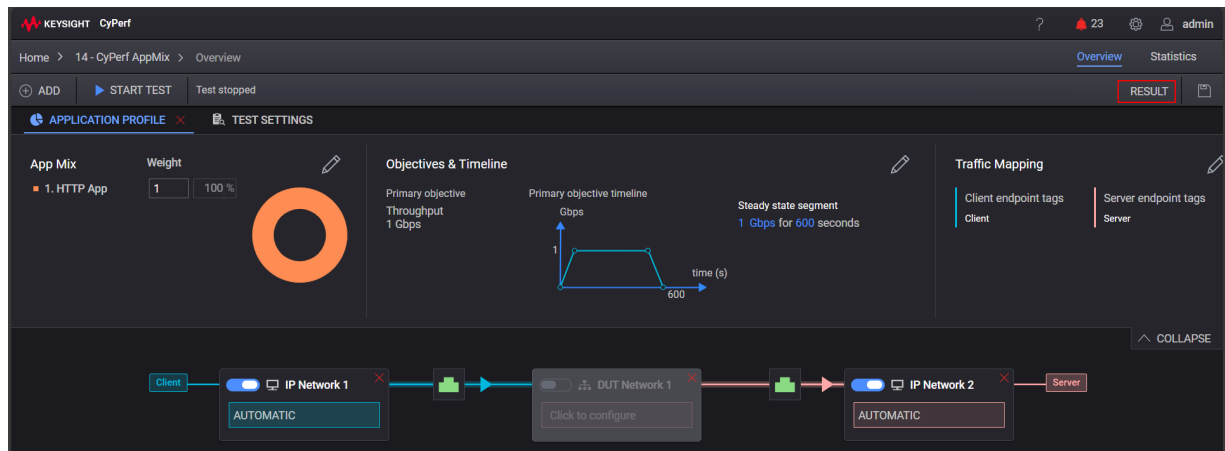
- Before starting a test, make sure the following two options are enabled:
 - Log Keys:** Select an **IP Network** tile in the test **Topology** section, and enable **Log Keys** under **IPsec Stacks** > **Stack Connections**.



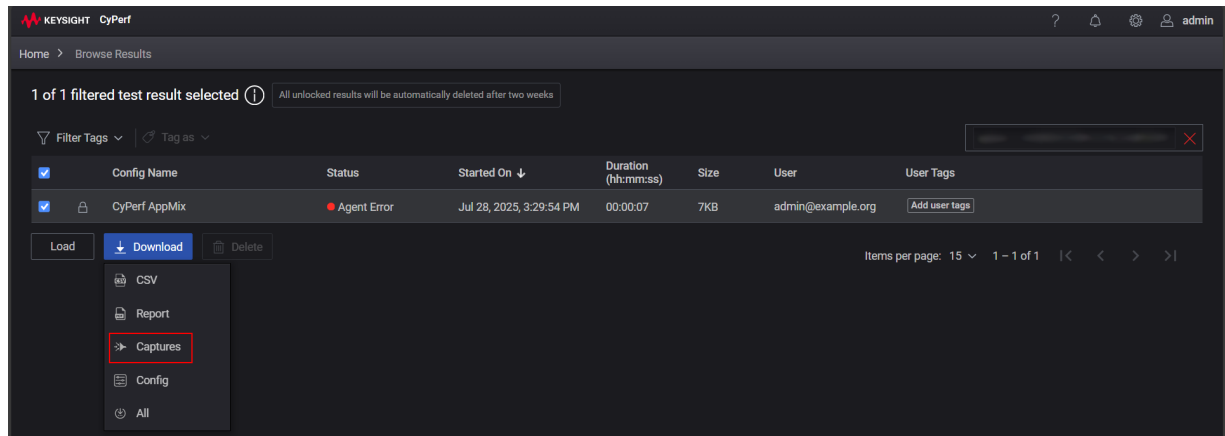
- Traffic Capture:** Select the **Assign Traffic Agents**  icon in the test **Topology** section. On the **Assign Traffic Agents** dialogue, use the toggle to enable **Traffic Capture** for the selected agent(s).



- Start the test and wait for the test run to complete.
- After the test run is complete, open the current test's results page by clicking **RESULT**.



4. Select the test results, click **Download**, and select the **Captures** option from the menu that opens.



The traffic capture .zip file is downloaded locally on your system. This archive contains the following files:

- **CyPerfPackets.pcap**: CyPerf capture file
 - **esp_sa file** and **ikev2_decryption_table**: log keys that you can use to decrypt the IPsec packets
5. Unzip the archive and copy the two log key files (i. e., **esp_sa file** and **ikev2_decryption_table**) under the following path on your system: **C:\Users\<user name>\AppData\Roaming\Wireshark**.
 6. In Wireshark, open the **CyPerfPackets.pcap** file and you can now analyze the decrypted IPsec packets.

CyberPackets.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter: <Ctrl>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	06:01:2c:0f:08:fb	Broadcast	ARP	42	Who has 172.16.3.163? Tell 172.16.3.111
2	0.000268	06:e1:f2:d9:1e:49	06:01:2c:0f:08:fb	ARP	42	172.16.3.163 is at 06:e1:f2:d9:1e:49
3	0.226436	30.0.0.1	172.16.3.163	ISAKMP	274	IKE_SA_INIT MID=00 Initiator Request
4	16.998193	30.0.0.2	172.16.3.163	ISAKMP	274	IKE_SA_INIT MID=00 Initiator Request
5	16.408043	172.16.3.163	30.0.0.2	ISAKMP	290	IKE_SA_INIT MID=00 Responder Response
6	16.404089	30.0.0.2	172.16.3.163	ISAKMP	246	IKE_AUTH MID=01 Initiator Request
7	16.407291	172.16.3.163	30.0.0.2	ISAKMP	262	IKE_AUTH MID=01 Responder Response
8	21.666462	172.16.3.163	30.0.0.2	ISAKMP	118	INFORMATIONAL MID=00 Responder Request
9	21.666920	50.0.0.2	172.16.4.186	TCP	134	25258 → 443 [SYN] Seq=0 Win=0 Len=0 MSS=1324 TSval=58231527 TSecr=0
10	21.667012	30.0.0.2	172.16.3.163	ISAKMP	118	INFORMATIONAL MID=00 Initiator Response
11	21.766274	50.0.0.2	172.16.4.186	TCP	134	[TCP Retransmission] 25258 → 443 [SYN] Seq=0 Win=0 Len=0 MSS=1324 TSval=58377071 TSecr=0
12	21.769536	172.16.4.186	50.0.0.2	TCP	134	443 → 25258 [SYN, ACK] Seq=0 Ack=1 Win=2896 Len=0 MSS=1387 TSval=58386727 TSecr=58377071
13	21.769762	50.0.0.2	172.16.4.186	TLSv1.2	262	Client Hello
14	21.772081	172.16.4.186	50.0.0.2	TLSv1.2	1254	Server Hello, Certificate, Server Hello Done
15	21.772713	50.0.0.2	172.16.4.186	TLSv1.2	454	Client Key Exchange, Change Cipher Spec, Finished
16	21.775608	172.16.4.186	50.0.0.2	TLSv1.2	182	Change Cipher Spec, Finished
17	21.775877	50.0.0.2	172.16.4.186	HTTP	470	GET /file.txt?name=val1 HTTP/1.1
18	21.777330	172.16.4.186	50.0.0.2	HTTP	422	HTTP/1.1 200 OK
19	21.777428	50.0.0.2	172.16.4.186	HTTP	470	GET /file.txt?name=val1 HTTP/1.1
20	21.779380	172.16.4.186	50.0.0.2	HTTP	422	HTTP/1.1 200 OK
21	21.779538	50.0.0.2	172.16.4.186	HTTP	470	GET /file.txt?name=val1 HTTP/1.1
22	21.780904	172.16.4.186	50.0.0.2	HTTP	422	HTTP/1.1 200 OK

< Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits)
 > Ethernet II, Src: 06:01:2c:0f:08:fb (06:01:2c:0f:08:fb), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 > Address Resolution Protocol (request)

```

0000  ff ff ff ff ff ff 06 01 2c 0f 08 fb 08 06 00 01  .....
0010  08 00 06 04 00 01 06 01 2c 0f 08 fb ac 10 03 6f  .....
0020  00 00 00 00 00 ac 10 03 a3  .....

```

CyberPackets.pcap Packets: 14287 - Displayed: 14287 (100.0%) Profile: Default

Decrypt TLS Packets

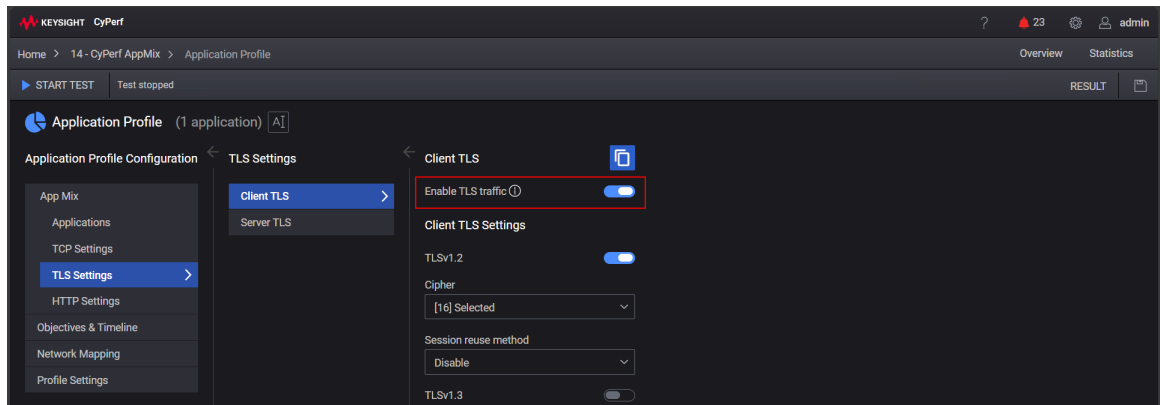
CyPerf allows you to download traffic captures and decrypt TLS packets for in-depth packet-level analysis using Wireshark.

To decrypt and analyze CyPerf TLS packets using Wireshark:

1. Before starting a test, make sure the following options are enabled:

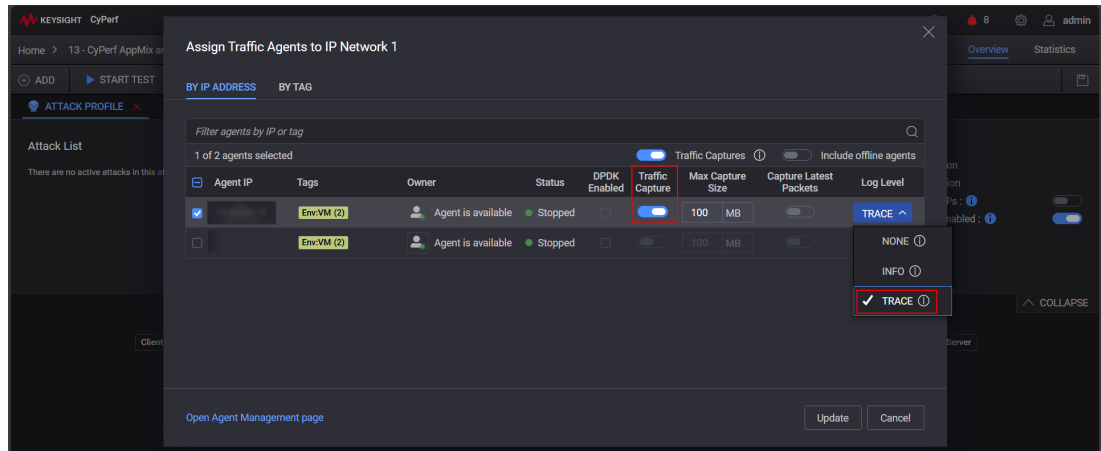
- **Enable TLS traffic:**

Make sure the **Enable TLS traffic** toggle is active under **TLS Settings > Client TLS** on the **Application Profile Configuration** page or the **Attack Profile Configuration** page respectively.



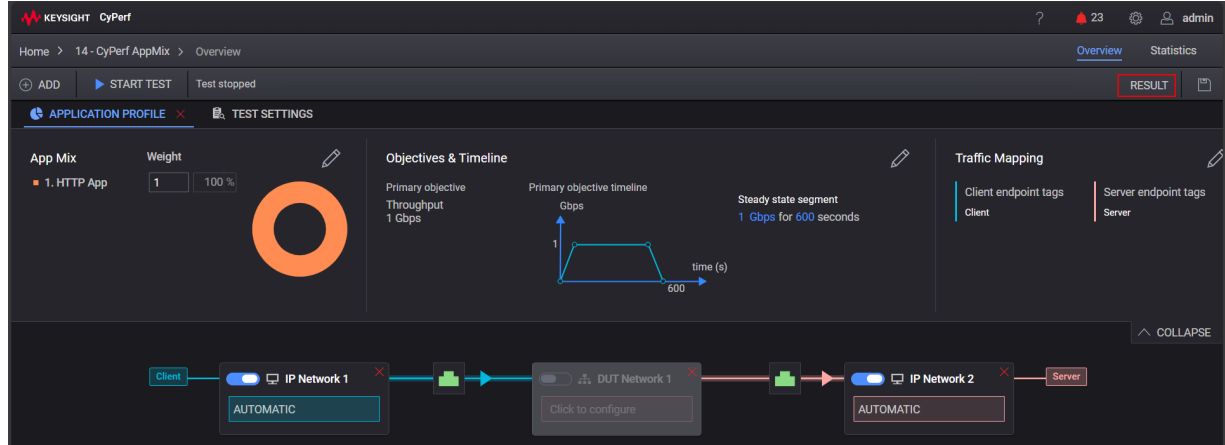
- **Traffic Capture and Trace log level:**

- a. Open the **Assign Traffic Agents** dialog by clicking  or  on the test **Topology** section.
- b. On the **Assign Traffic Agents** dialogue, select the agent(s) you want to use if not already selected.
- c. Then, use the toggle to enable **Traffic Capture** for the selected agent(s).
- d. Select **TRACE** under **Log Level**.

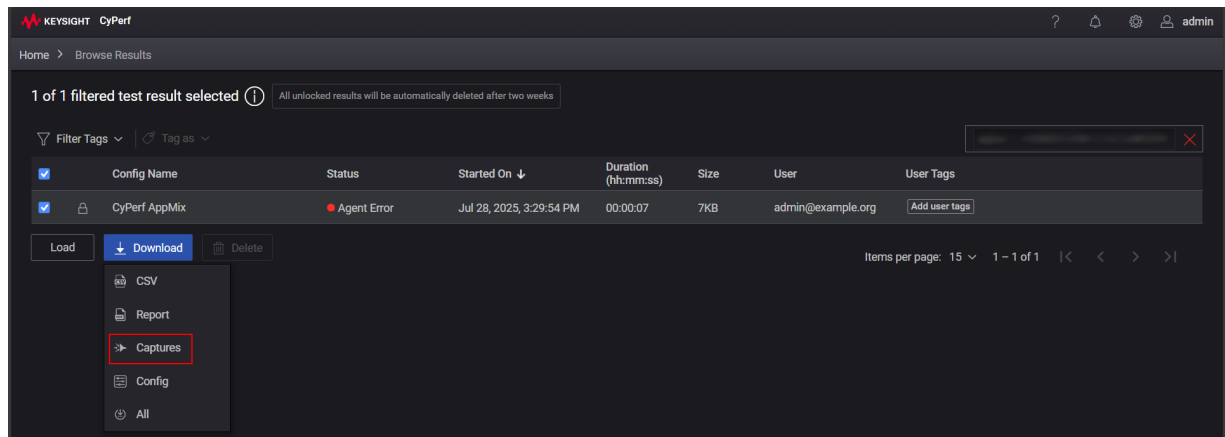


2. Start the test and wait for the test run to complete.

- After the test run is complete, open the **Browse Results** page for the current test by clicking **RESULT**.



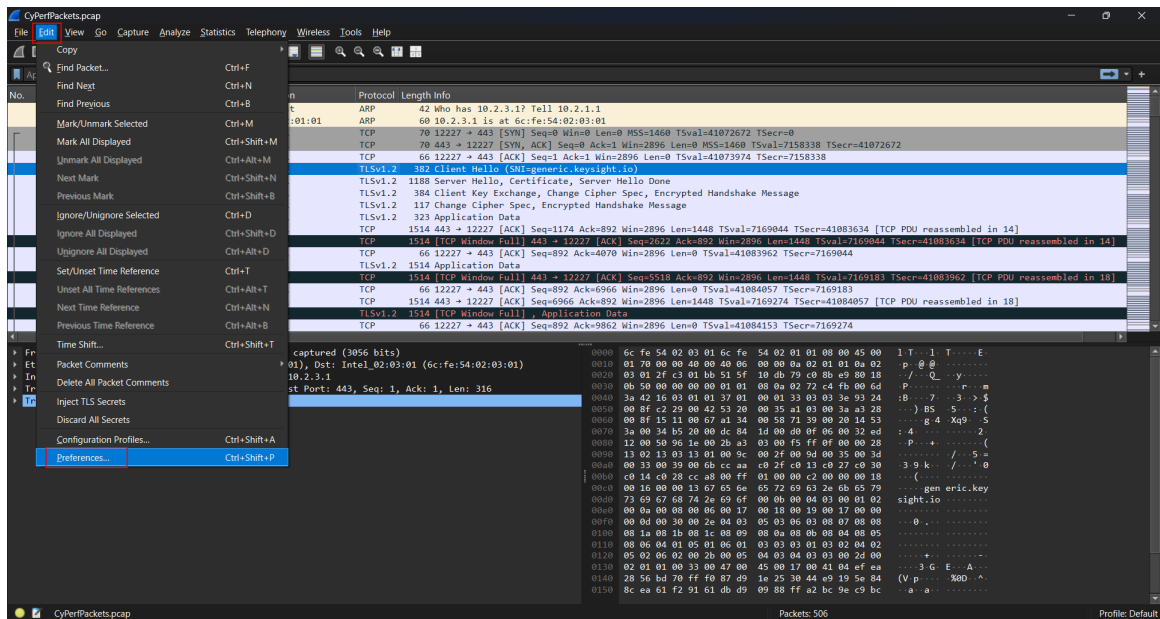
- Select the test results, click **Download**, and select the **Captures** option from the menu that opens.



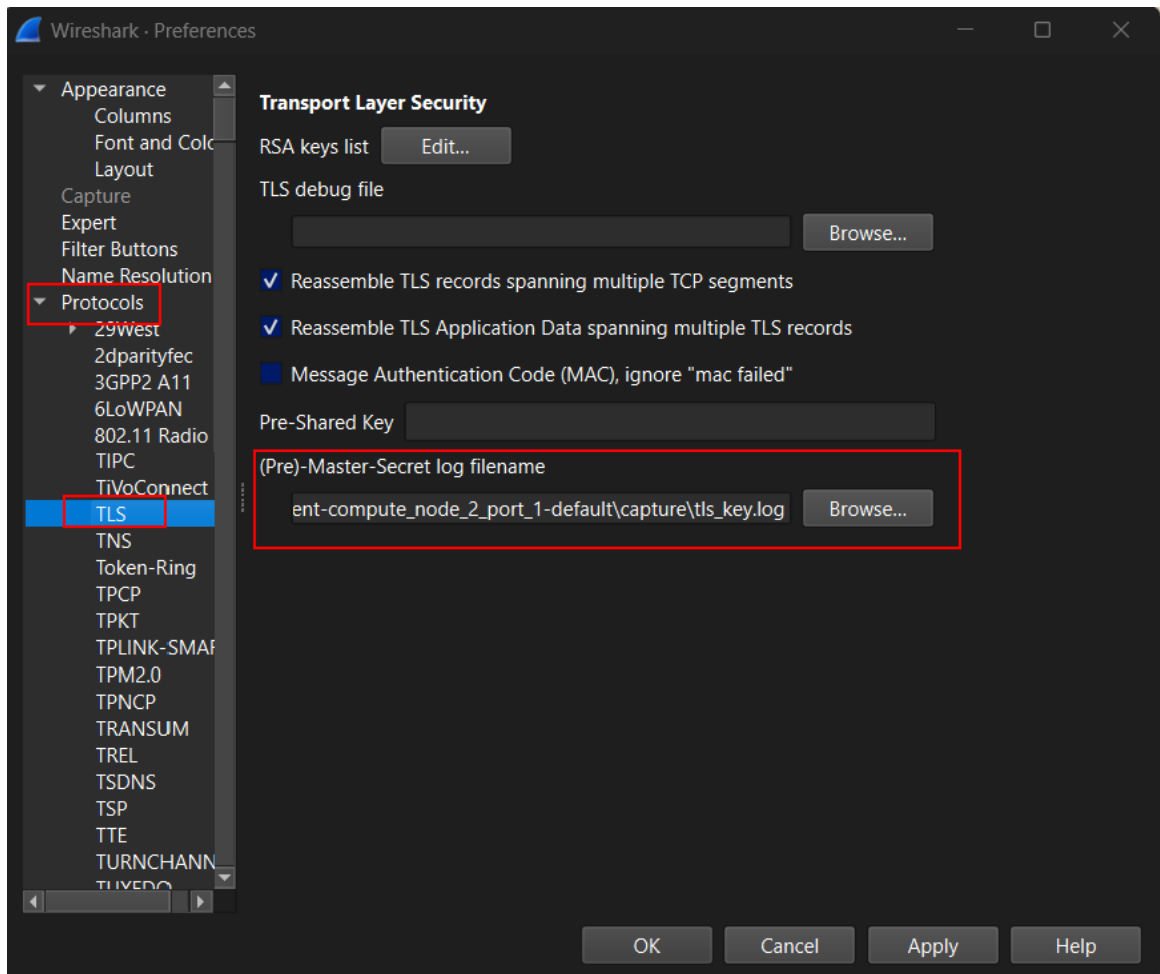
The traffic capture .zip file is downloaded locally on your system. This archive contains the following files:

- CyPerfPackets.pcap**: CyPerf capture file
 - tls_key.log**: log key that you can use to decrypt the TLS packets
- Unzip the archive and save the log key under an easy to find location on your system.
 - In Wireshark, open the **CyPerfPackets.pcap** file.
 - To decrypt the capture file, apply the **tls_key.log** file in Wireshark.

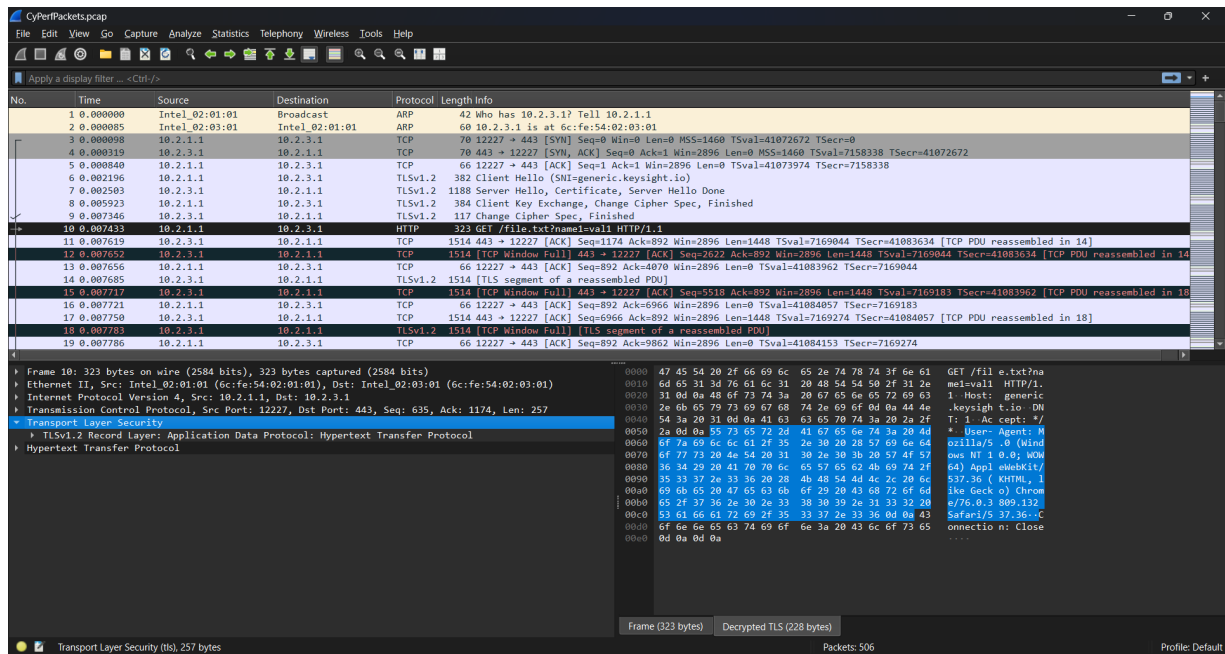
- a. Under **Edit**, select **Preferences**. The Preferences dialog opens.



- b. Expand the **Protocols** drop-down list, scroll down and select **TLS**.



- c. Under **(Pre)-Master-Secret log filename**, browse to and select the `tls_key.log` file you saved in step 5. Click **Apply**, then **OK** to confirm your actions.
8. Now you can see the decrypted TLS capture.



Collect Diagnostics

Keysight CyPerf diagnostics tool is used to collect debug logs and other essential information needed in troubleshooting any encountered issues. Data is provided both on agents and the entire system. For instance, the tool allows you to collect the agent packet captures.

To collect diagnostics, do the following:

1. Click the  menu and select **Collect Diagnostics**.
2. In the **Diagnostics** dialog box, select the optional components to include in the diagnostics files.
3. Click **Generate**, then wait for the diagnostics process to complete. You may close the dialog at any time without interrupting the process. When complete, the archive download link will appear in the list on the right side of the **Diagnostics** dialog.
4. To download a diagnostics archive, click the download link corresponding to the archive you want to download. You can download any diagnostics previously generated on the controller. Archives are listed in descending order by date.
5. To remove diagnostics, click the **Delete**  button for an individual archive, or click **Delete all archives**  to remove all collected diagnostics from the controller.

Logs Level

You have the option to change the internal application logs level:

1. Click the **Settings**  menu, and select the **Logs Level** option. The **Controller Logs Level** dialog opens.
2. Select an option from the **Current log level** drop down on the **Controller Logs Level** dialog. By default the **INFO** log level is enabled.

 **NOTE:** When selecting the **DEBUG** level, logging will take up a significant amount of disk space on the CyPerf controller. We recommend that you only enable **DEBUG** level logging for short periods of time, while reproducing issues that you wish to report.

See also [System Cleanup](#).

System Monitor

The **System Monitor** option under **Settings** (⚙️) enables you to:

- Perform [system cleanup](#) operations to free up space on the CyPerf controller.
- Monitor the [controller health](#)

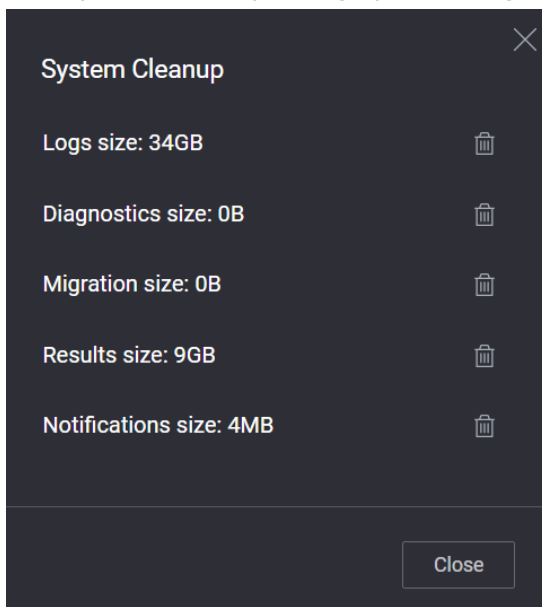
System Cleanup

System logs, diagnostics, migration archives, results archives, and notifications use available disk space on the CyPerf controller, which can trigger `log full` alerts for example.

To view the size of each of these file categories and free up disk space on the CyPerf controller:

1. Select **System Monitor** > **System Cleanup** under **Settings** (⚙️).

The System Cleanup dialog opens listing the size for each file category.

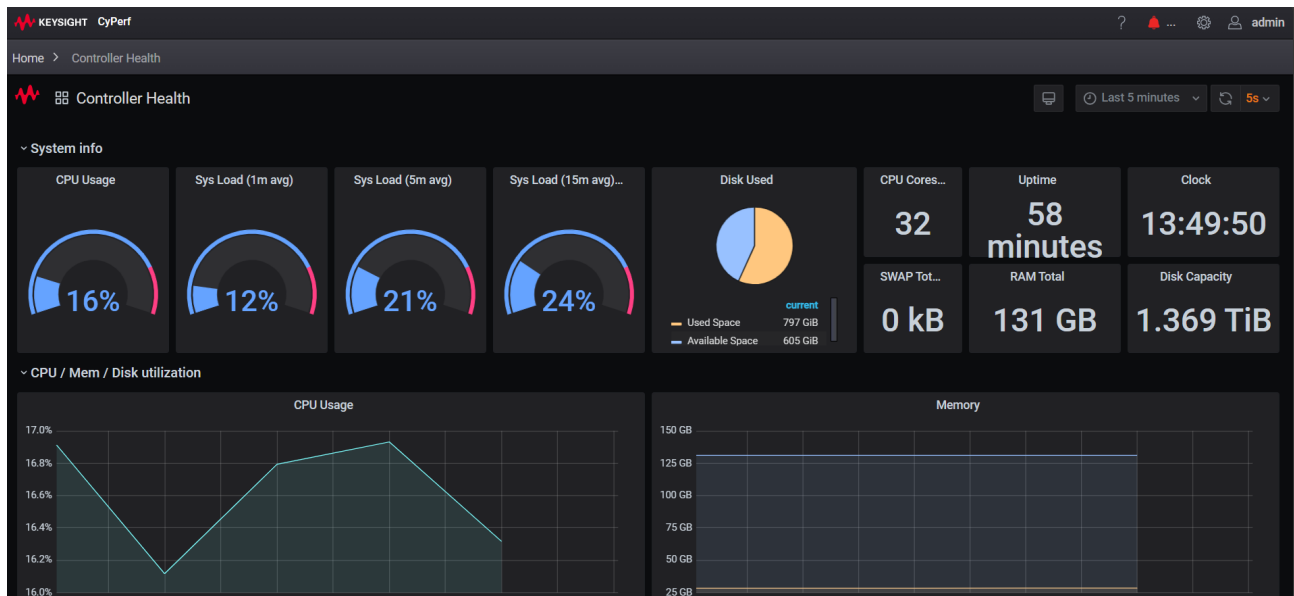


2. Click **Delete** (🗑️) next to the file category you want to remove. This action will delete the files from the controller, freeing up disk space for new logs to be collected.
3. Click **Close** to exit the dialog.

Controller Health

To view controller health statistics, under **Settings** (⚙️), select **System Monitor > Controller Health**.

The Controller Health dashboard opens displaying general system information as well as CPU, memory, and disk utilization graphs. You can select time range options for which to display data on the dashboard, you can refresh the data on the dashboard, or select the time interval at which data is collected and displayed.



CHAPTER 3

Web UI Overview

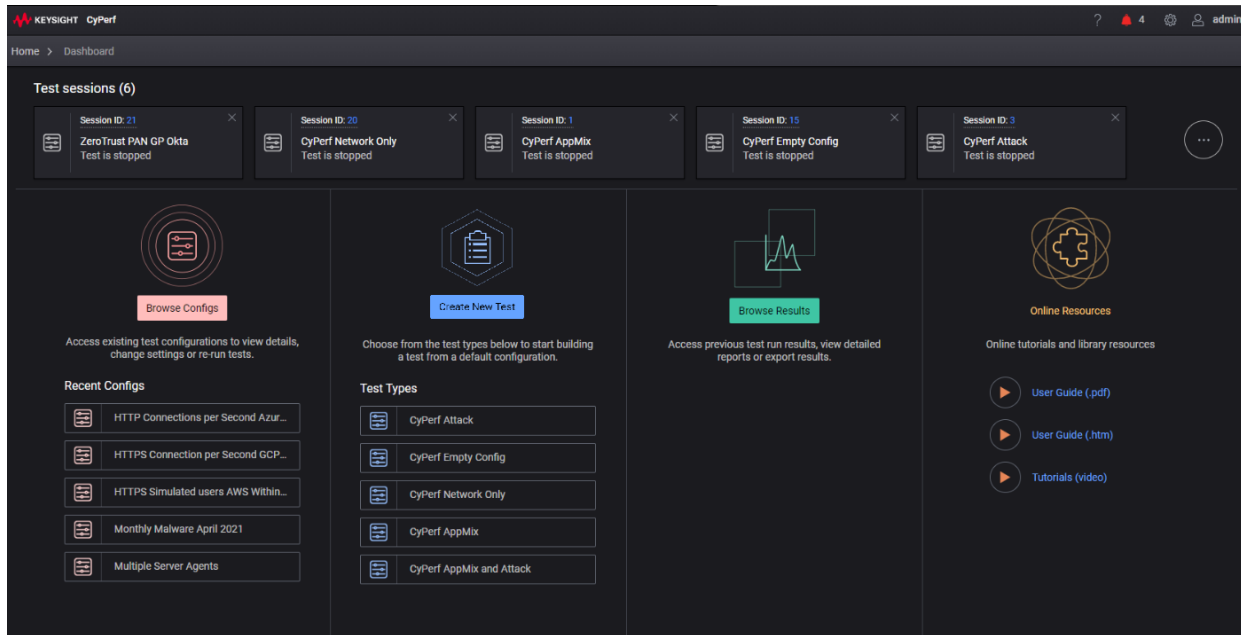
After you successfully log in, the **Dashboard** page opens. From this page you can navigate to the other Keysight CyPerf pages to view and customize test setups, view real-time statistics, view and export test results, view events, logs, and other application and test-specific information.

In this chapter you can find brief descriptions of the UI elements available on each Keysight CyPerf web UI page.

In this chapter:

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HTTP Settings	135
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Dashboard Page



The following UI elements are displayed on the **Dashboard** page:

UI Element	Description
Title bar	The Title bar is positioned at the top of the screen and is accessible from any view within Keysight CyPerf. In the top right corner of the title bar, the following UI elements are available: • Help menu : This menu allows you to access the Technical Support contact form, collect diagnostics, view EULA and Notices documents, access the Keysight CyPerf Software Downloads page, view the current version of the application. You can also view CyPerf Help articles and tutorials from the Help menu. • Notifications : The notifications icon allows you to view notifications triggered at system level, or at system and test session levels when inside a test session. Clicking the icon opens the Events dialog box listing all the notifications. To open the Event page and view more details on each notification, select the Go to events page button. See also View Notifications and Test Events. • Settings : This menu allows you to view and configure additional application settings, such as: ▪ Agent Management: View and manage CyPerf agents. ▪ Controller Security: ◦ Key Management: Copy or revoke server fingerprint. See Manage Server Fingerprints. ◦ Certificate Management: View, upload, generate a custom

UI Element	Description
	certificate, or auto-generate a certificate for the controller. See Manage Controller Certificates. ▪ Offline Tokens: Generate an offline token. ▪ User management: Manage user access to the system. ▪ Logs Level: Manage application logs for debugging purposes. ▪ Licensing: ◦ License Manager: Access and update the License Manager. ◦ License Servers: Add a license server in order to make additional Keysight CyPerf features available. See Manage Licenses. ▪ Library: ◦ Strikes Library: View available strikes. ◦ Resource Library (Beta): View and import available resources, such as payloads, TLS certificates, TLS keys, TLS DHS, playlists, media files, captures and custom applications. For more details, see Work with the Resource Library. ▪ Data Migration: Migrate data from one controller to another through an export controller procedure (click Export Controller) and an import controller procedure (click Import Controller). You can migrate authentication data and other resources (such as custom test configurations, external license servers, controller proxies, and results). See also Migrate Controller Data. ▪ System Monitor: ◦ Controller Health: View system information, CPU, memory and disk utilization, and other system statistics. ◦ System Cleanup: Clean the diagnostics, logs, or migration archives on the system. See System Monitor. ▪ Software Updates: Carry out software updates. See Update Software. • User Account : The user profile who is currently logged in is displayed next to the Settings menu. The User Account menu contains the following: ▪ User Profile: Opens the Keycloak Account Management screen that allows you to edit your profile, or manage user accounts on the system. ▪ Preferences: Switch between the light and dark themes to customize the way the application pages and options are displayed ▪ Log In/ Log Out: Log in or out of the system
Navigation bar	The Navigation bar is positioned at the top of the screen and is accessible from any view within Keysight CyPerf. You can return to the Dashboard page by

UI Element	Description
	selecting the Home button from anywhere in the Keysight CyPerf web UI. After you access the web UI, the Home > Dashboard page is displayed. After you create one or more test sessions, a drop down list of the test configurations loaded in each session will also be displayed on the Navigation bar. This drop down list allows you to easily navigate from one test configuration to another without needing to return to the Dashboard page every time you want to view or modify a test configuration. After you enter a test session the Overview and Statistics tabs become available.
Test sessions	This section displays the current test sessions. Each test session tile displays the name and status of the test. The test status can be: • Test is not started: The test session is created, the test configuration is loaded, but the test has not been started yet. • Test is initializing: After clicking the START TEST button on the test progress bar, the <i>initializing</i> state is displayed on the progress bar and the test session tile. During this phase the hardware resources are allocated and the test is prepared for starting. • Test is configuring: During this stage, the configuration is applied to the test. • Test is running: During this stage, the server and clients groups are connected, test iterations start one by one based on the configured parameters, traffic flows are connected, and traffic starts. • Test is stopping: During this stage, traffic stops, traffic flows disconnect, logs are collected, ports are released, and the hardware disconnects. • Test is stopped: The test is no longer running.
View More button 	Select this button to view all the sessions already created in your environment, which are listed on the Browse Sessions page. See also Browse Sessions Page.
Create New Test	This section allows you to create a new test from a list of predefined configurations. See Create and View a New Test Session.
Browse Configs	This sections allows you to manage previously run tests. You can access the recently run tests by selecting any of the tiles under the Recent tests list. When you select any of the recently run tests, the existing test configuration is loaded and a new test session is created based on this configuration. You can also browse through all the tests by selecting the BROWSE CONFIGS button. The newly open page also enables you to import configuration files using the Import button. See also Browse Configs Page.
Browse Results	This sections allows you to access, export, and delete the results of previously run tests. To browse through all the test results, select the BROWSE RESULTS button. You can also directly access test specific results of recently run tests by

UI Element	Description
	clicking the tiles under the Recent results list. See also Browse Results Page .
Online Resources	This sections allows you to access more Keysight CyPerf online resources such as user documentation and other resources.

Browse Configs Page

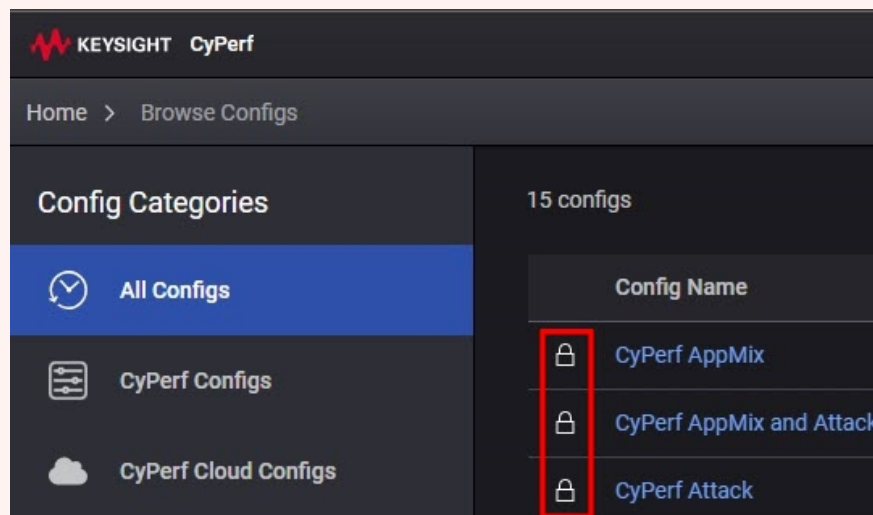
Configurations are grouped under the following tabs:

- All Configs
- CyPerf Configs
- CyPerf Cloud Configs
- [CyPerf DDoS Configs](#)
- CyPerf Features Configs
- CyPerf Performance Configs
- CyPerf Security Configs



IMPORTANT:

System configurations are locked by default and they cannot be deleted or exported.



The **Browse Configs** page allows you to:

- Search for specific configurations by using the search bar on the top right of the view
- Create a new test session based on the selected config, or open a recent test session if the selected config is already used in an existing session.
- Import or export configs: Click the **Export All** button to generate an archive containing all configurations in a .zip file. Click the **Import** button to import a .zip file containing configurations.

DDoS Configs

The following DDoS system configurations are available:

- **DDoS DNS Flood:** Simulates a scenario that includes regular DNS requests as well as a DDoS DNS flood.
- **DDoS HTTP RUDY:** Simulates an HTTP RUDY (R-U-Dead-Yet) attack, where the simulated attackers keep an HTTP connection alive by sending a few bytes at a time from a large HTTP POST body.
- **DDoS HTTP Slowloris:** Simulates an HTTP Slowloris scenario, where the simulated attackers keep an HTTP connection alive by sending a few bytes at a time from the HTTP headers area of GET or POST requests.

Browse Results Page

The **Browse Results** page allows you to access previous test run results, view detailed reports, or export results.

Config Name	Status	Started On	Duration (hh:mm:ss)	Size	User	User Tags
CyPerf AppMix	Manually Stopped	Feb 1, 2023, 7:41:38 PM	00:01:06	1MB	admin@example.org	Add user tags
CyPerf Attack	Completed	Feb 1, 2023, 3:52:47 PM	00:01:02	10MB	admin@example.org	Add user tags
CyPerf AppMix	Completed	Feb 1, 2023, 3:50:01 PM	00:02:13	1MB	admin@example.org	Add user tags

The following UI elements are available on the **Browse Results** page:

Column	Description
Config Name	The configuration name
Status	The status of the results: - Completed - In Progress - Manually Stopped, Agent Error, Internal Error, Config Error, None, Insufficient Licenses, Insufficient Agents, depending on the test issue type.
Started On	The time the test initially started
Duration (hh:mm:ss)	The total duration in hours, minutes and/or seconds of a test
Size	The size of the configuration file.
User	The name of the user who requested the result
User Tags	Displays custom tags previously saved by each user. Click the Add use tags field to enter one or multiple tags as comma-separated values.

You can lock the results of a test by selecting the **Lock**  icon.

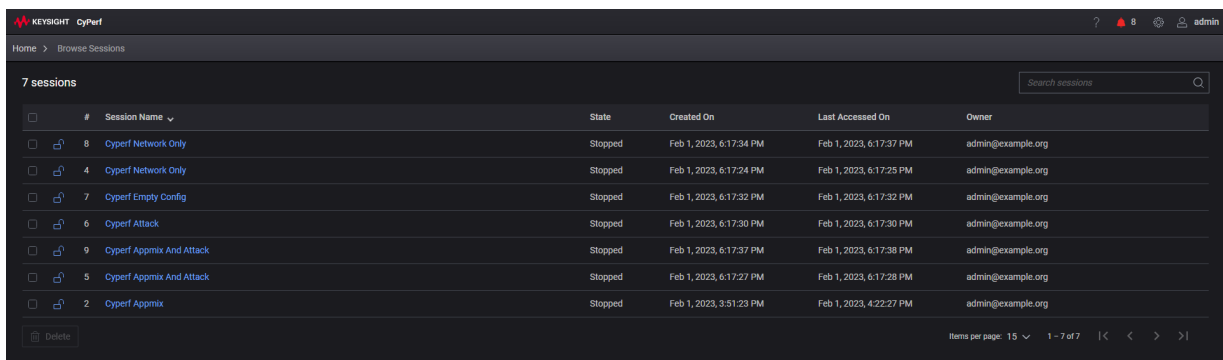


NOTE: All unlocked results are automatically deleted after a period of two weeks.

Browse Sessions Page

To access the **Browse Sessions** page:

1. Log in the web UI.
2. Select the **View More**  button on the right of the **Test sessions** panel. The **Browse Sessions** page opens listing all the available sessions on your environment. You can search and sort sessions based on different column filters, lock or unlock, and delete existing sessions.

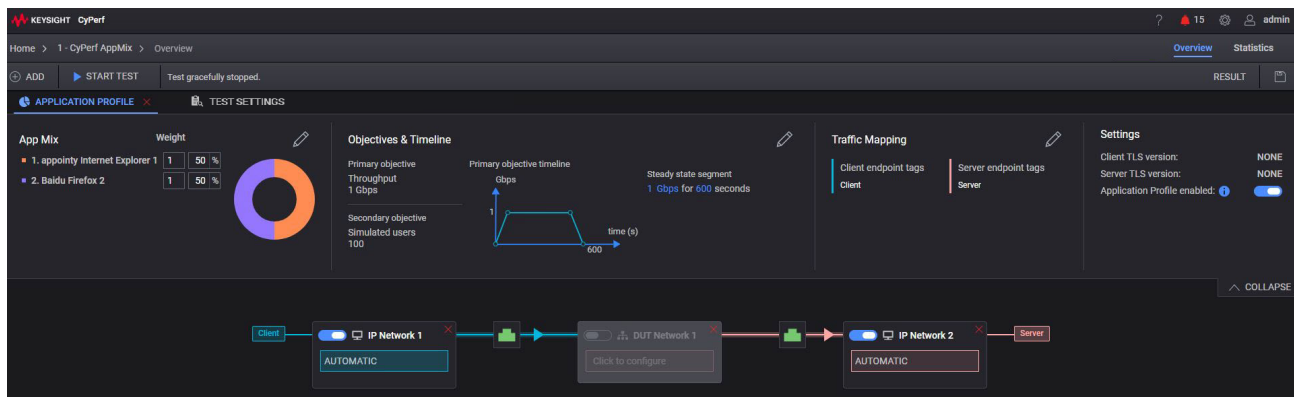


The following UI elements are available on the **Browse Sessions** page:

UI element	Description
Lock button 	Click this button to lock the selected test session.
Unlock button 	Click this button to unlock a test session.
#	Displays the number assigned to the session when it was first created.
Session Name	Displays the session name.
State	Displays the state of the session.
Created On	The date and time when the session was created.
Last Accessed On	The date and time the session was last accessed.

UI element	Description
Owner	The identification information for the user who created the session.
Delete button 	Select the sessions you want to delete, and click the Delete button. The test session is removed from the server and is no longer displayed on the Test sessions panel. <i>NOTE:</i> Deleting locked or pinned sessions is not currently supported from the web UI.

Test Overview Page



The **Overview** page displays a summary of the test configuration and traffic configuration, as well as a visual representation of the test topology. You can access the **Overview** page by creating a test session based on any predefined, newly created, or imported test configuration.

The following UI elements are available from the **Overview** page:

UI Element	Description
Navigation bar	See Dashboard Page .
Test progress bar	After you start a test from an active test session, the progress bar shows you how the test progresses. The progress bar is persistent across all the test session's views. After the test starts, the time it will take the test to complete and the time passed from the test start are also displayed on the progress bar. The following UI elements are displayed: • START TEST button: After you finish configuring the test, click this button to start the test • STOP TEST button: this button becomes available only after you start a test. • Test progress bar: Displays the progress of the test being run. • Amount of time that the test has been running

UI Element	Description
	- Estimated total amount of time for the test run Save button: It allows you to save the current test configuration or to save a copy of the test configuration. The newly saved test is displayed under the recent tests list on the Dashboard page as well as under the BROWSE CONFIGS page.
Application Profile	This tab is split into four sub-sections: App Mix panel Objectives & Timeline panel Traffic Mapping panel Settings panel See also Application Profiles.
Attack Profile	This tab is split into four sub-sections: Attack List panel Objectives & Timeline panel Traffic Mapping panel Settings panel See also Attack Profiles.
Test Settings	This tab allows you to: - Enable or disable an application or attack profile inside a configuration. - Enable Granular Statistics for any test where the configured applications or attacks use a Playlist file in their actions. See also Manage Granular Statistics. - View information about the total duration of a test, available disk space, and a total estimate of the disk usage for that particular configuration.
Test topology section	For each test, a visual representation of the test topology is displayed on the Overview page. See Test Topology.

Application Profiles

Keysight CyPerf allows you to create custom application profiles to employ in your tests.

The following sections describe the settings available for the application profile configuration:

- [App Mix](#)
- [Application Advanced Settings](#) (including [Connections Tab](#), [Application Advanced Settings](#), [HTTP Settings](#), and [Other Properties Tab](#))
- [Application Actions](#)
- [Application Objectives and Timeline](#)
- [Application Network Mapping](#)
- [Application Profile Settings](#)

App Mix

The following UI elements are available on the **Applications** tab under the **Application Profile Configuration>App Mix** section.

Field	Description
Edit Application 	Clicking the Edit button makes the Applications list editable. You can select the application(s) you want to remove from the list, and then click the Delete button . Alternatively, click Cancel to return to the initial list.
Add Application 	Select the application you want to add, for example: • eBanking Chrome to Apache, Firefox to IIS, Internet Explorer to Nginx, Microsoft Edge to Apache • eShop Chrome to Apache, Firefox to IIS, Internet Explorer to Nginx, Microsoft Edge to Apache • HTTP App • UDP Stream • Portal Chrome to Apache, Firefox to IIS, Internet Explorer to Nginx, Microsoft Edge to Apache • Social Network Chrome to Apache, Firefox to IIS, Internet Explorer to Nginx, Microsoft Edge to Apache • Video Platform Chrome to Apache, Firefox to IIS, Internet Explorer for Nginx, Microsoft Edge to Apache • Office365: ▪ Excel: Chrome, Firefox, Internet Explorer, Microsoft Edge ▪ OneDrive: Chrome, Firefox, Internet Explorer, Microsoft Edge ▪ Outlook: Chrome to IIS, Firefox to IIS, Internet Explorer to IIS, Microsoft Edge to IIS

Field	Description
	 NOTE: For the complete applications list, see the <i>ATI Release Notes</i> published with each CyPerf release.
Search bar	The search bar allows you to filter applications based on their names.
Active	Use the toggle to enable or disable the use of the application profile in the test.
Name	The application name.
TLS	Hover over the  icon to display TLS specific information such as <i>Inherits TLS</i> , <i>Client TLS: Disabled</i> , or <i>Server TLS: Disabled</i> .
Weight	Adjust the application weight by entering a value in the Weight field. If the primary objective of an Application Profile is set to Throughput , the selected weight distribution time depends on the types and number of applications added to the application list.
Action Buttons	• AI Rename button: Select to rename the application. •  Advanced Settings button: For details, see Application Advanced Settings. •  Delete button: Select to delete the application.
Reorder button	Click  and drag to change the item position in the list.

Application Advanced Settings

CyPerf allows you to customize the settings of each application added in an application mix.

To access each application's advanced settings:

1. Enter an existing test session, and navigate to the **Application Profile** page.
2. Under **Application Profile Configuration**, select the **Applications** tab.

Home > 1 - CyPerf AppMix > Application Profile

START TEST Test is not started

Application Profile (3 applications) AI

Application Profile Configuration

- App Mix
 - Applications**
 - TCP Settings
 - TLS Settings
 - HTTP Settings
- Objectives & Timeline
- Network Mapping

Applications Edit +

Search applications by name...

Active	Name	TLS	Weight	
☒	Adobe Reader ...		1 33.33%	AI
☒	AWS Console ...		1 33.33%	AI
☒	eBanking Chro...		1 33.33%	AI

If an application mix is already created, the available applications are listed under **Application**. If no applications are listed, add new applications by selecting the button.

3. Select the **Advanced Settings** gear menu () for any of the listed applications. The selected application's settings page opens.

Home > 1 - CyPerf AppMix > Application Profile > Adobe Reader Updates Firefox >

START TEST Test is not started

Overview Statistics

Adobe Reader Updates Firefox AI

Application Settings

- Action timeout (seconds) 30
- Connections**
- TLS Settings
- HTTP Settings
- Other properties

Connections

#	Name	Client Endpoint	Server Endpoint
1.	Adobe Reader Updates	Client	Server acroipm2.adobe.com
2.	Adobe Reader Updates	Client	Server amrmf.adobe.com
3.	Adobe Reader Updates	Client	Server ardownload.adobe.com

Connection Properties

- Name Adobe Reader Updates
- Client endpoint Client
- Server endpoint Server acroipm2.adobe.com
- Hostname User input acroipm2.adobe.com
- Automatic port settings ☒
- Resulting traffic destination port 80
- Resulting CyPerf server listen port

- Configure the application's connection properties, TLS, HTTP, and other settings under each corresponding tab.

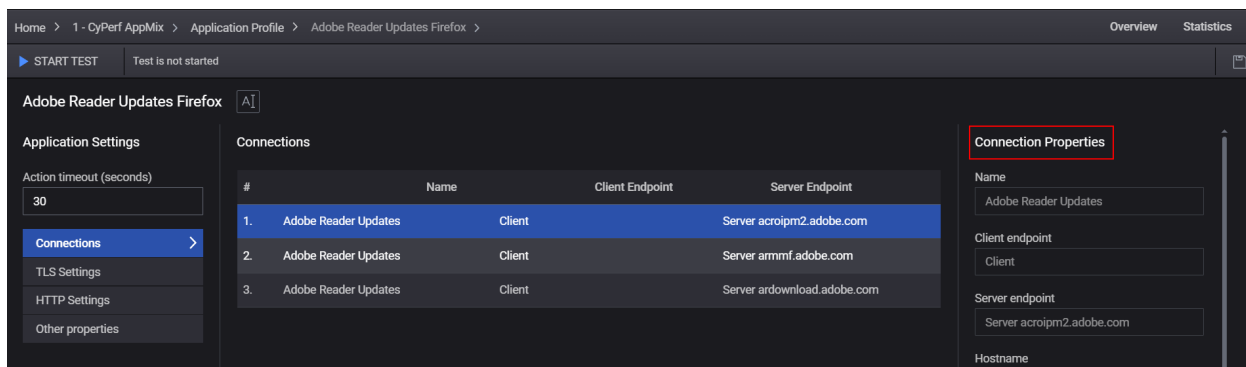
Application Settings

The following table describes the UI elements available on the **Application Settings** page for each application.

Field	Description
Action timeout (seconds)	Configure the length of time in seconds that the client waits for an HTTP request to complete.

Connections Tab

The following table describes the UI elements available under **Application Settings** > **Connections** tab > **Connection Properties** panel.



Field	Description
Name	The application name.
Client Endpoint	The client endpoint.
Server Endpoint	The server endpoint.
Hostname	The name of the host.
Destination Port	The TCP source port that the client endpoint is initiating connections from.
Server Port	The TCP port that the server endpoint is accepting connections on.
<i>HTTP forward proxy mode</i>	
Inherit DUT Configuration	CyPerf inherits the Forward Proxy configuration from the DUT network.
Force HTTP tunnel mode	This option overwrites the Forward Proxy configuration in the DUT Network. When you configured the Forward Proxy for the DUT Network and this

Field	Description
	option is enabled, then the Tunnel Mode is used for the selected application.
Force HTTP proxy mode	This option overwrites the Forward Proxy configuration in the DUT Network. When you configured the Forward Proxy for the DUT Network and this option is enabled, then the Proxy mode is used for the selected application.

Other Properties Tab

The **Other Properties** tab under **Application Advanced Settings** displays additional application-specific parameters that you can configure for the selected application.

See also:

- [TCP Settings](#)
- [TLS Settings](#)
- [HTTP Settings](#)

Application Actions

The following table lists some examples of the application actions and action parameters available in Keysight CyPerf.

Application Action	Action Parameters	Parameter Description
DNS	DNS Query	Query host: Displays the host for which the DNS lookup will be performed.
		Query type: Select the type of query to be executed.
DNS Flood	Simulates an attacker in a DDoS Flood that sends DNS requests. The DNS requests can be customized with any option available for regular DNS requests. Unlike the regular DNS app, the client will not wait to receive a DNS response before sending the next request.	
eBanking		
SignUp	SignUp username	Signup username.
	SignUp password	Signup password.
	SignUp confirm password	Signup password confirmation.
Login	Login username	Login username.
	Login password	Login password.
View Transactions	N/A	

Application Action	Action Parameters	Parameter Description
View Accounts	N/A	
Get Contact Page	N/A	
Logout	N/A	
<i>eShop</i>		
Search Product	Product name	Product name.
View Product	Product ID	Product ID.
Login	Login username	Login username.
	Login password	Login password.
Add to Cart	N/A	
Remove from Cart	N/A	
Buy	Full name	Full name.
	Address	Address
	Account number	Account number
Logout	N/A	
<i>HTTP</i>		
HTTP GET	Path	The value(s) of the requested path. When you choose the User input parameter type then you need to provide the value of the path. You can upload a .csv file containing multiple values by selecting the parameter type as Playlist file. For more information on the Playlist file, see Artifacts.
	Query	The value of the query requested.
	Request headers	Accept-Language.
		Sec-Fetch-User.
		Upgrade-Insecure-Requests.
		Sec-Fetch-Site.

Application Action	Action Parameters	Parameter Description
	Status code	The value of the response status code.
	Reason phrase	The value of the reason phrase.
	Response headers	Cache-Control.
		Etag.
	Response body	Payload profile.
		Synthetic data (bytes).
		Dynamic Payload .
HTTP POST	URL	The URL.
		Sec-Fetch-User.
		Upgrade-Insecure-Requests.
		Accept-Language.
		Sec-Fetch-Site.
	Request body	Payload profile.
		Synthetic data (bytes).
	Status code	The value of the response status code.
	Reason phrase	The value of the response reason phrase.
	Response headers	Etag.
		Cache-Control.
	Response body	Add a response message.
Sleep	Duration (milliseconds)	The value of the parameter in milliseconds. When enabled the next immediate action is delayed by the specified value. You can add a single value or a range of values. For example: 100 or 100 - 200 .

Application Action	Action Parameters	Parameter Description
		① NOTE: At the end of the test duration, the Sleep duration is shortened internally to a maximum of 1 second for the stopping simulated users.
Loop Begin	Number of iterations	Use Loop Begin and Loop End to repeat a defined sequence of actions within a CyPerf test workflow. Loop Begin marks the start of a loop block, and it must be paired with a Loop End action. The Number of iterations parameter specifies how many times a loop executes. An iteration number set to 0 configures the loop to run indefinitely (an infinite loop). An infinite loop must be positioned as the outermost loop in the action list, and its corresponding Loop End action must be the final action in that list. Infinite loops are subject to the following constraints: • When an AppMix includes only one application, infinite loops are not allowed if the test objective is Concurrent Connections (CC). • When an AppMix includes multiple applications, infinite loops are only permitted if the test objective is Simulated Users (SU).
Loop End	N/A	Loop End marks the end of a loop block, and it must be paired with a Loop Begin action.

Application Action	Action Parameters	Parameter Description
		</

Application Action	Action Parameters	Parameter Description
Create Excel Workbook	Workbook name	Workbook name.
Edit Workbook	Content	Editable content.
Pin Workbook	Workbook name	Workbook name.
Open Workbook In OneDrive	N/A	
Sign Out	N/A	
Office365 OneDrive		
Get Home Page	N/A	
Sign In	User Email	Login email.
	User Display Name	Displayed name.
	User Password	Login password.
Get OneDrive Tab	N/A	
Delete File	N/A	
Upload File	File Name	File name.
	Upload File	Synthetic data (bytes).
Create Folder	Folder Name	The name of the folder to be created.
Create Excel Workbook	Workbook Name	Workbook name.
Create Word Document	Document Name	Name of the word document.
Create Powerpoint Presentation	PowerPoint Name	Name of the power point.
Sign Out	N/A	
Office365 Outlook		
Sign In	Username	Login username.
	Password	Login password.

Application Action	Action Parameters	Parameter Description
View Inbox	N/A	
Send Message	Recipient	The email's recipient.
	Body	The email content.
Send Message With Attachment	Recipient	The email's recipient.
	Body	The email content.
	Attachment	Synthetic data (bytes).
Open Message	N/A	
Delete Message	N/A	
Sign Out	N/A	
<i>Portal</i>		
Login	Login Username	Login username.
	Login Password	Login password.
Search Image	Search Query	Query string.
Upload Image	Uploaded File Name	Filename to be saved on the server.
	Uploaded File	Payload profile.
		Synthetic data (bytes).
Logout	N/A	
<i>Social Network</i>		
Login	Login username	Login username.
	Login password	Login password.
News Feed	N/A	
View Profile	Member id	Member ID.
Like Post	N/A	
Unlike Post	N/A	
Create Post	Post content	Text string.

Application Action	Action Parameters	Parameter Description
Comment To Post	Comment content	Text string.
	Original post ID	Original post ID.
Logout	N/A	
Video Platform		
Login	Login username	Login username.
	Login password	Login password.
Search Video	Video name	Video name.
Download Video	Downloaded file name	Downloaded file name.
	Downloaded file	Payload profile.
		Synthetic data (bytes).
Upload Video	Uploaded file name	Uploaded file name.
	Uploaded file	Payload profile.
		Synthetic data (bytes).
Delete Video	N/A	
Like Video	N/A	
Unlike Video	N/A	
Logout	N/A	

The difference between Dynamic and Payload files

- If the chosen file is Payload (not Dynamic), the exact contents of the file can be seen on the wire.
- If the chosen file is Dynamic and the file does not contain Macros, then the behavior is the same as above.
- If the chosen file is Dynamic and the file contains Macros, then each Macro is evaluated during the test with the expected value that the Macro is meant to generate.

**NOTE:**

- Payload profile is complementary in all the Server file options. For example, if you upload a DH file, then it will also appear as a key or certificate file.
- For all the applications that have a **Login** or **Signup** action, you can also use the **Playlist file** option. You can upload a .csv file containing multiple usernames and their associated passwords, the playlist bypasses the single user login and uses all entries from the uploaded .csv file. The reference column will have the corresponding CSV column names, which can be chosen from the drop down menu. For example, the .csv file has two columns named **Username** and **Password**, add these exact names in the **Login username** and **Login password** fields.

For more information, see [Artifacts](#).

The difference between User Input and Dynamic User Input

- If the chosen input is User Input (not Dynamic), the configured value can be seen on the wire.
- If the chosen input is Dynamic and the configured value does not contain Macros, then the behavior is the same as above.
- If the chosen input is Dynamic and the configured value contains Macros, then each Macro is evaluated during the test with the expected value that the Macro is meant to generate.

Stateless UDP Stream Settings

You can use **UDP Stream** application profiles to send high-volume UDP traffic with controllable packet sizes for raw performance testing.

For Stateless Streaming traffic, the following traffic stream directions are supported:

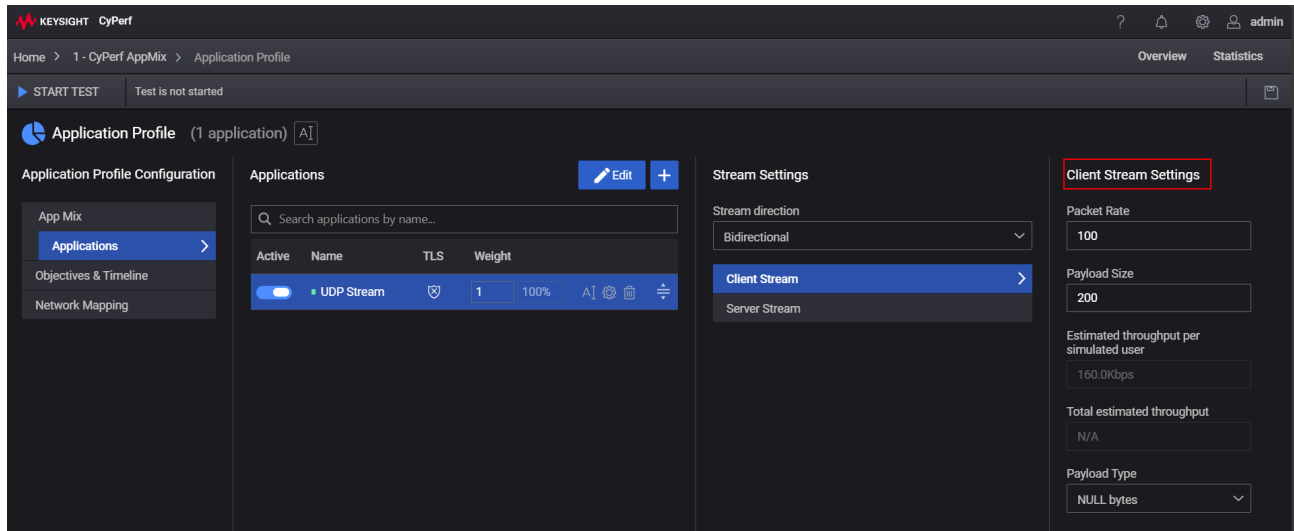
- **Client to Server**
- **Server to Client**
- **Bidirectional**

One-way and two-way traffic is supported, with computable overall throughput and computable throughput per simulated user. This is based on pre-computed stream count and known packet size.

A single agent can both send and receive application traffic, attack traffic, and stateless streaming traffic.

Client and Server Stream Settings

The following settings are available for configuration under **Application Profile Configuration > Applications > Stream Settings > Client Stream Settings** or **Server Stream Settings**:



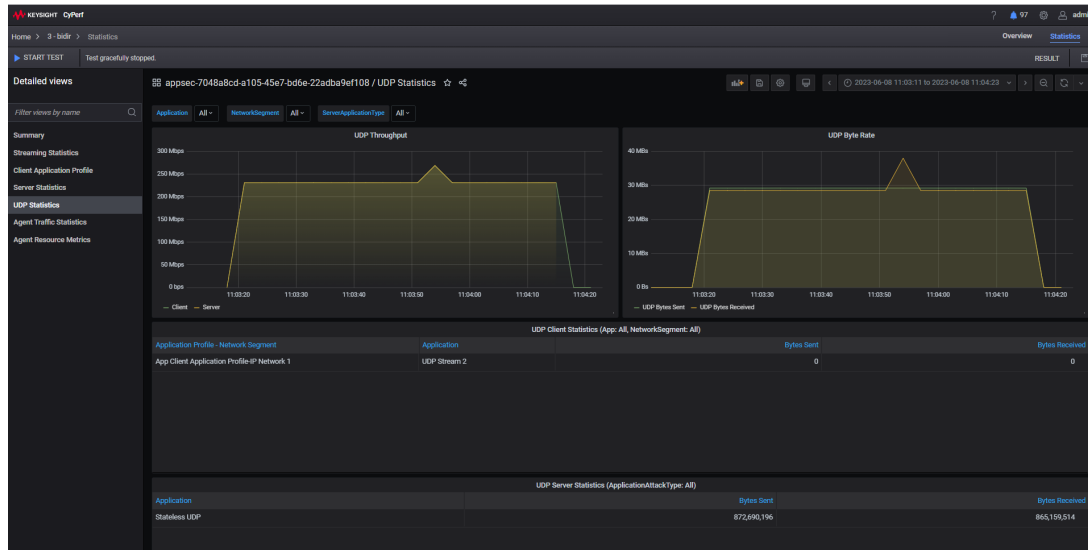
Field	Description
Packet Rate	Enter the packet rate for the selected client or server stream.
Payload Size	Enter the payload size for the selected client or server stream.
Estimated throughput per simulated user	Displays the estimated throughput per simulated user which is computed based on the configured Packet Rate and Payload Size values.
Total estimated throughput	Displays the total estimated throughput for the selected client or server stream.
Payload type	Select one of the available options: • Random: Generates a random payload size for each packet. • Pseudorandom: Generates a random data pool size. Each packet will be assigned a payload size from this random data pool. The Random data pool size parameter is enabled only after selecting the Pseudorandom payload type. • NULL bytes: Generates a payload full of 0 values.

Statistics

When you successfully start a test, CyPerf displays the **Statistics** page, where you can view real time statistics. For the **Stateless Streaming** two new statistics views are available under **Detailed views**:

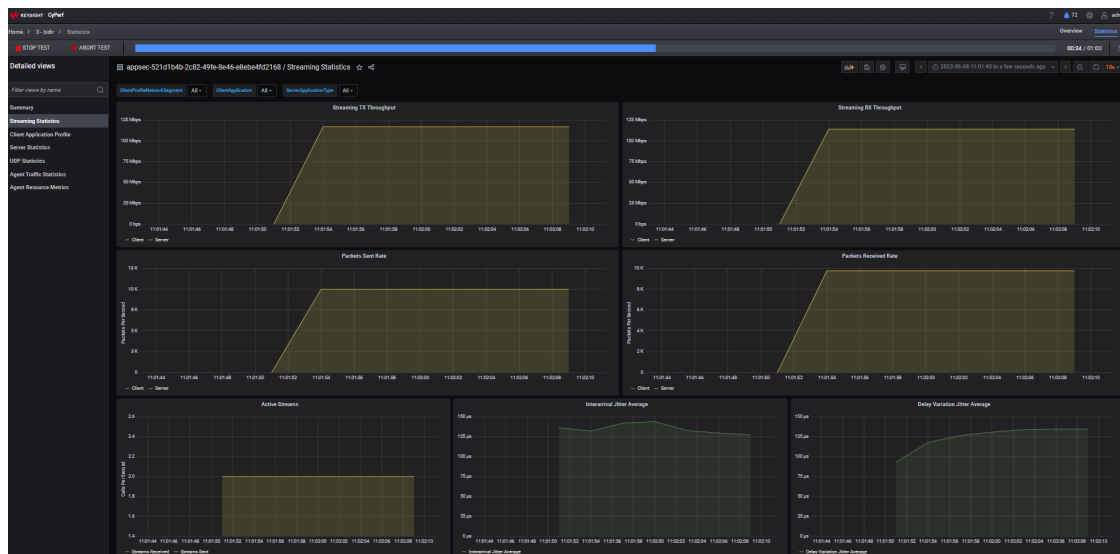
- **UDP Statistics**:

The following statistics are available on the **UDP Statistics** view: UDP Throughput, UDP Byte Rate, UDP Client Statistics (Bytes sent and Bytes received), and UDP Server statistics (Bytes sent and Bytes received).



- **Streaming Statistics:**

The following statistics are available on the **Streaming Statistics** view: Streaming TX Throughput, Streaming RX Throughput, Packets Sent Rate, Packets Received Rate, Active Streams, Interarrival Jitter Average, Delay Variation Jitter Average, Round Trip Time, Delay Variation Jitter Distribution, Packet Loss Distribution, Client or Server Packet Counters (Sent packets, Received packets, Sent bytes, Received bytes, Lost packets, Late packets, Duplicate packets, Misordered packets, Error Packets)



Application Objectives and Timeline

The following UI elements are available on the **Objectives and Timeline** tab under the **Application Profile Configuration** section.

① **NOTE:** For rate-based objectives, the TLS VPN tunnels that are created in a test could be less than the maximum possible configured tunnel count. This is because CyPerf uses a goal-seek algorithm to determine the maximum number of simulated users required to achieve the configured objective.

Field	Description
<i>Primary Objective</i>	
Objective type	Select a primary objective type from the available options: • Throughput • Connections per second • Simulated users • Concurrent connections • Prompts per second ① NOTE: Streaming applications (such as Zoom or VoIP) support only the Simulated Users primary objective. Starting with release 26.0.0, UDP applications also support the Secondary objective. See also About Primary Objectives.
<i>Secondary Objectives</i>	
Secondary objective enabled	Allows you to enable or disable the secondary objective.
Objective type	Select a secondary objective type from the available options: • Throughput • Connections per second • Simulated users

Field	Description
	 NOTE: Concurrent Connections is not currently supported as a secondary objective. - When the <i>primary</i> objective is set to Concurrent Connections, <i>none</i> of the <i>secondary</i> objectives are supported.
Steady state segment	For both primary and secondary objectives, you can configure the steady state segment value (in Gbps) to set the data-transfer type, and the segment duration. For the Throughput objective, you can choose from additional units: • Gbps • Mbps • Kbps • bps
<i>Timeline</i>	
Step up segment	Enable and configure the step duration and the number of steps for the segment.
Step down segment	
Max simulated users per second	Configure the maximum number of simulated users per second that you want to use. The default value is 0, which means that there is no limit of simulated users per second.
Max pending simulated users	Configure the maximum amount of pending simulated users that you want to use. You can use either a fixed value or a percentage of the total users.
Timeline representation	Represents a preview of the set duration and value for the application timelines.
<i>Advanced Settings</i>	
Warm up period (seconds)	The time taken by the devices in the middle to finish the server health check. Configure the warm up period option to force clients to wait until the end of the configured timer before requests start being sent.
Optimization Mode	Configure the type of optimization: • Balanced: Optimized for both rate-based test objectives (throughput and connections per second) and capacity-based test objectives (simulated users). This is the default option. • Rate Mode: Optimized for rate-based test objectives (throughput and connections per second) at the cost of capacity-based objectives.

Field	Description
Streaming Purpose CPU allocation (%)	Enter the CPU percentage that the agent allocates for generating UDP streaming traffic (e.g., Zoom traffic). The default value is 25%. Values can range from 0% to 99%. For tests that are configured with a high UDP streaming traffic proportion, it is recommended that the Streaming purpose CPU allocation (%) parameter is set to higher values, such as 25% or more. An automatic calibration option is also supported for automatically finding and configuring the optimal streaming purpose CPU allocation value. See the Auto Calibrate option below for more details. See also Inherit streaming CPU allocation.
Auto calibrate	 NOTE: For tests containing UDP streaming traffic (e.g., Zoom), the Auto calibrate option is the recommended method for configuring the Streaming purpose CPU allocation (%) parameter. The Auto calibrate option is based on an automatic algorithm, and it involves at least two calibration steps. At the end of the calibration process, the system automatically finds and configures the optimal Streaming purpose CPU allocation (%) value for the particular test configuration and deployment environment in use. On average, the Auto calibrate process takes approximately 22 minutes. However, in certain scenarios (such as, unpredictable and unreliable network paths) the process can take up to 45 minutes. For details on how to use the Auto calibrate feature, see Auto Calibrate Streaming Purpose CPU Allocation.
Connections graceful stop timeout (seconds)	Number of seconds the agent waits for flows to time out before gracefully closing the open connections. The default value is 120 seconds. See more details in Start and Stop Tests.

About Primary Objectives

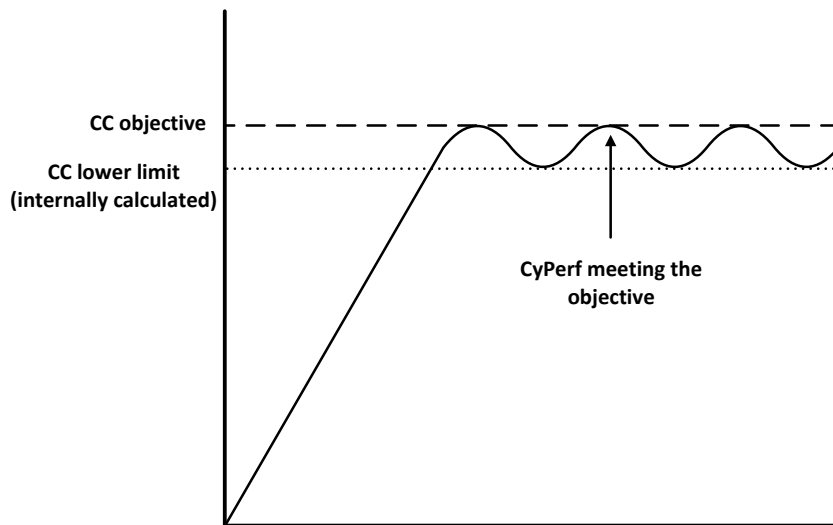
Consider the following notes when configuring the primary objective for the test:

Simulated Users as primary objective

The configured Simulated Users value is distributed among the applications based on each application's configured weight.

Concurrent Connections as primary objective

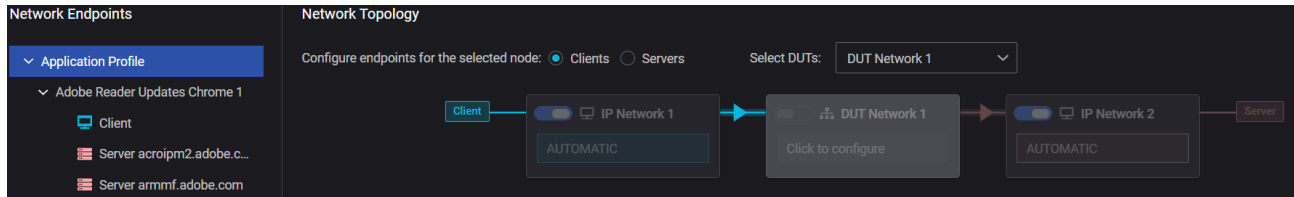
Similar to the Simulated Users objective value, the Concurrent Connections (CC) value is also distributed among the Applications based on each application's configured weight. CyPerf will not exceed the configured value and tries to maintain a close enough concurrent connections level during the test run. Depending on the application's connection count requirement, there might be a gap between the configured limit and the achieved concurrency. The mechanism for maintaining the Concurrent Connections objective uses this gap to allow closing connections from running application flows and reopening connections for new flows. Typically, this gap will be two times the connection count for the application (or a fixed buffer of 40, whichever is greater) for every core being used for the CyPerf client agent simulating the application traffic.



NOTE:

1. The **Concurrent Connection** objective prioritizes persistent connection behavior. Therefore, high **max transaction per connection** values might result in application failure due to starvation for some simulated users waiting for execution to continue the application sessions.
2. For CyPerf client agents using the **Concurrent Connections** objective, you cannot explicitly control **ramp up** or **ramp down** rate for **Simulated Users**. In this case the **Simulated Users** value will be internally calculated, and 10% of that will be used as **Max pending simulated users** without a specific limit per second.

Application Network Mapping



The Network Mapping section enables you to adjust the traffic flow for each application profile.

The following UI elements are available on the **Network Mapping** tab under the **Application Profile Configuration** section.

Field	Description
Network Endpoints	Displays the network endpoints in the currently configured application profile. A yellow dot is shown next to an item mapped differently than its parent.
Network Topology	Displays the network topology for each application profile. You can configure endpoints for the selected node, which can be either clients or servers.

For more information on how to configure the traffic flow for an application, refer to [Manage Application Network Mapping](#).

Application Profile Settings

The following UI elements are available on the **Profile Settings** tab under the **Application Profile Configuration** section.

Field	Description
Active	Use the toggle to enable or disable the use of the application profile in the test. You can also enable or disable an application profile by switching the Application Profile enabled toggle on or off in the Settings panel under the Application Profile tab on the Test Overview page.
Use all source IPs	By default, this setting is <i>disabled</i>. When Use all source IPs is <i>disabled</i>, each Simulated User will be bound to a specific set of source IP addresses (IPv4 and/or IPv6) for the entire duration of the test. As a result, the number of source IPs used in the test will be limited by the number of Simulated Users. When Use all source IPs is <i>enabled</i>, Simulated Users will collectively try to use all source IP addresses configured in the test. They do this by switching source IPs as needed across different iterations of each application throughout the entire test duration. If there are multiple DUT destinations, or if the value of Max transactions (over a persistent connection) is greater than 1, source IP switching will occur only after the application execution meets those conditions. In such cases, source IP switching might happen after a few iterations.  NOTE: • This feature is supported only for the IP Stack and MAC DTLS Stack. • Source IP switching may not occur if a single iteration of an application continues for the entire test duration. Examples include UDP Stream applications, Zoom applications using Sleep durations comparable to the test duration, or applications using infinite Max transactions over a persistent connection. You can also enable or disable Use all source IPs from the Settings panel under the Application Profile tab on the Test Overview page.

Attack Profiles

Keysight CyPerf allows you to create attack profiles to employ in your tests.

The following sections describe the settings available for the attack profile configuration:

- [Attack List](#)
- [Attack Advanced Settings](#) (including [Connections Tab](#), [Other Properties Tab](#), and [Application Advanced Settings](#))
- [Attack Actions](#)
- [Attack Strikes](#)
- [Attack Objectives and Timeline](#)
- [Attack Network Mapping](#)
- [Attack Profile Settings](#)

Attack List

The following UI elements are available on the **Attack List > Attacks** tab under the **Attack Profile Configuration** section.

Field	Description
Edit button 	Use this button to edit the list of attacks or the list of strikes and actions. Click the Edit button and select the items on the list that you want to delete; then click the Delete  button.
Add Attack button/ Add Strikes and Actions button 	You can filter attacks based on category name, for example Browser, Date, Direction, Intent , and others. From the filtered attack list, select the attack you want to add, for example: • eBanking Attack on Chrome, Firefox, Internet Explorer and Microsoft Edge browsers • eShop Attack on Chrome, Firefox, Internet Explorer and Microsoft Edge browsers • Portal Exploit Attack with JSP file upload on Nginx server and PHP file upload on IIS server • Portal RCE Attack on Apache server and XSS Attack on Apache server • Social Network Attack on Chrome, Firefox, Internet Explorer and Microsoft Edge browsers • Top browser attacks • Top Google Chrome, Internet Explore, Microsoft Edge and Mozilla Firefox browser attacks • Top RCE attacks • Top Safari browser attacks • Top webshell attacks

Field	Description
	• Video Platform Exploit Attack with JSP file upload on Apache or Nginx server • Video Platform Exploit Attack with PHP file upload on Apache or Nginx server  NOTE: For the complete attacks list, see the ATI Release Notes published with each CyPerf release.
Search bar	The search bar allows you to filter attacks based on their names.
Active	Use the toggle to enable or disable the use of the attack profile in the test.
Name	The attack name.
TLS	Hover over the  icon to display TLS specific information such as <i>Inherits TLS, Client TLS: Disabled, or Server TLS: Disabled.</i>
Action buttons	Select to access the following additional settings: • AI Rename - Select to rename the attack. •  Advanced Settings - for further information, refer to Attack Advanced Settings •  Delete - Select to delete the attack.
Change Item Position button 	Click this button to change the position of the attack in the Attacks list, or the position of the strike and action in the Strike and Actions list.
Connect Strikes to Server Endpoints button 	Select the strike you want to connect to the server endpoint. Then, click the Connect Strikes to Server Endpoints button.

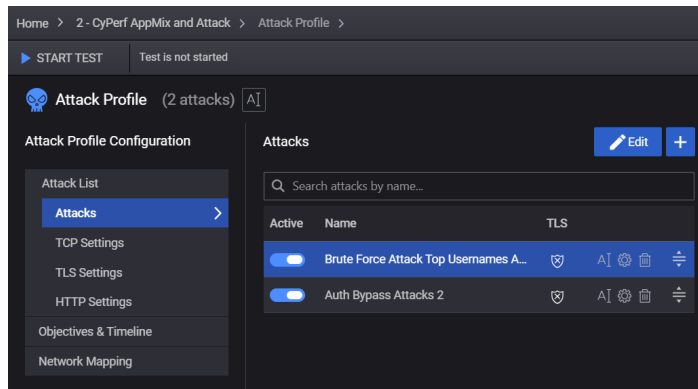
 **NOTE:** You can also create and use an Office 365 by using the **Customize Attack** tab in **Configure Attack Profile > Attacks**.

Attack Advanced Settings

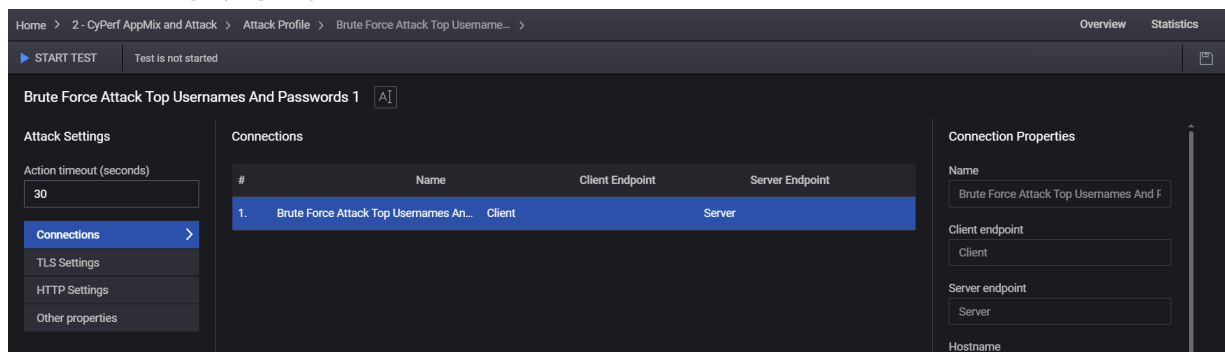
CyPerf allows you to customize the settings of each attack added in an attack profile.

To access each attack's advanced settings:

1. Enter an existing test session, and navigate to the **Attack Profile** page.
2. Under **Attack Profile Configuration**, select the **Attacks** tab. If an attack mix is already created, the available attacks are listed under **Attacks**. If no attacks are listed, add new attacks by selecting the **+** button.



3. Select the **Advanced Settings** gear menu (⚙️) for any of the listed attacks. The selected attack's settings page opens.



4. Configure the attack properties, connection properties, TLS, and HTTP settings, under each corresponding tab.

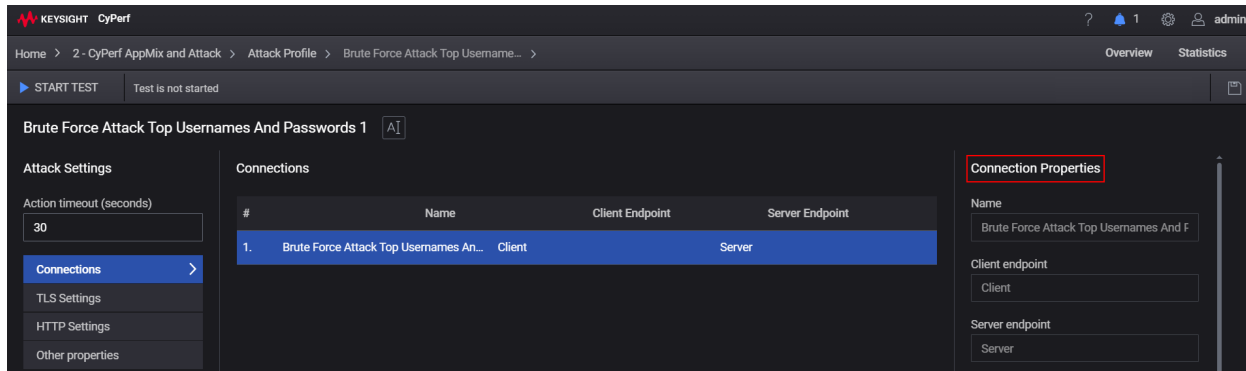
Attack Settings

The following table describes the UI elements available on the **Attack Settings** page for each attack.

Field	Description
Action timeout (seconds)	Configure the length of time in seconds that the client waits for an HTTP request to complete.

Connections Tab

The following table describes the UI elements available under **Attack Settings > Connections** tab > **Connection Properties** panel.



Field	Description
Name	Set the connection name.
Client Endpoint	Enter the client endpoint.
Server Endpoint	Set the server endpoint.
Hostname	The name of the host.
Destination port	The TCP source port that the client endpoint is initiating connections from.
Server port	The TCP port that the server endpoint is accepting connections on.
Encryption disabled	Disables the decryption for the selected application. If TLS is enabled, all applications from the Traffic/Attack profile have their data encrypted, except for the applications which have Encryption disable on.
<i>HTTP forward proxy mode</i>	
Inherit DUT Configuration	CyPerf inherits the Forward Proxy configuration from the DUT network.
Force HTTP tunnel mode	This option overwrites the Forward Proxy configuration in the DUT Network. When you configured the Forward Proxy for the DUT Network and this option is enabled, then the Tunnel Mode is used for the selected application.
Force HTTP proxy mode	This option overwrites the Forward Proxy configuration in the DUT Network. When you configured the Forward Proxy for the DUT Network and this option is enabled, then the Proxy mode is used for the selected application.



NOTE: A default port value of 0 means that a random value is picked between the **Min source port** and **Max source port** parameters configured under the **TCP Settings** profile tab.

Other Properties Tab

The **Other Properties** tab under **Attack Advanced Settings** displays additional attack-specific parameters that you can configure for the selected attack. The following table lists as an example the parameters available for *Strike Adobe Acrobat JBIG2 Stream Indexing Overflow (SMTP Quoted Printable)*.

Field	Description
Email recipient	The username of the recipient at the email domain.
Email sender	The username of the sender at the email domain.
Email sender domain	The internet TLD of the email sender.
SMTP username	Enter the username used for SMTP authentication.
SMTP password	Enter the password used for SMTP authentication.

See also:

- [TCP Settings](#)
- [TLS Settings](#)
- [HTTP Settings](#)

Attack Actions

The following table lists some examples of the attack actions and action parameters available in Keysight CyPerf.

Attack Action	Action Parameters	Parameter Description
eBanking		
Sign Up	SignUp Username	Signup username.
	SignUp Password	Signup password.
	SignUp Confirm Password	Signup password confirmation.
Login	Login Username	Login username.
	Login Password	Login password.
View Transactions	N/A	
View Accounts	N/A	
Get Contact page	N/A	
Logout	N/A	
eShop		
Search Product	Product name	Product name.
View Product	Product ID	Product ID.
Login	Login password	Login password.
	Login username	Login password.
Add to Cart	N/A	
Remove from Cart	N/A	
Buy	Full name	Full name.
	Address	Address.
	Account number	Account number.
Logout	N/A	
Portal Exploit		
Login	Login username	Login username.
	Login password	Login password.

Attack Action	Action Parameters	Parameter Description
Search Image	Search query	Query string.
Upload Image	Uploaded file name	Uploaded file name.
	Uploaded file	Uploaded file.
Logout	N/A	
Social Network		
Login	Login username	Login username.
	Login password	Login password.
News feed	N/A	
View Profile	Member ID	Member ID.
Like Post	N/A	
Unlike Post	N/A	
Create Post	Post content	Text string.
Comment to Post	Original post ID	Original post ID.
	Comment content	Text string.
Logout	N/A	
Video Platform Exploit		
Login	Login username	Login username.
	Login password	Login password.
Search Video	Video name	Video name.
Download Video	Downloaded file name	The name of the downloaded file.
	Downloaded file	Downloaded file.
Upload Video	Uploaded file name	The name of the uploaded file.
	Uploaded file	Uploaded file.
Delete Video	N/A	
Like Video	N/A	

Attack Action	Action Parameters	Parameter Description
Unlike Video	N/A	
Logout	N/A	
Portal XSS		
Login	Login username	Login username.
	Login password	Login password.
Search Image	Search query	Query string.
Upload Image	Uploaded file name	The name of the uploaded file.
	Uploaded file	Uploaded file.
Logout	N/A	

Attack Strikes

You can customize your attacks by adding strikes to simulate various vulnerabilities. You can insert one or multiple strikes at the beginning of an attack or after an attack action.

The following table includes some attack strikes for illustrative purposes. Note that this is not a comprehensive list. Keysight CyPerf Web interface offers full descriptions for all attack strikes.

Attack Strike	Description
Strike Cybozu Injection	This strike exploits an SQL injection flaw in the Cybozu web application.
Strike WordPress XSS	The WordPress plug-in "UserPro" uses an Instagram library that is vulnerable to Reflected Cross-Site Scripting (XSS).
Strike ZTE CPE Webshell	This strike exploits a command execution vulnerability in ZTE F460/F660 cable modem Web Interface. The vulnerability is due to improper access checks of the web platform resources. Successful exploitation can result in arbitrary commands on the target system.
Strike Tomcat File Upload	This strike exploits a remote command execution vulnerability in Apache Tomcat. The vulnerability allows attackers to upload arbitrary files to the Tomcat application server by utilizing the HTTP PUT method. By uploading a .JSP file to the Tomcat Application Server, an attacker can execute malicious code on the remote machine.
Strike SQL Injection and XSS Vector	This strike uses malicious SQL code for backend database manipulation to access information that was not intended to be displayed.
Strike Brute Force Attack	This is a standard brute force attack.
Strike Slowloris GET	The Slowloris attack opens multiple connections to the targeted web server and keeps them open for as long as possible by periodically sending partial HTTP headers/Request body for each request. The targeted server's maximum concurrent connection pool is ultimately filled, and any additional (legitimate) connection attempts are denied. ① NOTE: If there is a "\$" character in Request URL, you need to escape with a "\". For example, "\$abc" needs to be written as "\\\$abc". The Slowloris strikes are different from other application actions or strikes in that they are never supposed to be completed unless they are aborted by the DUT or server. Thus there exists no graceful stop mechanism for Slowloris in CyPerf. You can use the Force Stop button to stop the test, otherwise it automatically aborts the ongoing requests after a 2 minute grace period once the configured test duration is over.
Strike Slowloris POST	

For the **Strike SQL Injection and XSS Vector** and the **Strike Brute Force Attack** actions, you can also use the Playlist file option. You can upload a .csv file containing multiple usernames and their associated passwords, the playlist bypasses the single user login and uses all entries from the uploaded .csv file. The reference column will have the corresponding csv column names, which can be chosen from the dropdown menu. For example, the .csv file has two columns named Username and Password, add these exact names in the Login username and Login password fields. For more information, see [Artifacts](#).

The table below contains the number of entries (how many HTTP exchanges a strikes generates for a playlist) available for each of the CSV files that Strikes use:

CSV Playlists	Rows
Brute Force Attack Top Usernames And Passwords.csv	1000 rows
Strike SQLi Vector Blind MSSQL INSERT.csv	106 rows
Strike SQLi Vector Blind MSSQL WHERE.csv	39 rows
Strike SQLi Vector Blind MySQL INSERT.csv	89 rows
Strike SQLi Vector Blind MySQL ORDER BY.csv	18 rows
Strike SQLi Vector Blind MySQL WHERE.csv	44 rows
Strike SQLi Vector Detect Generic Blind.csv	31 rows
Strike SQLi Vector Detect Generic Error Based.csv	154 rows
Strike SQLi Vector Detect Generic SQLi.csv	263 rows
Strike SQLi Vector Detect Generic Time Based.csv	95 rows
Strike SQLi Vector Detect Generic Union Select.csv	424 rows
Strike SQLi Vector Detect MSSQL Blind.csv	8 rows
Strike SQLi Vector Detect MSSQL.csv	12 rows
Strike SQLi Vector Detect MySQL.csv	9 rows
Strike SQLi Vector Detect MySQL MSSQL.csv	10 rows
Strike SQLi Vector Detect NoSQL.csv	20 rows
Strike SQLi Vector Detect Oracle.csv	55 rows
Strike SQLi Vector Detect xPlatform.csv	193 rows
Strike SQLi Vector Exploit DB2 Enumeration.csv	12 rows
Strike SQLi Vector Exploit MSSQL Enumeration.csv	11 rows

CSV Playlists	Rows
Strike SQLi Vector Exploit MySQL Injection Login Bypass.csv	5 rows
Strike SQLi Vector Exploit MySQL Read Local Files.csv	1 row
Strike SQLi Vector Exploit PostgreSQL Enumeration.csv	19 rows
Strike XSS Vector Reflected Fuzzers.csv	1433 rows
Strike XSS Vector Reflected Group1.csv	200 rows
Strike XSS Vector Reflected Group2.csv	200 rows
Strike XSS Vector Reflected Group3.csv	200 rows
Strike XSS Vector Reflected Group4.csv	200 rows
Strike XSS Vector Reflected Group5.csv	200 rows
Strike XSS Vector Reflected Group6.csv	200 rows
Strike XSS Vector Reflected Group7.csv	200 rows
Strike XSS Vector Reflected Group8.csv	33 rows
Strike XSS Vector Reflected Html5.csv	184 rows
Strike XSS Vector Reflected Rsnake.csv	138 rows
Strike XSS Vector Reflected Sample.csv	20 rows



NOTE: To find out more on how to add strikes to your attack, refer to [Work with Attack Actions and Strikes](#).

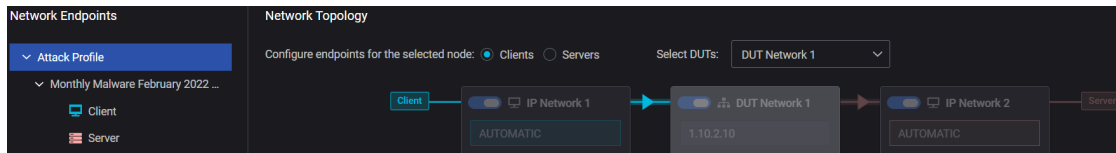
Attack Objectives and Timeline

The following UI elements are available on the **Objectives and Timeline** tab under the **Configure Attack Profile** section.

Field	Description
<i>Objectives</i>	
Attacks per second	The number of attacks carried out per second.
Max concurrent attacks	The maximum number of concurrent attacks.

Field	Description
<i>Timeline</i>	
Iteration count	With Iteration Count = 1 the entire list of attacks is executed once; after that, the Attack profile stops generating attacks. If an attack contains one or more configured playlists, instead of executing once, that attack iterates over all the rows of the playlist that contain the maximum number of entries (rows). Entries from other playlists (if any) are iterated in a round-robin manner.
Duration (seconds)	This field is available only when Iteration count is set to 0, and it represents the attack duration in seconds. The default value is 600 seconds.
Warm up period (seconds)	Enter the number of seconds it takes the attack to start.
Connections graceful stop timeout (seconds)	Number of seconds the agent waits for flows to time out before gracefully closing the open connections. The default value is 120 seconds. See more details in Start and Stop Tests .

Attack Network Mapping



The Network Mapping section enables you to adjust the traffic flow for each attack. The following UI elements are available on the **Network Mapping** tab under the **Attack Profile Configuration** section:

Field	Description
Network Endpoints	Displays the network endpoints in the currently configured attack profile. A yellow dot is shown next to an item mapped differently than its parent.
Network Topology	Displays the network topology for each attack profile. You can configure endpoints for the selected node, which can be either clients or servers.

NOTE: For more information on how to configure the traffic flow for an attack, refer to [Manage Attack Network Mapping](#).

Attack Profile Settings

The following UI elements are available on the **Profile Settings** tab under the **Attack Profile Configuration** section.

Field	Description
Active	Use the toggle to enable or disable the use of the attack profile in the test. You can also enable or disable an attack profile by switching the Attack Profile enabled toggle on or off in the Settings panel under the Attack Profile tab on the Test Overview page.
Use all source IPs	By default, this setting is <i>disabled</i>. When Use all source IPs is <i>disabled</i>, each Simulated User responsible for Concurrent Attacks will use a specific set of source IP addresses (IPv4 and/or IPv6) throughout the entire test duration. As a result, the source IPs used for the attack traffic will be limited by the number of Max concurrent attacks. When Use all source IPs is <i>enabled</i>, Concurrent Attacks will collectively try to use all source IP addresses configured in the test. They do this by switching source IPs as needed across different iterations of each attack throughout the entire test duration.  NOTE: The Use all source IPs setting may have no effect when single iteration mode is selected (as opposed to allowing multiple iterations during the test duration), or when a single iteration of an Attack (e.g. Slowloris) continues until the end of the test duration. You can also enable or disable Use all source IPs from the Settings panel under the Attack Profile tab on the Test Overview page. For more details, refer to the same setting under Application Profile Settings.

TCP Settings

The following UI elements are available on the **TCP Settings** tab under the **Application Profile Configuration** and the **Attack Profile Configuration** section. These parameters are configurable for both clients and servers.

Field	Description
Min retransmission timeout	The lowest value that the computed RTO timer value can be set to. Expiry of the RTO timer indicates that the sender has not received an acknowledgment for the transmission, which triggers a retransmission of the segment. Upon each retransmission, the RTO timer value is doubled, up to the Max value. The default Min value is 2000 milliseconds and the permissible range of values is between 50 to 120000 milliseconds.
Max retransmission timeout	The highest value that the computed RTO timer value can be set to. The default value is 3200 milliseconds and the permissible range of values is between 1000 to 120000 milliseconds.
Min source port	The source port specifies which ports are used for client connections. The Min value specifies the lower bound (the lowest permissible port number). The default value is 1024.
Max source port	The Max value specifies the upper bound (the highest permissible port number). The default value is 65535.
Receive buffer size	The default size of the receive buffer (in bytes). This parameter affects the TCP receive window size. If you increase the size of the receive buffer, then the receive window is prolonged. If you are experiencing high latency on your test network, increase the size of the receive buffer to improve the throughput.
Transmit buffer size	The default size of the transmit buffer (in bytes). This parameter affects the TCP transmit window size. If you increase the size of the transmit buffer, the transmit window is prolonged. If you are experiencing high latency on your test network, increase the size of the transmit buffer to improve the throughput.
Enable RFC1323 TCP timestamps	Enable or disable RFC1323 TCP timestamps using the toggle button. If enabled, the client or server inserts an RFC 1323 timestamp into each packet.  NOTE: Enabling the TCP Timestamp option adds 12 bytes to the TCP header. This reduces the effective configured MSS.
Selective acknowledgments	This option helps prevent CyPerf endpoints to retransmit unnecessary packets when packet loss occurs in the network. When <i>enabled</i>, the data receiver can inform the sender about all segments

Field	Description
	that have arrived successfully. Therefore, the sender will retransmit only the segments that have actually been lost. When <i>disabled</i>, the exchange of selective acknowledgments between the endpoints is no longer permitted. This is the default value.  NOTE: Make sure both endpoints support selective acknowledgments before enabling this option.
Explicit Congestion Notification	When <i>enabled</i>, CyPerf uses standard ECN markings in the TCP/IP headers to detect network congestion without packet drops. This allows network devices to signal congestion by marking packets rather than dropping them, enabling endpoints to reduce their transmission rate proactively. When <i>disabled</i>, congestion is detected only through packet loss events.  NOTE: Make sure both endpoints support ECN before enabling this option.

TLS Settings

The TLS settings described in the following table define the TLS communication for *all* the applications and attacks which were configured to have the **Inherit TLS** option enabled. To enable TLS settings only on *specific* applications or attacks, you can access the **Advanced Settings** of the selected application or attack profile. See [Application Advanced Settings](#) or [Attack Advanced Settings](#).

To access the **TLS Settings** tab:

1. In an existing test session, select the **App Mix** or the **Attack List** configuration tile. The **Application Profile** page or the **Attack Profile** page opens.
2. Select the **TLS Settings** tab under the **Application Profile Configuration** section or the **Attack Profile Configuration** section respectively. You can now configure the client and server TLS settings.

 **NOTE:** Multiple TLS versions are available for configuration for both clients and servers; for example, you can configure both TLS 1.2 and TLS 1.3 versions from **Application Profile Configuration/ Attack Profile Configuration > TLS Settings > Client TLS / Server TLS**. You can choose multiple ciphers for each different version. The client first sends these TLS versions and ciphers in a client HELLO message, and then the server chooses one of the versions and ciphers and replies back to the client with a server HELLO message. The client then proceeds with the handshake.

Field	Description
Inherit TLS	This option is enabled by default, and is available only under the Advanced Settings of the selected application or attack profile. The option is not available at Application or Attack profile level. When Inherit TLS is enabled, the client and server agent inherit the SSL profile configured under the TLS Settings tab on the Application Profile or Attack Profile page. When Inherit TLS is disabled, and the Client TLS is enabled, the client receives an SSL profile for each application or attack. When Inherit TLS is disabled, and the Server TLS is enabled, the server receives a single SSL profile.  NOTE: The server TLS settings must be identical across all profiles for the same agent. If multiple server TLS profiles are configured at different levels for the same agent, an error is triggered informing you that there are differences between the profiles.
<i>Client TLS Settings</i>	
	Select this button to apply the client TLS settings configured at the currently selected application or attack level to the other existing application or attack profiles. When selecting this button, you are prompted to choose one or multiple client TLS profiles from the current configuration to which the current TLS settings will be applied.
Enable TLS traffic	If enabled, the client will initiate all connections to the destination over TLS, based on the settings below. If disabled, connections will be established in plaintext. Even if Enable TLS traffic is not active, the client agent may use the TLS settings based on responses it receives from the server. For example, if the server sends an HTTP Redirect (3xx) response with a https:// URL, when the client agent follows that redirect, it will initiate a TLS connection based on the configured TLS settings.
TLSv1.2	When enabled, this option allows you to customize TLS 1.2 settings for the client (such as selecting one or multiple ciphers, a session reuse method, and a session reuse count). This option is enabled by default to support the Enable TLS traffic scenarios described above.
TLSv1.3	When enabled, this option allows you to customize TLS 1.3 settings for the client (such as selecting one or multiple ciphers to be used in the client HELLO message). This option is enabled by default to support the Enable TLS traffic scenarios described above.
Cipher	Specifies a prioritized list of supported ciphers, which you can choose from by clicking the corresponding check boxes. At the TLS handshake stage, only

Field	Description
	selected ciphers in the given precedence order are advertised as being available.
Supported Groups	For TLSv1.3, select the cryptographic group(s) to be used. The following groups are supported, with the first three selected by default: • P-256 (<i>default</i>) • P-384 (<i>default</i>) • P-521 (<i>default</i>) • X25519 • MLKEM512 • MLKEM768 • MLKEM1024 • P256_MLKEM512 • SecP256r1MLKEM768 • P384_MLKEM768 • SecP384r1MLKEM1024 • P521_MLKEM1024 • X448_MLKEM768 • X25519_MLKEM512 • X25519_MLKEM768  NOTE: When the client initiates a connection, it sends the server a list of supported cryptographic groups along with a key share for one of these groups (typically the first one on the list). If the server does not support this initial group, it may respond with a Hello Retry Request and asks the client to provide a new key share for another group from the list of supported groups in the client Hello. If the server supports any of the other groups apart from the first one, then it will send the Hello Retry request; otherwise, it will terminate the connection.
Session Reuse Method	Select one of the Session Reuse Methods in the drop-down menu: • Disable • Session ticket • Session ID
Session Reuse Count	The number of simultaneous connections that can share the same Session ID or Session ticket. This option is enabled only for client TLSv1.2 settings, after you choose either the Session ticket or the Session ID method.
Middlebox Compatibility	This option is enabled by default. It allows for compatibility with Middleboxes, which do not support TLSv1.3. For more details, refer to this document.
Immediate Close	If enabled, the endpoint closes the TCP connection immediately after sending the TLS CLOSE message (i.e., the endpoint does not wait for a confirmation from the other end).
<i>Server TLS Settings</i>	

Field	Description
	Select this button to apply the server TLS settings configured at the currently selected application or attack level to the other existing application or attack profiles. When selecting this button, you are prompted to choose one or multiple server TLS profiles from the current configuration to which the current TLS settings will be applied.  NOTE: The server TLS settings must be identical across all profiles for the same agent. If multiple server TLS profiles are configured at different levels for the same agent, an error is triggered informing you that there are differences between the profiles.
TLSv1.2	When enabled, this option allows you to customize TLS 1.2 settings for the server (such as selecting one or multiple ciphers, and a session reuse method). This option is disabled by default.
TLSv1.3	When enabled, this option allows you to customize TLS 1.3 settings for the server (such as selecting one or multiple ciphers to be used by the server). This option is disabled by default.
Cipher	Specifies a prioritized list of supported ciphers, which you can choose from by clicking the corresponding check boxes. At the TLS handshake stage, only selected ciphers in the given precedence order are advertised as being available.
Supported Groups	For TLSv1.3, select the cryptographic group(s) to be used. The following groups are supported, with the first three selected by default: • P-256 (<i>default</i>) • P-384 (<i>default</i>) • P-521 (<i>default</i>) • X25519 • MLKEM512 • MLKEM768 • MLKEM1024 • P256_MLKEM512 • SecP256r1MLKEM768 • P384_MLKEM768 • SecP384r1MLKEM1024 • P521_MLKEM1024 • X448_MLKEM768 • X25519_MLKEM512 • X25519_MLKEM768

Field	Description
	NOTE: When the client initiates a connection, it sends the server a list of supported cryptographic groups along with a key share for one of these groups (typically the first one on the list). If the server does not support this initial group, it may respond with a Hello Retry Request and asks the client to provide a new key share for another group from the list of supported groups in the client Hello. If the server supports any of the other groups apart from the first one, then it will send the Hello Retry request; otherwise, it will terminate the connection.
Session Reuse Method	Select one of the Session Reuse Methods in the drop-down menu: • Disable • Session ticket • Session ID This option is enabled only for server TLSv1.2 settings.
Middlebox Compatibility	This option is enabled by default. It allows for compatibility with Middleboxes, which do not support TLSv1.3. For more details, refer to this document.
Immediate Close	If enabled, the endpoint closes the TCP connection immediately after sending the TLS CLOSE message (i.e., the endpoint does not wait for a confirmation from the other end).
SNI Enabled	Enable the server name indicator. When disabled, the server file options described below become available.
Certificate file	Upload your certificate file. NOTE: To retrieve the content of a certificate file, use the following REST API call: <pre>GET /api/v2/resources/tls-certificates/{tlsCertificateId}/contentFile</pre> For example, <code>https://{controller IP address}/api/v2/resources/tls-certificates/Secured_RSAdes3_cert_2048.pem/contentFile</code>
Key file	Upload your key file.

Field	Description
	 NOTE: To retrieve the content of a key file, use the following REST API call: GET /api/v2/resources/tls-keys/{tlsKeyId}/contentFile For example, https://{controller IP address}/api/v2/resources/tls-keys/Secured_RSAdes3_key_2048.pem/contentFile
Key file password	Enter your key file password.
DH file	Upload your DH file.  NOTE: - This option is available only for TLSv1.2. - Uploading a file is complementary in all of the above server file options. For example, if you upload a DH file, then it will also appear as a key or certificate file.

HTTP Settings

HTTP settings can be configured both at the traffic profile level for all the available applications or attacks, and at a per-application or per-attack level.



NOTE:

The HTTP settings set at the *traffic profile* level apply to all the available applications/attacks defined for the current profile.

The HTTP settings set at the *per-application* or *per-attack* level override the traffic profile settings.

The following table describes the UI elements on the **HTTP Settings** tab under the **Application Settings** or the **Attack Settings** panels respectively.

Field	Description
<i>HTTP Settings</i>	
Inherit HTTP	When Inherit HTTP is enabled, the client and server agent inherit the HTTP settings configured at the traffic profile level. This option is enabled by default, with the following exceptions: - Configuration imported from older software versions, which had different values than the default ones in the HTTP profile for the parameters in the Parameters section under Application advanced settings - Applications that come from ATI with profiles that differ from the default predefined profiles. For example: Facebook Chrome which had the Inherit HTTP option enabled, or Facebook Mozilla which had the Inherit HTTP option disabled. Other limitations for 3.0 and newer versions: - For the HTTP App, the headers defined at both the application level and traffic profile level will be ignored when the HTTP App has <i>request</i> and <i>response</i> headers that are exposed as Action parameters, in which case those headers will be used. Overriding this behavior and using the HTTP profile headers instead is not currently supported.
<i>Client HTTP Settings</i>	
HTTP profile	Select the client HTTP profile from the available options: Android Chrome (default option) Microsoft Edge Firefox Internet Explorer Opera

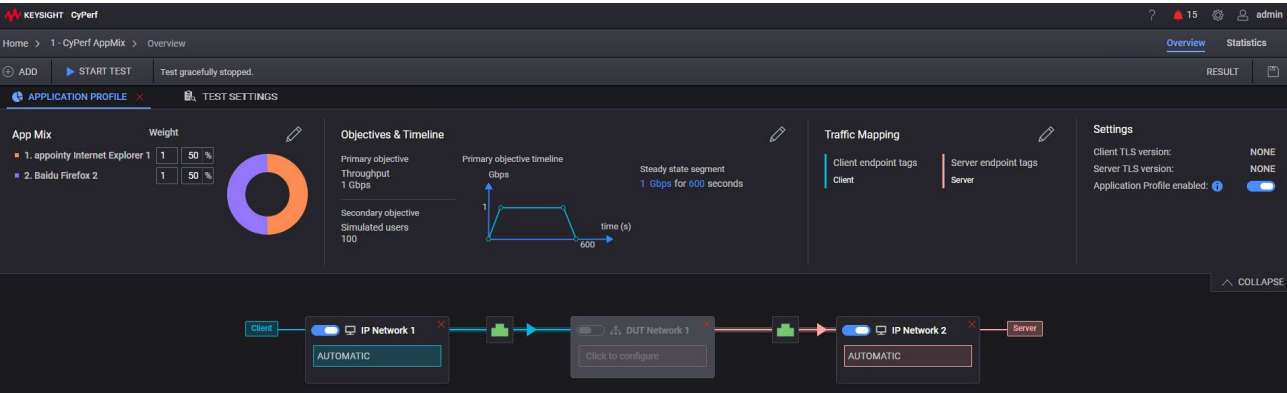
Field	Description
	• Safari • Custom client profile
Max transactions	The maximum number of application iterations per connection. The default value is 1 .
HTTP version	Select the version of the HTTP protocol to be used on the client side from the available options: • HTTP/1.1 (default option) • HTTP/1.0 • HTTP/2  NOTE: • The HTTP/2 version is supported only for the HTTP App. It is not supported for the other types of applications, nor for attacks. • To configure HTTP/2 version for the HTTP App, select the HTTP/2 version at the application profile level (Application Profile Configuration > HTTP Settings > HTTP Version). To run a test with applications that have both HTTP/2 version (for the HTTP App) and HTTP/1.X version enabled (for other applications), make sure the Inherit HTTP option is disabled at the level of the other applications. To disable the Inherit HTTP option at an application level: - Under Application Profile Configuration, select the Applications tab. - Select the Advanced Settings gear menu () for the other application(s) used in the test. - On the Application Settings page that opens, under the HTTP Settings tab, disable the Inherit HTTP option using the toggle.
Connection persistence	Configure the connection persistence mode. The available options are: • Standard (default option): This mode inherits the behavior as described by the selected HTTP version standard. • Disabled: This mode forces the connection to close after every HTTP message. • Enabled: This mode enforces connection persistence through explicit keep-alive frames.
Headers	Add, delete, or customize the headers that the client will send.

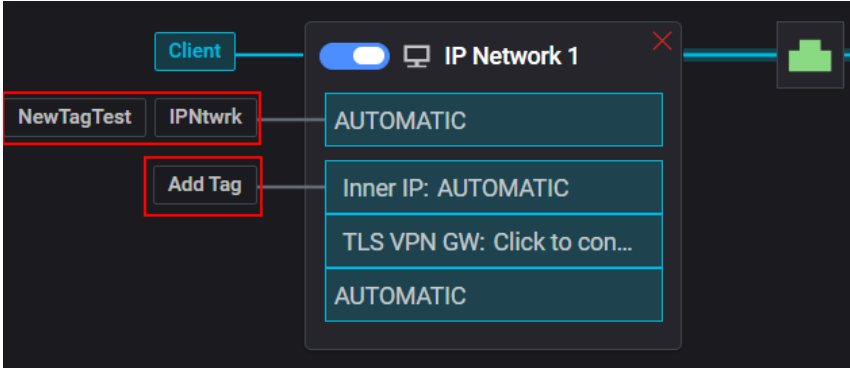
Field	Description
<i>Server HTTP Settings</i>	
HTTP profile	Select the server HTTP profile from the available options: • nginx • Apache (default option) • IIS • Custom server profile
HTTP version	Select the version of the HTTP protocol to be used on the client side from the available options: • HTTP/1.1 (default option) • HTTP/1.0 • HTTP/2  NOTE: • The HTTP/2 version is supported only for the HTTP App. It is not supported for the other types of applications, nor for attacks. • To configure HTTP/2 version for the HTTP App, select the HTTP/2 version at the application profile level (Application Profile Configuration > HTTP Settings > HTTP Version). To run a test with applications that have both HTTP/2 version (for the HTTP App) and HTTP/1.X version enabled (for other applications), make sure the Inherit HTTP option is disabled at the level of the other applications. To disable the Inherit HTTP option at an application level: a. Under Application Profile Configuration, select the Applications tab. b. Select the Advanced Settings gear menu () for the other application(s) used in the test. c. On the Application Settings page that opens, under the HTTP Settings tab, disable the Inherit HTTP option using the toggle.
Connection persistence	Configure the connection persistence mode. The available options are: • Standard (default option): This mode inherits the behavior as described by the selected HTTP version standard. • Disabled: This mode forces the connection to close after every HTTP message. • Enabled: This mode enforces connection persistence through explicit keep-alive frames.

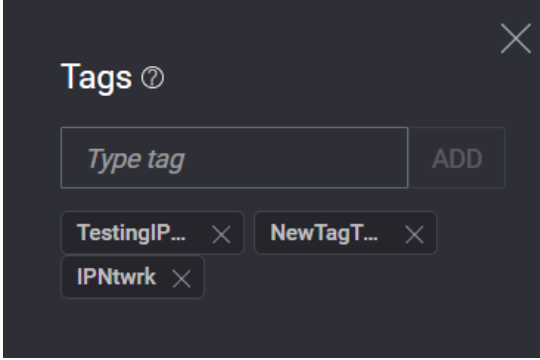
Field	Description
Additional headers	Configure other headers that will be sent alongside the application specific headers.
Use application server headers	This option is enabled by default. When enabled, all the servers to which the current profile applies will use their specific headers from the ATI definition. When disabled, the servers will use the list of headers specified in the Headers parameter.
Headers	Add, delete, or customize the headers that the server will send.

Test Topology

The **Overview** page displays the test topology used in the current configuration. The DUT, client, and server networks can be configured on each of the network element's respective pages.



Feature	Description
Client	The default configuration of any Keysight CyPerf test includes one client. In the network topology, the client is positioned on the DUT's left side. To add a new client, go to Add > Network Elements. For more details on how to configure this network segment, see Configure Network Segments.
Server	The default configuration of any Keysight CyPerf test includes one server. In the network topology, the server is positioned on the DUT's right side. To add a new server, go to Add > Network Elements. For more details on how to configure this network segment, see Configure Network Segments.
Tag	Each network element can be assigned one or more tags which can be used for filtering. To add new tags, either click an existing tag or hover over the network element, and click Add Tag. Example  The example screenshot shows the 'IP Network 1' configuration dialog. It has a 'Client' label and a 'NewTagTest' button. Below the 'NewTagTest' button is an 'Add Tag' button. The dialog also shows 'IPNtwrk' and 'AUTOMATIC' buttons. Below these are 'Inner IP: AUTOMATIC', 'TLS VPN GW: Click to con...', and another 'AUTOMATIC' button.

Feature	Description
	In the Tags dialog that opens, type one or multiple comma-separated values, and click the ADD button or press the ENTER key. Example  To remove an existing tag, click the tag to open the Tags dialog. Then, click the Delete filter (X) button corresponding to the tag you want to delete. You can also set a tag for the IP Range in a similar fashion.
	Hover over this icon to see how many traffic agents have been assigned.
	This icon signals that one or more traffic agents are unavailable or unmapped to middleware.
	This icon shows that no agents have been assigned.
Configure DUT	Enable the toggle button  to configure the DUT.

DUT Properties

The following UI elements are available on each DUT's **Configuration** page, under **DUT Network** > **DUT Configuration** section:

Field	Description
Enable to configure	Use the toggle button  to enable the configuration of the selected DUT's IP address or hostname, as well as the simple or advance configuration settings.
Config settings	
Hostname	The IP address or hostname of the DUT. Hostname resolution is only supported for IPv4 addresses.
Simple Configuration	
<i>HTTP Health Check Properties</i>	
Enabled	Use the toggle button  to enable the HTTP health check.
Port	The HTTP port number used for the interface communications.
Target URL	The HTTP health check URL.
Payload File	Select one of the available options: • Payload profile: Select one of the available profiles in the drop-down box, or select a custom payload file. Then, select the Upload button, and upload the required file. • Synthetic data (bytes): Enter synthetic data in bytes that the DUT will use as payload.
HTTP Version	Select one of the available HTTP version options, and enter the HTTP version in the corresponding field: • User input • Dynamic user input
<i>HTTPS Health Check Properties</i>	
Enabled	Use the toggle button  to enable the HTTPS health check.
Port	The HTTPS port number used for the interface communications.
Target URL	The health check URL.
Payload File	Select one of the available options: • Payload profile: Select one of the available profiles in the drop-down box, or select a custom payload file. Then, select the Upload button, and upload the required file.

Field	Description
	• Synthetic data (bytes): Enter synthetic data in bytes that the DUT will use as payload.
HTTP Version	Select one of the available HTTP version options, and enter the HTTP version in the corresponding field: • User input • Dynamic user input
<i>TCP Health Check Properties</i>	
Enabled	Use the toggle button  to enable the TCP health check.
Port	Set the TCP port number used for the interface communications.
Advanced Configuration	
Forward Proxy	
Enable to configure	Use the toggle button  to enable Forward Proxy configuration.
Hostname	The IP address or hostname of the proxy.
Client DUT port	Set the client DUT port number.  NOTE: When enabling the ZeroTrust PEP for Forward Proxy, the port is automatically set to 443.
HTTP forward proxy mode	Select one of the available HTTP forward proxy modes: • Proxy mode • Tunnel mode When you enable ZeroTrust PEP , the HTTP forward proxy mode is automatically set to Tunnel mode , and the option becomes read-only.
Domain suffix (optional)	If configured, all hostnames in all traffic using the selected DUT will have the current suffix automatically added. For example: Given an Application or Attack that uses <code>example.com</code> as hostname, and a Domain suffix is set to <code>test-suffix.io</code> , then the actual hostname that will be used when generating traffic will be <code>example.com.test-suffix.io</code> . This option is useful for setting up a custom DNS rule for testing purposes. For example, the DNS server used by the DUT could be configured with a rule like <code>*.test-suffix.io A 10.0.20.1</code> ,

Field	Description
	where <code>10.0.20.1</code> would be the CyPerf Server Test IP.
Non-Proxied Hosts	Set up hostnames that will be contacted directly, i.e. without routing traffic through a proxy server. Click the Add button and enter the hostname in the empty field added to the hostname list. Then, click outside the text box to confirm adding the new hostname to the list. Click the Delete button to remove a hostname from the list.
Use real hostname	If <i>enabled</i>, the client agent will send requests to the HTTP forward proxy with hostnames that match the HTTP Host headers in each simulated application. For example, it will send requests like <code>CONNECT generic.keysight.io</code> or <code>GET http://generic.keysight.io</code> when running the default HTTP application. If using this mode, the real Forward Proxy device under test must be configured to use a DNS server that can resolve these simulated hostnames to the IP of the CyPerf server. The DNS Server feature in the Network topology can be used to help achieve this configuration, by setting the CyPerf server agent to act as the DNS server for the Forward Proxy device. If <i>disabled</i>, the client agent will use the IP of the server agent in requests to the forward proxy (for example, <code>CONNECT 10.0.0.20</code> or <code>GET http://10.0.0.20</code>).
<i>ZeroTrust PEP</i>	
Enable to configure	Use the toggle button  to enable ZeroTrust PEP configuration on the DUT for forward proxy. See also Configure a ZeroTrust DUT .
Authentication profile	Select one of the available authentication profiles: • Juniper Zerotruster Device: Juniper Zerotruster Device PEP, with SAML authentication through the Okta IdP • PAN EP: PAN Explicit Proxy PEP, with SAML authentication through the Okta IdP, or the Simulated IdP Depending on the authentication profile you select, different parameters become available for configuration on the ZeroTrust PEP panel (for example, Authentication method, Identity provider, Username and Password, Max authentication token reuse, and others). See also Configure a ZeroTrust DUT.
Simulated IDP	When selecting the PAN Explicit Proxy PEP, with SAML authentication through the Simulated IdP, Click the Simulated IdP Properties button becomes available. Use the parameters under the Simulated

Field	Description
	IdP Properties panel to simulate the identity provider (IdP) in the authentication flow when testing a ZeroTrust PEP. When simulated IdP configuration is enabled on a HTTP Redirect response (3xx) from the PEP, the CyPerf agent will internally generate the response that will otherwise have been received from the IdP. To enable simulated IdP: 1. Under DUT Configuration > Advanced Configuration > Forward Proxy > ZeroTrust PEP, use the toggle to enable configuration of the ZeroTrust PEP. Then, configure the PEP hostname and PEP port, and select the authentication profile as PAN EP. 2. Under Identity Provider, select the Simulated IdP option. The Simulated IdP Properties panel is displayed allowing you to configure the options below: • Single sign on URL • Audience URI • Name ID format • Response signature • Assertion signature • Certificate file • Key file • Key file password • Download Simulated IdP metadata When running a test with simulated IdP configuration enabled: • A new IdP Requests Succeeded statistics column is displayed under Client Authentication Flow Statistics. • The Client IdP Request Tools panel under the ZeroTrust Statistics will not be available because there will be no requests to an external IdP.
IDP	Under the IDP section, configure the following options for Forward Proxy: • SAML HTTP Binding for SSO Request to IdP: Select the HTTP Post binding to send the SAML AuthRequest via a HTTP POST request. • Username • Password • Max Authentication Token Reuse Count
SGW	Set up hostnames that will be contacted directly, i.e. without routing traffic through a proxy server.

Field	Description
	Click the Add button and enter the hostname in the empty field added to the hostname list. Then, click outside the text box to confirm adding the new hostname to the list. Click the Delete button to remove a hostname from the list.
Reverse Proxy	
See Simple Configuration for descriptions that also apply to the Reverse Proxy settings.	
Enable to configure	Use the toggle button  to enable ZeroTrust PEP configuration on the DUT. See also Configure a ZeroTrust DUT .
Hostname	The IP address or hostname of the ZeroTrust gateway.
Port	The port number of the ZeroTrust gateway. NOTE: - When enabling the ZeroTrust PEP for Reverse Proxy, the port is automatically set to 443. - The port number configured in this field applies to all the connections as a destination, unless overridden at the connection level.
ZeroTrust PEP	
Authentication profile	Select one of the available authentication profiles: Cisco Firepower: Cisco Firepower PEP with SAML authentication through the Simulated IdP Cisco Umbrella: Cisco Umbrella PEP with SAML authentication through the Okta IdP, or the Simulated IdP F5 BIG-IP Identity Aware Proxy: F5 BIG-IP Identity Aware Proxy PEP with SAML authentication through the Okta IdP, or the Simulated IdP PAN GP Clientless VPN: PAN GlobalProtect Clientless VPN PEP with SAML authentication through the Okta IdP, or the Simulated IdP Generic PEP: Generic PEP with SAML authentication through the Simulated IdP Depending on the authentication profile you select, different parameters become available for configuration on the ZeroTrust PEP panel (for example, Authentication method, Identity provider, Username and Password, Max authentication token reuse, and others). See also Configure a ZeroTrust DUT.

Field	Description
NOTE: - The F5 BIG-IP Identity Aware Proxy, PAN EP, and PAN GP Clientless VPN profiles use the POST binding with IdP. The DUT should also be configured to use the POST binding for interoperability reasons. See also SAML HTTP Binding for SSO Request to IdP below. - For compatibility with the Generic (PEP) profile, the DUT should be configured to use the HTTP redirect binding. - The Generic PEP profile offers customization options through the parameters described under the SGW section below, such as Application Base Path, Logout Request , and Logout Request Path.	
<i>Simulated IdP Properties</i>	
Click the Simulated IdP Properties button to display the parameters available under the Simulated IdP Properties panel. Use these parameters to simulate the identity provider (IdP) in the authentication flow when testing a ZeroTrust PEP. When simulated IdP configuration is enabled on a HTTP Redirect response (3xx) from the PEP, the CyPerf agent will internally generate the response that will otherwise have been received from the IdP. To enable simulated IdP: - Under DUT Configuration > Advanced Configuration > Reverse Proxy > ZeroTrust PEP, use the toggle to enable configuration of the ZeroTrust PEP. Then, configure the PEP hostname and PEP port, and select the authentication profile as either Cisco Firepower, Cisco Umbrella, F5 BIG-IP Identity Aware Proxy, PAN GP Clientless VPN, or Generic PEP. - Under Identity Provider, select the Simulated IdP option. The Simulated IdP Properties panel is displayed allowing you to configure the options below. When running a test with simulated IdP configuration enabled: - A new IdP Requests Succeeded statistics column is displayed under Client Authentication Flow Statistics. - The Client IdP Request Tools panel under the ZeroTrust Statistics will not be available because there will be no requests to an external IdP.	
Single sign on URL	URI provided by the PEP. The CyPerf client sends the request to this URI after generating the SAML response.
Audience URI	URI provided by the PEP, which needs to be included in the generated SAML response required by the PEP.
Name ID format	Select the name identifier format used to identify the subject.
Response signature	Use the toggle button to enable the signature of the entire authentication response.
Assertion signature	Use the toggle button to enable the signature of the attribute statement within the response.

Field	Description
Signature algorithm	Select the algorithm used for signing the authentication response.
Certificate file	Select the certificate used for signing the authentication response.
Key file	Select the certificate used for checking the authentication response.
Key file password	Enter the password of the key file.
Download Simulated IdP metadata	Downloads the IdP metadata which will include the selected certificate file. The IdP metadata downloaded from the CyPerf UI must be used to configure the IdP settings on the PEP device.
<i>IDP</i>	
SAML HTTP Binding for SSO Request to IdP	Depending on the selected authentication profile, choose one of the following options: • HTTP Post: Select this binding to send the SAML AuthRequest via a HTTP POST request.  NOTE: This option is available for the Cisco Umbrella, F5 BIG-IP Identity Aware Proxy, PAN EP, and PAN GP Clientless VPN authentication profiles. • HTTP Redirect: Select this binding to encode the SAML AuthnRequest as a URL parameter and send it via an HTTP redirect.  NOTE: This option is available only for the Cisco Firepower and Generic PEP authentication profiles.
Username	Select one of the available options: • Dynamic user input: Enter one or multiple comma-separated username and password pairs in the corresponding fields. When establishing communication, the username and password are picked from the list in round-robin fashion for authentication purposes. • Playlist file: You can upload a .csv file containing multiple usernames and their associated passwords.
Password	
Max Authentication Token Reuse Count	Enter the maximum number of times that the authentication token can be reused by the DUT before it is considered invalid or expired.
<i>SGW</i>	

Field	Description
Application Base Path	Enter the base path that will be used for the simulated application. NOTE: This option is available for the Cisco Firepower, Generic PEP, and PAN GP Clientless VPN authentication profiles.
Logout Request	Select Enabled to generate and send a Logout request. If you do not want to use an explicit Logout request, then select Disabled. NOTE: The Logout Request option is available only for the Generic PEP authentication profile.
Logout Method	Select one of the available options: • Clear Cookies: Authentication cookies are cleared on the client side. • Logout Request: A logout request is made from the client side. NOTE: This option is available only for the Cisco Firepower authentication profile.
Logout Request Path	Choose User input to configure the logout request path when the Logout Request option is enabled. NOTE: This option is available only for the Cisco Firepower and Generic PEP authentication profiles.

Network Segments Properties

The following UI elements are available in the **IP Network > Network Configuration** section.

Field	Description
Properties	
Network tags	You can add one or multiple comma-separated tags.  NOTE: Alternatively, you can manage network tags on the Overview page. To do that, you need to select a network segment's tag in the network topology, and type one or multiple comma-separated tags or remove existing ones.
Minimum agents	Enter the minimum number of agents.
Inherit streaming CPU allocation	When streaming applications (such as Zoom or UDP Stream) are enabled as part of an App mix, the Inherit streaming CPU allocation setting becomes available for both the client and server network segments. This setting is enabled by default. When <i>enabled</i>, the current network segment inherits the global Streaming purpose CPU allocation configured under Objective & Timeline > Advanced Settings in the streaming application settings. When <i>disabled</i>, the current network segment will override the global setting and allocate custom CPU resources based on its role in the test. In this case, the Streaming Purpose CPU Allocation (%) field becomes available for configuration. Streaming Purpose CPU Allocation (%) setting specifies the CPU percentage allocated for each network segment individually. This provides more granular control than the global Streaming purpose CPU allocation setting. This option is particularly useful in asymmetric traffic scenarios (for example, high-throughput UDP stream tests), where one segment requires more CPU resources to generate packets at line rate, while another segment requires fewer resources to monitor incoming traffic.

IP Stacks & Properties

The table below defines the UI elements found on the **Network Configuration** page under the **Network Stacks > IP Stacks** tab:

Field	Description
IP Stacks	
Add Range	Use this button to add a new IP range.

	
	Use this button to rename an existing IP range.
	Use this button to delete an existing IP range.
IP Range Properties	
Automatic IP	Use the toggle to enable or disable automatic IP. You can use automatic IP when a test interface already has one or more IP addresses assigned to it. When Automatic IP is enabled, select one of the available options for Automatic IP type: IPv4 and IPv6, IPv4 only, or IPv6 only . When Automatic IP is disabled, you can configure the IP manually. IPv6 addresses are also supported.
IP start	The first IP address in the range. This is the base address used for enumerating all the addresses in the range. The default address is 10.0.0.10 .
IP increment	The value used to enumerate all the addresses in the range. The default value is 0.0.0.1 .
Count	The number of addresses to be created for this range. The default value is 1 and the maximum value is 10.000.000 .
Max count per agent	The number of IP addresses that are assigned to each agent. The default value is 1 and the maximum is 1.000.000 .
Automatic IP prefix length	Use the toggle to enable automatic IP prefix length. If Automatic IP prefix length is disabled, you can configure it manually.
IP prefix length	Determines the size of the subnet and is the number of 1 bits in the subnet mask. The default value is 16 .
Automatic gateway	Use the toggle to enable automatic gateway. If Automatic gateway is disabled, you can configure it manually.
Gateway	The gateway address to be associated with all the addresses in the range. The default value is 10.0.0.1 .
Network tags	You can add one or multiple comma-separated tags.
Automatic MSS	Use the toggle to enable automatic MSS. The Maximum Segment Size (in bytes) is automatically calculated using the following formula: MSS = MTU - IP Header Size - TCP Header Size When this option is disabled, you can manually configure the MSS value. The default MSS value is 1460 bytes.
VLAN Range	Click the VLAN Range tab to display the VLAN Range panel.

VLAN enabled	Use the toggle to enable VLAN on the IP Network Segment.
VLAN ID	The first VLAN ID to be used for VLAN. The minimum value is 1 , the maximum value is 4094 .
Increment by	The value to be added to the VLAN ID for each new assignment. The maximum value is 4094 . The default value is 1 .
Count	The number of unique VLAN IDs that will be created. The default value is 1 , the maximum value is 4094 .
Count per agent	The number of VLAN IDs that are assigned to each agent in round-robin order. The default value is 1 , the maximum value is 1.000.000 .
Tag Protocol Identifier	Set to 0x8100 by default. The TPID can not be set to another value.
802.1Q Priority	Set to 0 by default.
<i>Static ARP</i>	
The Static ARP properties allow you to configure a list of ranges that the agent will use to map IP addresses to MAC addresses instead of relying on ARP requests. Static ARP entries at per IP range can only be added if VLAN is enabled. NOTE: - Static IP-to-MAC mappings are supported for both IPv4 and IPv6. - Statistics for IPv6 Neighbor Discovery are not currently available in the UI; use packet captures to verify related behavior.	
Static ARP Table	Click the Add range (+) button to add a new entry to the static ARP table.
<i>Static ARP Entry Properties</i>	
Remote IP start	The base value used when the network stack element creates IP addresses for the selected ARP entry.
Remote IP increment	The value to use when incrementing the IP address range. The default value is 0.0.0.1.
Remote MAC start	The base value used when the network stack element creates MAC addresses for the selected ARP entry.
Remote MAC increment	The increment value used to create a range of MAC addresses for the selected static ARP entry. The default value is 00:00:00:00:00:01.
Count	Enter the total number of IP-to-MAC mappings that will be created in the selected ARP entry. The default value is 1.

TLS VPN Stacks

The table below defines the UI elements found on the **Network Configuration** page under the **Network Stacks > TLS VPN Stacks** tab:

Field	Description
<i>TLS VPN Stacks</i>	
	Use this button to add a new tunnel stack.
	Use this button to rename an existing tunnel stack.
	Use this button to delete an existing tunnel stack .
<i>Stack Connections</i>	
<i>Inner IP Range Settings</i>	
Network tags	Add one or multiple comma-separated tags to identify a network.
<i>Tunnel Range Settings</i>	
Tunnel count per outer IP	Configure the amount of tunnels for each outer IP. The default amount is set to 1 tunnel per outer IP.
Tunnel establishment timeout (seconds)	Configure the timeout request in seconds. The DCP timeout default value is set to 120 seconds. It is recommended to configure a higher timeout value if the tunnel establishment timeout happens with the default value set.
DNS name servers	Add one or multiple comma-separated DNS names.
Vendor type	Choose one of the available tunnel types: • Cisco AnyConnect • PAN GP • FORTINET • F5
TCP Destination Port	Configure CyPerf's Client TCP destination port. The default value is 443 . Values can range from 1 to 65535 .
VPN Gateway	Configure the VPN gateway.
<i>Cisco AnyConnect</i>	
VPN gateway	Add or configure the IP or hostname of a gateway. Only a single gateway is supported per each tunnel range.
DTLS Handshake Timeout	The time (in seconds) allowed for the DTLS handshake to complete successfully; if the configured value is exceeded, the tunnel is declared

Field	Description
	failed. The default value is 60. Values can range from 0 to 65535 seconds.
DTLS Retransmission Timeout	The time (in seconds) after which a DTLS retransmission will occur. The default value is 65. Values can range from 0 to 65535 seconds.
Connection Profiles	Configure one or multiple comma-separated profiles. The profiles are chosen by using round robin if multiple connection profiles are configured. For example: If two connection profiles are configured as c1 and c2 and three usernames and passwords are configured as u1-p1, u2-p2 and u3-p3 with the tunnel count set to 4, then the behavior is the following: • Tunnel1 uses c1 and the username and password of u1-p1 • Tunnel2 uses c2 and the username and password of u2-p2 • Tunnel3 uses c1 and the username and password of u3-p3 • Tunnel4 uses c2 and the username and password of u1-p1

Field	Description
<i>PAN GP</i>	
 NOTE: HIP Report Support for PAN GP CyPerf now supports HIP (Host Information Profile) report submission during SSL VPN tunnel setup for PAN GlobalProtect. HIP reports are submitted only if requested by the DUT, and CyPerf respects the HIP report interval configured on the firewall. Statistics track HIP submission success or failure, evaluation failures, and HIP submission due to HIP-related issues. See also PAN GP HIP Report Statistics. Limitation: TCP-in-TCP is not supported.	
Use Portal	Use this toggle to enable or disable direct gateway access. This option is enabled by default. When enabled, this option allows you to configure a portal hostname and one or multiple VPN gateways. When disabled, the Portal hostname option is no longer available.
Portal Hostname	Configure the IP or hostname of a portal. Only a single portal is supported per each tunnel range.
VPN Gateways	Configure one or multiple comma-separated VPN gateways. Multiple gateways are picked by using round robin. For example: If two gateways are configured as gw1 & gw2 and three usernames and passwords are configured to u1-p1, u2-p2 and u3-p3 with the tunnel count set to 4, then the behavior is the following : • Tunnel1 uses gw1 and the username and password of u1-p1 • Tunnel2 uses gw2 and the username and password of u2-p2 • Tunnel3 uses gw1 and the username and password of u3-p3 • Tunnel4 uses gw2 and the username and password of u1-p1
ESP Probe Retry Timeout	The time (in seconds) after which the ESP Probe is retransmitted. The default value is 60 seconds.
ESP Probe Timeout	The time (in seconds) after which the tunnel is declared failed (ESP Connect Failure) if an ESP Probe response is not received. The default value is 65 seconds.
<i>FORTINET</i>	
VPN Gateway	Add or configure the IP or hostname of a gateway. Only a single gateway is supported per each tunnel range.

Field	Description
<i>F5</i>	
VPN Gateway	Add or configure the IP or hostname of a gateway. Only a single gateway is supported per each tunnel range.
<i>Outer TLS Profile (read-only)</i>	
TLSv1.2	This option is configured to <i>Read-only</i> and cannot be modified.
Cipher	This option is configured to <i>Read-only</i> and cannot be modified.
<i>Auth Settings</i>	
Authentication method	To configure TLS VPN authentication for all vendor types, select the IP Network 1 tile on the test topology section. The IP Network 1 page opens. Under the Network Configuration panel, go to Network Stacks and select TLS VPN Stacks. Add a new stack if none created, or select an existing one, e.g. Tunnel Stack 1. Then, select Tunnel Range > Auth Settings, and choose an authentication method from the available options: • AAA • AAA+Certificate • Certificate Depending on the selected authentication method, more authentication settings become available for configuration as described below.
Authorization	Select an authorization method from the available options : • User input: Enter one or multiple comma-separated Usernames and Passwords pairs. • Playlist file: Select or upload a playlist file, in the form of a .csv file containing multiple usernames and their associated passwords. Then, make sure to properly select the respective column names for usernames and passwords. For more information on the Playlist file option, see Artifacts. When establishing TLS VPN tunnels, the username and password are picked from the provided list in round-robin fashion. The Authorization settings are available only for the following authentication methods: • AAA • AAA and Certificate
Certificate file	Select the certificate used for authentication. Available only for the following authentication methods: • AAA and Certificate • Certificate

Field	Description
Key file	Select the key file used for authentication. Available only for the following authentication methods: • AAA and Certificate • Certificate This is an <i>optional</i> parameter; however, if you enter a password in the Key file password field, the parameter becomes <i>required</i>.
Key file password	Enter the password of the key file. This is an <i>optional</i> parameter, and is available only for the following authentication methods: • AAA and Certificate • Certificate
<i>Data Encapsulation</i>	
Encapsulation	Select the encapsulation method depending on the configured vendor type. For <i>Cisco AnyConnect</i>, the following two options are available: • DTLS • TLS For <i>PAN GP</i>, the following two options are available: • ESP over UDP • TLS For <i>FORTINET</i>, the following two options are available: • PPP over DTLS • PPP over TLS For <i>F5</i>, the following two options are available: • PPP over DTLS • PPP over TLS When selecting DTLS, ESP over UDP, or PPP over DTLS, the UDP client port can also be configured.
UDP Client Port	Configure CyPerf's Client UDP Port. The default value is 1024 for all vendor types. The value is incremented by 1 for each added tunnel.
Outer IP Range Settings	
<i>Automatic IP</i>	<i>You can use automatic IP when a test interface already has one or more IP addresses assigned to it.</i> <i>If you disable Automatic IP you are allowed to configure the IP manually. IPv6 addresses are supported.</i>

Field	Description
IP start	The first IP address in the range. This is the base address used for enumerating all the addresses in the range. The default address is 10.0.0.10 .
IP increment	This value is used to enumerate all the addresses in a range. The default value is 0.0.0.1 .
Count	The number of addresses to be created for a range. The default value is 1 and the maximum value is 10.000.000 .
Max count per agent	The number of IP addresses that are assigned to each agent. The default value is 1 and the maximum value is 1.000.000 .
Automatic IP prefix length	<i>Use the toggle to enable automatic IP prefix length. If Automatic IP prefix length is disabled, you can configure it manually.</i>
IP prefix length	Determines the size of the subnet and is the number of 1 bits in the subnet mask. The default value is 16 .
Automatic gateway	<i>Use the toggle to enable automatic gateway. If Automatic gateway is disabled, you can configure it manually.</i>
Gateway	The gateway address to be associated with all the addresses in the range. The default value is 10.0.0.1 .
Automatic MSS	Use the toggle to enable automatic MSS. When this option is disabled, you can manually configure the MSS value. The default MSS value is 1460 bytes.  NOTE: Jumbo frames are not supported for TLS VPN stacks.

Other TLS VPN notes

For TLS VPN, when the **Outer IP Range** is configured with multiple IPs in automatic mode, and when a CSV file with multiple entries has been uploaded under the **Auth Settings** of the tunnel range, all of the outer IPs iterate over the single row of the CSV file. The same behavior occurs when multiple **Connection Profiles** for CISCO and multiple **VPN gateways** for PAN GP have been configured.

For example:

- A configuration with 10 outer IPs is set to automatic mode, and the **Tunnel count per outer IP** set to 1.
- A CSV file is uploaded that has 10 rows of credentials, and the configured objective type is **Simulated User** set to 10.

In the example above, all of the 10 outer IPs use the entries from the first row of the uploaded CSV file.

PAN GP HIP Report Statistics

The following PAN GP HIP Report statistics are available:

Statistic	Description
HIP Reports Submitted	Indicates the number of HIP reports sent by CyPerf after successful tunnel establishment. HIP report submission is triggered only if the PAN is configured to request it.
HIP Reports Submission Successful	Represents the count of HIP reports for which the DUT responded with a SUCCESS status, confirming that the report was accepted and processed.
HIP Reports Submission Failed	Counts the number of HIP report submissions where the DUT returned a FAILURE response, indicating an issue during the submission process.
HIP Reports Evaluation Failed	Reflects cases where the HIP report was successfully submitted but failed to meet the security policy configured on the DUT. This typically results in access restrictions.

IPSec Stacks

To work with IPSec Stacks settings:

1. In an existing test session, create a new network segment or select an existing network segment whose configuration you want to update. The **Network Configuration** page opens.
2. Under **Network Stacks**, select **IPSec Stacks**.
3. Add a new IPSec stack using the **Add** button. A new IPSec stack is added to the configuration.
4. Select the newly added IPSec stack, and start configuring each setting as required.

Refer to the following sections for definitions of the global and per-range IPSec settings available in the CyPerf user interface:

- [Global IPSec Stack Settings](#)
- [Per IPSec Stack Range Settings](#)

Global IPSec Stack Settings

The following options are available under **IPSec Stacks > Stack Connections**:

Parameter	Description
Stack Role	Sets the role for your test. • Initiator: Select this option if the network segment will function as the initiator in the test (i.e., the selected network segment initiates the tunnels). • Responder: Select this option if the network segment will function as the responder in the test (i.e., the selected network segment listens for

Parameter	Description
	and responds to connection requests from its peer). NOTE: The Responder role is not currently supported.
Max Initiation Rate	Specifies the rate per second at which CyPerf initiates, retries and tears down IPsec tunnels. NOTE: This setting is available only for <i>IPSec initiator</i> on the <i>server</i> side. The setting is not supported for <i>IPsec initiator</i> on the <i>client</i> side, or for <i>IPsec responder</i>.
Max Pending Tunnels	Specifies the targeted maximum number of pending tunnels. When this maximum number is reached, CyPerf maintains the backlog of pending tunnels by issuing one new setup or tear-down request for each tunnel that is successfully set up or torn down. NOTE: This setting is available only for <i>IPSec initiator</i> on the <i>server</i> side. The setting is not supported for <i>IPsec initiator</i> on the <i>client</i> side, or for <i>IPsec responder</i>.
Enable Rekey	Enables or disables renegotiation of Phase 1 and Phase 2 SAs on expiry of tunnel lifetimes: - When disabled, tunnels are torn down when their lifetimes expire. - When enabled, the tunnels' Phase 1 and Phase 2 options are renegotiated before their lifetimes expire, and the tunnels stay up. The rekey parameters control the renegotiation process. This option is enabled by default.
Rekey Retry Count	The total number of rekey retries permitted. The value you enter defines the number of attempts that the IPsec plug-in makes to renegotiate the Phase 1 and 2 SAs. If all attempts at renegotiation fail, the IPsec plug-in drops the tunnel. The valid range of values is from 0 through 100,000. The default value is 0.
Rekey Margin	The rekey margin, in seconds. The value you enter represents the number of seconds subtracted from the connection expiration time, in order to ensure that the creation of new IPsec SAs begins before the current IPsec SAs expire. The valid range of values is from 0 through 3600. The default value is 10.
Log Keys	Enables or disables log keys.

Parameter	Description
	When enabled, two log key files are downloaded with the CyPerf capture file that you can use to decrypt IPSec packets for Wireshark analysis. For more details, see Decrypt IPSec Packets .
Setup Timeout	The number of seconds to wait for a response from the DUT before declaring that a tunnel setup attempt has failed. Values can range from 1 to 600. The default value is 30.
Retransmission Timeout	Time to wait before declaring the packet lost and begin retransmission. Values can range from 1 to 600 seconds. The default value is 30.
Tunnel Reattempt Count	The number of attempts that the IPsec plug-in makes to renegotiate the Phase 1 and Phase 2 SAs. If all attempts at renegotiation fail, the plug-in drops the tunnel. Values can range from 0 to 100,000. The default value is 0.
Tunnel Reattempt Interval	The number of seconds to wait before retrying the tunnel creation. Values can range from 0 to 60. The default value is 10.
Tunnel Reattempt Interval Increment	Increment value (in seconds) to be used as the delay between subsequent tunnel creation retry attempts. Values can range from 0 to 60. The default value is 10.

See also [Understanding the IPSec Timeout and Retransmission Settings](#).

Understanding the IPSec Timeout and Retransmission Settings

To better understand the correlation between the IPSec timeout and retransmission settings available under **IPSec Stacks > Stack Connections**, we will consider the example below.

The following two concepts are involved our example:

Packet retransmission:

- The timeout is controlled by the **Retransmission Timeout** setting.
- The number of packet retransmissions is calculated as $\text{floor}(\text{Setup Timeout} / \text{Retransmission Timeout}) - 1$. This represents the number of times a packet can be retransmitted until the current establishment attempt times out. In this example, this number is 1.

Setup reattempt:

- The timeout is calculated as $\text{floor}(\text{Setup Timeout} / \text{Retransmission Timeout}) * \text{Retransmission Timeout}$. In this example, the timeout for each tunnel attempt is 60.
- The number of **Setup reattempts** is controlled by the **Tunnel Reattempt Count** setting.

- The interval between a **timeout** and the next **reattempt** has a base value **Tunnel Reattempt Interval**, and it grows with **Tunnel Reattempt Interval Increment** after each reattempt.

Given the above values for the timeout and retransmission settings, you can find below the results triggered on the initiator when the IKE_AUTH RX is blocked on the responder:

3 0.048588	30.0.0.1	40.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Initiator Request
4 0.051033	40.0.0.1	30.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Responder Response
5 0.052194	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request
10 30.052366	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request
14 70.054475	30.0.0.1	40.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Initiator Request
15 70.056245	40.0.0.1	30.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Responder Response
16 70.057398	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request
18 100.058984	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request
22 147.059198	30.0.0.1	40.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Initiator Request
23 147.060908	40.0.0.1	30.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Responder Response
24 147.062087	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request
25 177.062105	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request
30 231.062949	30.0.0.1	40.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Initiator Request
31 231.064597	40.0.0.1	30.0.0.1	ISAKMP	274 IKE_SA_INIT MID=00 Responder Response
32 231.065753	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request
36 261.066264	30.0.0.1	40.0.0.1	ISAKMP	246 IKE_AUTH MID=01 Initiator Request

0:00 -> init

0:30 -> retransmit ike_auth	(+30)	(Retransmission Timeout)
1:00 -> timeout1	(+30)	(Retransmission Timeout)
1:10 -> tunnel init retry1	(+10)	(Tunnel Reattempt Interval)
1:40 -> retransmit ike_auth	(+30)	(Retransmission Timeout)
2:10 -> timeout2	(+30)	(Retransmission Timeout)
2:27 -> tunnel init retry2	(+10+7)	(Tunnel Reattempt Interval + Tunnel Reattempt Interval Increment)
2:57 -> retransmit ike_auth	(+30)	(Retransmission Timeout)
3:27 -> timeout3	(+30)	(Retransmission Timeout)
3:51 -> tunnel init retry3	(+10+7+7)	(Tunnel Reattempt Interval + 2 * Tunnel Reattempt Interval Increment)
4:21 -> retransmit ike_auth	(+30)	(Retransmission Timeout)
4:51 -> timeout4	(+30)	(Retransmission Timeout)

-> tunnel declared failed to application

This means that the setup is reattempted three times after the first attempt, which means there are four attempts in total. The IKE_AUTH packet is retransmitted once for each attempt, which means it is transmitted eight times in total.

Per IPSec Stack Range Settings

For each IPSec stack range connection, a complex suite of settings are available under the following tabs:

- [Local Subnet](#) tab
- **IPSec Ranges** tab with the its corresponding sub-sections and tabs:
 - [IPSec Range basic settings](#)
 - [Remote Subnet](#) settings (available for Initiator stack role only)
 - [Authentication](#) settings
 - [IKE Phase 1](#) settings
 - [IKE Phase 2](#) settings
 - [Timers](#) settings
 - [Remote Access](#) settings (available for Responder stack role only)
- [Outer IP Range](#) tab

Local Subnet Tab

The following options are available under **IPSec Stacks > Local Subnet**:

Parameter	Description
Start	The base address for enumerating all the local subnets in the range.
Increment	The increment to be used for enumerating all the local subnets in the range. The default IPv4 value is 0.0.1.0. For each address in the IP range, a subnet will be generated by incrementing the local subnet field with the increment value. For example, if you have an IP range with a count of 5, the following subnets will be created on the port: • 40.0.0.0/24 • 40.0.1.0/24 • 40.0.2.0/24 • 40.0.3.0/24 • 40.0.4.0/24 The number of hosts created on each subnet is defined in the Host Count field.  NOTE: The Local Subnet Increment By value must be the same on the initiator side and the responder side. If there is a mismatch (for example, 0.0.1.1 versus 0.0.1.0), the tunnels will come up, but traffic will fail.
Prefix	The length (in bits) of the subnet mask to be applied to all the addresses created in the range. For <i>IPv4</i>, values can range from 0 through 32; the default value is 24.

Parameter	Description
	This option is disabled when Multi P2 over P1 is enabled.
Hosts Prefix	The length (in bits) of the subnet mask to be applied to all the addresses created in the range. For <i>IPv4</i>, values can range from 0 through 32; the default IPv4 value is 32. This option is available when Multi P2 over P1 is enabled.
Hosts Increment	The increment to be used for enumerating all the local subnets of the phase 2 tunnels that belong to each phase 1. The default <i>IPv4</i> value is 0.0.0.1. For each IPsec P2, a subnet will be generated by incrementing the local subnet field with the increment value. For example: If you have an IP range with a count of 3 with Host count of 2, Local Subnet Increment By 0.0.1.0, Local Subnet Hosts Prefix 32, and Local Subnet Hosts Increment By 0.0.0.1, then the following IP addresses will be used: • 40.0.0.1/32 • 40.0.0.2/32 • 40.0.1.1/32 • 40.0.1.2/32 • 40.0.2.1/32 • 40.0.2.2/32 The number of hosts created on each subnet is defined in the Host Count field. This option is available when Multi P2 over P1 is enabled.
Host count per tunnel	The total number of hosts to be created for each of the local subnets. Values can range from 1 to 65,534. The default value is 1.
Total Host Count	The total number of hosts to be created.
Network tags	Enter one or multiple custom network tags. The tags entered in this field will be displayed under the network topology section of the test Overview page for the selected network element's IPsec Stack.

IPSec Range Basic Settings

The following settings are available under **IPSec Stacks > IPSec Ranges > Stack Range Connection**:

Parameter	Description
Test Scenario	Type of IPsec test scenario for which you are defining the current configuration. The following options are available: • Site-to-Site: Two sites are connected through a pair of IPsec Secure Gateways. When this option is selected, the fields pertaining to Remote Access are disabled. Site-to-Site is the default option. • Remote- Access: An individual client is connected to a LAN through a secure tunnel. In this scenario, the client is operating as its own Secure Gateway. When this option is selected, on the Initiator side, the fields pertaining to Local Subnet are disabled.
Public Peer	Public IP address of the peer.
Public Peer Increment	Increment value to be used for generating the Public IP addresses of the peer.
Multi P2 over P1	Select this option to enable the multiple phase 2 tunnels for each phase 1 SA feature. This parameter determines whether your test will establish a single phase 2 tunnel, or multiple phase 2 tunnels, for each phase 1 SA: • When <i>enabled</i>, the IPsec plug-in can establish multiple phase 2 SAs over a single phase 1 SA. In this case, the plug-in creates a phase 2 tunnel for each emulated host in the range. Therefore, the number of phase 2 tunnels per range equals the number of Emulated Gateways multiplied by the number of hosts in the local subnets. For example: ▪ If you have one Emulated Gateway with four emulated hosts, the IPsec plug-in will establish four phase 2 tunnels (1 x 4). ▪ If you have two Emulated Gateways, each with four emulated hosts, the IPsec plug-in will establish eight phase 2 tunnels (2 x 4). • When <i>disabled</i>, the plug-in establishes a single phase 2 tunnel for each phase 1 SA. Additional configuration requirements: • This feature supports both site-to-site and remote access scenarios. • For remote access scenarios, the configuration payload is sent inside each <code>CREATE_CHILD_SA</code> request. • The SA association is host-to-host; therefore, ensure that the number of local subnets configured in your test is equal to the number of remote subnets.

Remote Subnet Tab

This tab is available only for the **Initiator** stack role.

The following settings are available under **IPSec Stacks > IPSec Ranges > Remote Subnet > Remote Subnet Config**:

Parameter	Description
Start	The base address for enumerating all the remote subnets in the range.
Increment	The increment to be used for enumerating all the remote subnets in the range. The generated subnets will be used as traffic selectors. The default IPv4 value is 0.0.1.0. NOTE: The Remote Subnet Increment By value must be the same on the initiator side and the responder side. If there is a mismatch (for example, 0.0.1.1 versus 0.0.1.0), the tunnels will come up, but traffic will fail.
Prefix	The length (in bits) of the subnet mask to be applied to all the addresses created in the range. For <i>IPv4</i>, values can range from 0 through 32; the default IPv4 value is 24. This option is disabled when Multi P2 over P1 is enabled.
Hosts Prefix	The length (in bits) of the subnet mask to be applied to all the addresses created in the range. For <i>IPv4</i>, values can range from 0 through 32; the default IPv4 value is 32. This option is available when Multi P2 over P1 is enabled.
Hosts Increment	The increment to be used for enumerating all the remote subnets in the range. The generated subnets will be used as traffic selectors. The default <i>IPv4</i> value is 0.0.0.1. This option is available when Multi P2 over P1 is enabled.
Single Remote Subnet	Select this option if you want to configure a single remote subnet on the responder side. Selecting this field inhibits the generation of remote subnets. NOTE: If you want to configure a single remote subnet for all phase 2 tunnels in a multi P2 over P1 scenario, you need to configure the Remote Subnet Hosts Increment By field value to 0.0.0.0 for IPv4.

Authentication Tab

The following settings are available under **IPSec Stacks > IPSec Ranges > Authentication > Authentication Settings**:

Parameter	Description
Pre-Shared key	The Pre-Shared Key value. This is a string value with a maximum of 4096 characters. The default is ipsec.  NOTE: Ensure that the Pre-Shared Key value does not include a trailing space. CyPerf will treat the trailing space as part of the value. Some DUTs will drop an Authentication Failed notification payload, while others will issue a Payload_Malformed notification payload. In both cases, the tunnel will be dropped by the DUT.

IKE Phase 1 Tab

The following settings are available under **IPSec Stacks > IPSec Ranges > IKE Phase 1**:

Parameter	Description
Lifetime	Specifies the Phase 1 Security Association (SA) lifetime, in seconds. Values can range from 0 through 31,557,600.
Encryption Algorithm	Specifies the encryption algorithm used to protect communications during the phase 1 message exchange.
Hash Algorithm	Specifies the hashing algorithm to use for phase 1.
DH Group	Specifies the Diffie-Hellman (DH) Group.
PRF Algorithm	Specifies the algorithm used to perform Pseudo-Random Functions (key derivations).
Initial Contact	When enabled, the IPsec plug-in will send the INITIAL_CONTACT notification payload as part of the IKE SA establishment. An IKE SA can use an INITIAL_CONTACT notification message in order to assert that it is the only SA currently active between the authenticated identities. This message may be sent when an SA is established following a crash, thereby allowing the recipient to delete any other IKE SAs it has, to the same authenticated identity, rather than waiting for a timeout. This parameter is disabled by default.

Parameter	Description
	 NOTE: The IPsec plug-in always ignores the INITIAL_CONTACT notification payload, if it is received.

IKE Phase 2 Tab

The following settings are available under **IPSec Stacks > IPSec Ranges > IKE Phase 2:**

Parameter	Description
Lifetime	Specifies the Phase 2 Security Association (SA) lifetime, in seconds. Values can range from 1 to 31,557,600.
Encryption Algorithm	Specifies the encryption algorithm used to protect communications during phase 2 message exchange.
Hash Algorithm	Specifies the hashing algorithm to use for Phase 2.
PFS Enabled	When enabled, the CyPerf uses Perfect Forward Secrecy (PFS) for all the traffic in the range. This option is disabled by default.
PFS Group	Select the Perfect Forward Secrecy (PFS) Group to be used.
NAT-T Enabled	When enabled, CyPerf implements NAT-T for all the traffic in the range. Enable this parameter when running IPsec over NAT devices. NAT-T is configurable in IPv4 environments only. IPsec NAT-T NAT-T (Network Address Translation Traversal) was developed to address the problem of using IPsec over NAT devices. Because NAT devices modify addresses in the IP header of packets, these packets fail the checksum validation when IPsec is in use. For IPsec, the packets appear to have been modified in transit, something IPsec is intended to prevent. NAT-T detects the presence of NAT devices between two hosts, switches the IPsec function to a non-IPsec port, and encapsulates the IPsec traffic within UDP packets. To preserve the original source and destination port numbers, NAT-T inserts an additional header containing the port numbers between the IP header and the ESP header. For example, after IKE peers initiate negotiation on port 500, detect support for NAT-T, and detect a NAT device along the path, they can negotiate to switch the IKE and UDP-encapsulated traffic to another port, such as port 4500 (the IPsec plug-in listens on port 4500 to establish a connection for IKEv2.). The main RFCs for NAT-T implementations are: • RFC 3715, IPsec-Network Address Translation (NAT) Compatibility Requirements

Parameter	Description
	• RFC 3947, Negotiation of NAT-Traversal in the IKE • RFC 3948, UDP Encapsulation of IPsec ESP Packets

Timers Tab

The following settings are available under **IPSec Stacks > IPSec Ranges > Timers > Timers Config**:

Parameter	Description
DPD Enabled	When enabled, each IKE peer in the range uses the Dead Peer Detection (DPD) protocol to determine proof of liveness of the other peer. The peers send DPD HELLO messages according to the interval that you specify under the DPD Idle Period option. When disabled, the IKE peers do not send DPD HELLO messages. An IPsec endpoint uses DPD to confirm that its peer is still up. DPD is implemented in IKE through the use of an asynchronous, bidirectional message exchange: • DPD HELLO • DPD HELLO ACK A complete DPD exchange (transmission of DPD HELLO and receipt of the corresponding DPD HELLO ACK) serves as proof of liveness. If a VPN device does not receive a response to a DPD HELLO within a specified time, it assumes that the peer is dead or unreachable, and tears down the tunnel.  NOTE: • If DPD is enabled, CyPerf always sends the DPD messages regardless of the traffic that is being sent over the tunnel. • The IPsec plug-in always responds to DPD messages received from the DUT whether or not DPD is enabled. • The IPsec plug-in implementation of DPD does not use an explicit retry mechanism. For example, if you set the idle period to 5 seconds and the timeout to 14 seconds, the plug-in will send two DPD HELLOs (at 5 and 10 seconds) within the timeout period. If at least one of those DPD HELLO messages receives a DPD HELLO ACK, the timer will be reset and the tunnel will remain up.
DPD Idle Period	The interval for sending DPD messages, in seconds. For example, if you set this to 60, the IPsec plug-in sends DPD HELLO messages every 60 seconds to each peer defined for the range. This value must be smaller than the tunnel lifetimes. Values can range from 1 through 2,147,483,647; the default value is 1000.
DPD Timeout	The amount of time allowed for receiving a response to a DPD HELLO message, in seconds. If the IPsec plug-in sends one or more DPD HELLO messages and does

Parameter	Description
	not receive a response within the DPD Timeout period, it assumes that the peer is down and terminates the tunnel. Values can range from 1 through 2,147,483,647; the default value is 10.

Remote Access Tab



NOTE:

The Remote Access tab is available only for the **Responder** stack role. These options are not currently supported.

The following settings are available under **IPSec Stacks > IPSec Ranges > Remote Access**:

Parameter	Description
ModeCfg Start Address	Defines the base address to be used for the ModeCfg address pool (the IP addresses that the server port will assign to the clients). The three ModeCfg Address parameters are used only by responder ports, i.e., only for a Responder Stack Role test or for the responder port in a port-to-port test.
ModeCfg Address Suffix	Defines the IP address suffix for the ModeCfg address pool. Values can range from 1 to 128; the default value is 24.
ModeCfg Address Increment	Defines the increment value for the ModeCfg address pool. The default value is 0.0.0.1.

Outer IP Range

The following settings are available on the **Network Configuration** page under **Network Stacks > IPSec Stacks > Outer IP Range**:

Parameter	Description
Automatic IP	You can enable this option when a test interface already has one or more IP addresses assigned to it. When this option is disabled, you can configure the IP manually by setting appropriate values for the following fields: • IP start: The first IP address in the range. This is the base address used for enumerating all the addresses in the range. The default address is 10.0.0.10. • IP increment: The value used to enumerate all the addresses in the range. The default value is 0.0.0.1. • Count: The number of addresses to be created for this range. The default value is 1 and the maximum value is 10.000.000.

Parameter	Description
	• Max count per agent: The number of IP addresses that are assigned to each agent. The default value is 1 and the maximum is 1.000.000.
Automatic IP prefix length	When this option is enabled, CyPerf will use an automatic IP prefix length. When disabled, you can enter a custom value in the IP prefix length field.
IP prefix length	Determines the size of the subnet, and is the number of 1 bits in the subnet mask. The default value is 16.
Automatic gateway	When this option is enabled, CyPerf will use an automatic gateway value. When disabled, you can enter a custom value in the Gateway field.
Gateway	The gateway address to be associated with all the addresses in the range. The default value is 10.0.0.1.
Automatic MSS	Use the toggle to enable automatic MSS. When this option is disabled, you can manually configure the MSS value. The default MSS value is 1460 bytes.  NOTE: Jumbo frames are not supported for IPSec stacks.

IPSec Statistics

The following statistics are available for IPSec Stack Range Connections:

Stat Name	Description
Sessions initiated	The total number of tunnels that were initiated.
Sessions succeeded	The total number of tunnels that were successfully established.
Sessions failed	The total number of tunnel initiations that failed.
Sessions active	The number of tunnels that are up at a given time. The sessions may have been successfully established, but because of the dynamic nature of IPsec, some tunnels may be torn down over time (due to a lack of DUT responsiveness or a lack of successful rekeys).
Phase1 total rekeys	The total number of rekey attempts (both successful and unsuccessful) during Phase 1.
Phase1 successful rekeys	The total number of rekey attempts that succeeded during Phase 1.

Stat Name	Description
Phase1 failed rekeys	The total number of rekey attempts that failed during Phase 1.
Phase2 total rekeys	The total number of rekey attempts (both successful and unsuccessful) during Phase 2.
Phase2 successful rekeys	The total number of rekey attempts that succeeded during Phase 2.
Phase2 failed rekeys	The total number of rekey attempts that failed during Phase 2.
DPD hellos sent	The total number of Dead Peer Detection (DPD) Hello packets (R-U-THERE messages) sent to the other side.
DPD hellos received	The total number of Dead Peer Detection (DPD) ACK packets (R-U-THERE-ACK messages) received from the other side.
DPD acks received	The total number of Dead Peer Detection (DPD) ACK packets (R-U-THERE-ACK messages) received from the other side.
Total retries	The total number of retries. This is a count of all of the attempts to establish a tunnel, when the initial tunnel establishment failed.
Start client timeout	The total number of tunnel creation timeouts.
Stop client timeout	The total number of tunnel tear down timeouts.

**NOTE:**

For the CyPerf *client* agents, since the tunnel setup rate can only be controlled by their indirect relationship with **Simulated Users**, you can use the [Max pending simulated users](#) and [Max simulated users per second](#) options. However, for the CyPerf *server* agent initiating IPSec tunnels, you can use the [Max Initiation Rate](#) and [Max Pending Tunnels](#) options.

For CyPerf *client* agents using the **Concurrent Connections** objective, you cannot control the tunnel setup rate explicitly. In this case the **Simulated Users** value will be internally calculated and 10% of that will be used as **Max pending simulated users** without a specific limit per second. This derived tunnel setup rate may lead to tunnel failures if the Device Under Test is unable to handle the rate effectively.

MAC DTLS Stacks

To configure MAC DTLS stacks for any given IP Network segment:

1. In an existing test session, create a new network segment or select an existing network segment whose configuration you want to update. The **Network Configuration** page opens.
2. Under **Network Stacks**, select **MAC DTLS Stacks**.
3. Add a new MAC DTLS stack using the **Add**  button. A new MAC DTLS stack is added to the configuration.
4. Select the newly added MAC DTLS stack, and start configuring each setting as required. See the table below for details about each setting.

When running a test with MAC DTLS, a new **Instantaneous MAC DTLS Tunnels** statistics graph will be displayed on the **Statistics** page under the **MAC DTLS Tunnels** panel.



IMPORTANT:

- MAC DTLS stacks cannot be used in combination with other network stacks in the same IP Network.
- When MAC DTLS stacks are enabled, the Ethernet range cannot run with [Automatic MAC](#) or [One MAC per IP/ Tunnel](#).

The following settings are available on the **Network Configuration** page under **Network Stacks** > **MAC DTLS Stacks**:

Field	Description
<i>MAC DTLS Stacks</i>	
	Use this button to add a new MAC DTLS stack.
	Use this button to rename an existing MAC DTLS stack.
	Use this button to delete an existing MAC DTLS stack.
<i>MAC DTLS Properties</i>	
Tunnel destination MAC start	The first tunnel destination MAC address in the range. This is the base address used for enumerating all the Tunnel destination MAC addresses in the MAC DTLS range. The default address is AA:BB:CC:DD:EE:FF .
Tunnel destination MAC increment	The value used to increment all the Tunnel destination MAC addresses in the range. The default value is 00:00:00:00:00:01 .
Tunnel count	Displays the number of tunnels for the DTLS range calculated as <code>ethCount*vlanCount</code> . This option cannot be configured.

In key	The starting 256-bit key used to decrypt incoming data for the DTLS range.
In key increment	The step size or the amount by which the In key is incremented during the decryption process (256 bits).
Out key	The starting 256-bit key used to encrypt outgoing data for the DTLS range.
Out key increment	The step size or the amount by which the Out key is incremented during the encryption process (256 bits).
In IV	The starting 32-bit Initialization Vector (IV) used to decrypt incoming data for the DTLS range.
In IV increment	The step size or the amount by which the starting In IV is incremented during the decryption process (32 bits).
Out IV	The starting 32-bit Initialization Vector (IV) used to encrypt outgoing data for the DTLS range.
Out IV increment	The step size or the amount by which the starting Out IV is incremented during the encryption process (32 bits).
Epoch	The 16-bit starting Epoch value for the DTLS range. The Epoch value is used to differentiate between different sessions or periods of communication. Valid values can range between 0 and 65535.
Epoch increment	The step size or the amount by which the starting Epoch value is incremented within the DTLS range (16 bits). Each subsequent Epoch value in the range will be incremented by this amount. Valid values can range between 0 and 65535.
<i>VLAN Range</i>	
VLAN enabled	When enabled, this option allows you to apply VLAN settings on top of the existing DTLS configuration.
VLAN ID	The first VLAN ID to be used for VLAN. The minimum value is 1 , the maximum value is 4094 .
Increment by	The value to be added to the VLAN ID for each new assignment. The maximum value is 4094 . The default value is 1 .
Count	The number of unique VLAN IDs that will be created. The default value is 1 , the maximum value is 4094 .
Tag protocol identifier	Set to 0x8100 by default.
802.1Q Priority	Set to 0 by default.
<i>IP Range</i>	
IP start	The first IP address in the range. This is the base address used for enumerating

	all the addresses in the range. The default address is 10.0.0.10 .
IP increment	The value used to enumerate all the addresses in the range. The default value is 0.0.0.1 .
Total host count	Displays the total number of inner IPs calculated as tunnelCount*hostCount . This option cannot be configured.
IP prefix length	Determines the size of the subnet and is the number of 1 bits in the subnet mask. The default value is 16 .
Gateway	The gateway address to be associated with all the addresses in the range. The default value is 10.0.0.1 .
MSS	Configure the Maximum Segment Size (in bytes). The default MSS value is 1460 bytes.
Host count per tunnel	The number of inner IP addresses per tunnel. The default value is 1 , the maximum value is 10,000,000 .

Emulated Router

You can use the Emulated Router settings to emulate a single device endpoint for multiple IPs. When the Emulated Router is enabled, multiple IPs will establish/ receive connections via the Emulated Router's MAC address. You can configure Emulated Router with plain IP stack for both IPv4 and IPv6, but TLS VPN stack and IPsec stack can only be configured with IPv4. The Emulated Router and VLAN settings are configurable from the UI for IP stack, but the VLAN settings will not take any effect in the presence of an Emulated Router.

To work with the Emulated Router settings:

1. In an existing test session, create a new network segment or select an existing network segment whose configuration you want to update. The **Network Configuration** page opens.
2. Under **Network Configuration**, select **Emulated Router**. The **Emulated Router Properties** section opens.
3. Use the toggle to enable or disable the Emulated Router settings, and configure the available IPv4 and IPv6 settings.

See also [Emulated Router Properties](#) for definitions of the Emulated Router settings available in the CyPerf user interface.

Emulated Router Properties

The following settings are available for both IPv4 and IPv6, under **Network Configuration > Emulated Router > Emulated Router Properties > IPv4 Emulated Router Range** or **IPv6 Emulated Router Range**:

Parameter	Description
Automatic IP	You can enable this option when a test interface already has one or more IP addresses assigned to it. When this option is disabled, you can configure the IP manually by setting appropriate values for the following fields: • IP start: The first IP address in the range. This is the base address used for enumerating all the addresses in the range. The default IPv4 address is 10.0.0.10. The default IPv6 address is fc00::0:10 • IP increment: The value used to enumerate all the addresses in the range. The default IPv4 value is 0.0.0.1. The default IPv6 value is ::1. • Count: The number of addresses to be created for this range. The default value is 1 and the maximum value is 10.000.000. • Max count per agent: The number of IP addresses that are assigned to each agent. The default value is 1 and the maximum is 1.000.000.
Automatic IP prefix length	When this option is enabled, CyPerf will use an automatic IP prefix length. When disabled, you can enter a custom value in the IP prefix length field.
IP prefix length	Determines the size of the subnet, and is the number of 1 bits in the subnet mask. Values can range from 1 through 128. The default IPv4 value is 16, and the default IPv6 value is 64.
Automatic gateway	When this option is enabled, CyPerf will use an automatic gateway address to be associated with all the addresses in the range.

Ethernet Range

Field	Description
<i>Ethernet Range</i>	
Automatic MAC	Use the toggle to enable automatic MAC. When enabled, this option allows the test to use the current MAC address found on the agent. When disabled, you can manually configure the MAC start address and MAC address increment options below.
MAC start address	The base value used when the network stack element creates MAC addresses. This address is associated with the first interface on the port.
MAC address increment	The value to use when incrementing the MAC address.
One MAC per IP/	When enabled, only one MAC address per IP is created. When disabled,

Tunnel	you can manually configure the Count and Max count per agent options below.
Count	The total number of MAC addresses to be created for the range. The default value is 10 .
Max count per agent	The maximum number of MAC addresses to be created per agent.
Static ARP	
The Static ARP properties allow you to configure a list of ranges that the agent will use to map IP addresses to MAC addresses instead of relying on ARP requests. NOTE: • Static IP-to-MAC mappings are supported for both IPv4 and IPv6. • Statistics for IPv6 Neighbor Discovery are not currently available in the UI; use packet captures to verify related behavior.	
Static ARP Table	Click the Add range (+) button to add a new entry to the static ARP table.
Static ARP Entry Properties	
Remote IP start	The base value used when the network stack element creates IP addresses for the selected ARP entry.
Remote IP increment	The value to use when incrementing the IP address range. The default value is 0.0.0.1.
Remote MAC start	The base value used when the network stack element creates MAC addresses for the selected ARP entry.
Remote MAC increment	The increment value used to create a range of MAC addresses for the selected static ARP entry. The default value is 00:00:00:00:00:01.
Count	Enter the total number of IP-to-MAC mappings that will be created in the selected ARP entry. The default value is 1.

DNS Resolver

Field	Description
DNS Resolver	
Cache timeout (seconds)	The amount of time (in seconds) that the resolver will wait for a response from a remote name server before retrying from another name server.
Name servers	Indicate the server name.

DNS Server

You can configure a DNS server at the server network segment level, which will respond to DNS queries from external clients on the port configured under the **DNS Server Properties** in the CyPerf web UI.

For each hostname configured in the application at the connection level (whether configured as a single entry or a playlist), the DNS server will respond with the first IP from the computed destination with respect to the traffic mapping.

For example, given the following configuration with:

- Two server network segments
- An application with a connection hostname `generic.keysight.io`
- Traffic mapped only on the second server network segment

Then, the DNS server will respond with the first IP found for that second server network segment. If there is a DUT active with a reverse proxy in front of the second server network segment, the hostname will resolve to the IP of the reverse proxy.



NOTE:

The following limitation applies to hostname resolution:

- The hostname will resolve only to the first IP found for the network segment. More specifically, the DNS server itself responds for all IPs configured in the test (either automatic or manual); however, the queries will respond only with the first IP.

The table below lists the UI elements found on the **Network Configuration** page under the **DNS Server** tab:

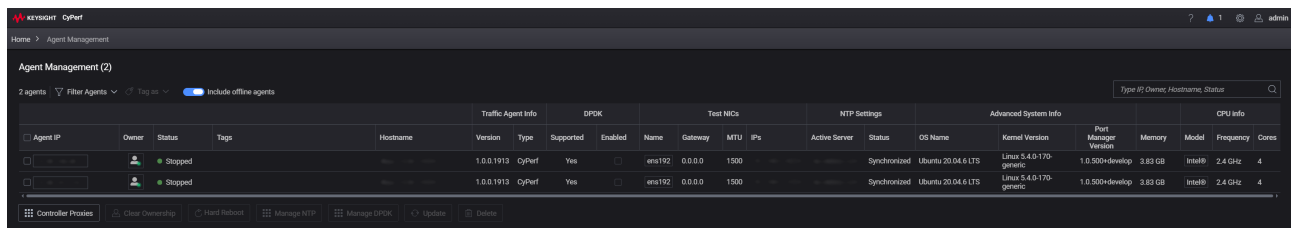
Field	Description
DNS Server Properties	
Enabled	Use the toggle to enable or disable the configuration of a DNS server.
Port	Set the port that will be used for DNS communication. The default value is 53, i.e. the standard port for DNS communication.

DUT Connections

Field	Description
DUT Connections	
Select connected DUTs	Select one, multiple or all connected DUTs.

Agent Management

The **Agent Management** page allows you to view and manage CyPerf traffic agents. To open the **Agent Management** page, select **Agent Management** under the **Settings** (⚙️) menu.



Agent IP	Owner	Status	Tags	Hostname	Version	Type	Supported	Enabled	Name	Gateway	MTU	IPs	Active Server	Status	OS Name	Kernel Version	Port Manager Version	Memory	Model	Frequency	Cores
10.0.1913		Stopped		...	1.0.0.1913	CyPerf	Yes		ens192	0.0.0.0	1500			Synchronized	Ubuntu 20.04.6 LTS	Linux 5.4.0-170-generic	1.0.500+develop	3.83 GB	Intel®	2.4 GHz	4
10.0.1913		Stopped		...	1.0.0.1913	CyPerf	Yes		ens192	0.0.0.0	1500			Synchronized	Ubuntu 20.04.6 LTS	Linux 5.4.0-170-generic	1.0.500+develop	3.83 GB	Intel®	2.4 GHz	4

The following actions are available for managing agents:

- Filter agents by tag names and remove filtering results:
To filter agents:
 - Select **Filter agents**.
 - Type the name of the tag or select it from the available list.
 - Select **Close**. The content on the **Agent Management** page is updated with the filtering results.

To remove filtering results, click the **Clear** button.

- Search for agents:
Use the search bar in the upper-right corner of the **Agent Management** page to search for agents based on IP, owner, hostname, or status.
- Add or remove custom tag names for each agent:
To add a tag:

1. Select one or more agents from the ones available.
2. Select **Tag as**.
3. Type the name of the tag in the **Search or add tag** field, and select **Add**. Select **Update** to add the tag name.

To remove a tag:

1. Select one or more agents from the ones available.
2. Select **Tag as**.
3. Select **Remove tags**.
4. Use the search functionality to identify the tag name or select it from the list. Select **Update** to remove the tag name.

You can also clear ownership, manage NTP servers, manage DPDK, update, reboot, and delete agents. See the table below for more details.

The following UI elements are available under **Agent Management**:

UI Element	Description
Agent IP	Indicates the agent's IP.
Owner	Indicates the agent's owner.
Status	Indicates the current status of the target.
Tags	Displays the custom tags associated to each agent.
Hostname	Displays the hostname for each agent.
<i>Traffic Agent Info</i>	
Version	Displays the traffic agent version. If an agent is running a different traffic agent version than the recommended version, a warning icon is displayed next to the current agent version. When hovering over the warning icon, a tooltip displays the recommended version. See also Update button and Updating Agents from the Controller UI .
Type	Displays the traffic agent type.
<i>DPDK</i>	
Supported	Shows whether the DPDK feature is supported on the selected agent.
Enabled	Shows whether the DPDK feature is enabled on the selected agent. To enable or disable the DPDK feature on a specific agent, select the agent and click the Manage DPDK button. Then, choose Enable for all selected agents or Disable for all selected agents .
<i>Test NICs</i>	
Name	Displays the test NIC's name. When hovering over the name, the MAC address is

UI Element	Description
	displayed for each test NIC.
Gateway	Displays the test NIC gateway.
MTU	Displays the maximum transmission unit (MTU) of the test NIC. To change the MTU: 1. First, manually configure the MTU on the CyPerf agent test interfaces. 2. Then, reload the configuration [cyperf configuration reload] to take effect. The same procedure of MTU customizations generally applies to most cloud and VM environments with possible exceptions applicable to some proprietary tunnel interfaces.
IPs	Displays the test NIC IP.
<i>NTP Settings</i>	
Active Server	Displays the NTP server to which the agent has connected.
Status	Displays the NTP server status, for example Synchronized .
<i>Advanced System Info</i>	
OS Name	Displays the name of the OS used by the agent.
Kernel Version	Displays the system kernel version for the agent.
Port Manager Version	Displays the version of the port manager.
<i>CPU Info</i>	
Memory	Displays the memory allocated to each agent on deployment.
Model	Displays the CPU model.
Frequency	Displays the CPU frequency in GHz.
Cores	Displays the number of CPU cores for each agent.
Controller Proxies button	Allows you to register a controller proxy. See Add a Controller Proxy .
Clear Ownership button	Releases the ownership of the selected agent(s). The button is enabled only when a test is running.

UI Element	Description
Hard Reboot button	Initiates a hard reboot on the selected agent(s).
Manage NTP button	Allows you to manage NTP server tags for the selected agent.
Manage DPDK button	Click to enable or disable DPDK for the selected agents.
Update button	Initiates an update operation for the selected agents supporting this feature. See also Version and Update Agents from the Controller UI .
Delete button	Removes the selected agent(s) from the Agent Management list.

Statistics Page

When you successfully start a test, CyPerf displays the **Statistics** page, where you can view the test's real time statistics. The specific groups of statistics that are collected and displayed on the statistics dashboard depend on the types of tests, traffic, and other settings that you have chosen in your configuration.

You can filter statistics based on various filtering options. For example, you can filter statistics by application, attack or network segment:

The screenshot shows the CyPerf Statistics page. The top navigation bar includes 'Home > 2 - CyPerf Empty Config > Statistics'. The main header has 'START TEST' (with a play icon) and 'Test gracefully stopped.' on the left, and 'RESULT' (with a document icon) on the right. Below the header, there's a 'Detailed views' sidebar on the left with a search bar 'Filter views by name' and a list of views: Summary, Client Application Profile, Client Aggregated HTTP Statistics, Client Statistics per Action (highlighted), Server Statistics, Server HTTP Statistics, TCP Statistics, Agent Traffic Statistics, Agent Resource Metrics, Yahoo Mail, and Youtube.

The main content area is titled 'Client Statistics per Action'. It features a filter bar with 'ClientProfileNetworkSegment' and an 'All' dropdown menu (highlighted with a red box). Next to it is an 'ApplicationAttack' filter (also highlighted with a red box). Below the filter bar, there's a table with columns 'Profile - Network Segment', 'Action Status', and 'Action'. The 'Action Status' column has a dropdown menu open, showing 'Selected (1)' and 'All' (checked). The 'Action' column lists various actions like 'Upload Video', 'Unsubscribe from C...', 'Unlike Video', 'Subscribe to Channel', 'Sign Out', and 'Sign In'.

Below the main table, there's a section titled 'HTTP Statistics (AppAttack: All)' with a table showing 'Profile - Network Segment', 'Application & Attack', and 'Action' columns. The data rows show various actions like 'Upload Video', 'Unsubscribe from Channel', 'Unlike Video', 'Subscribe to Channel', 'Sign Out', and 'Sign In'.

Home > 2 - CyPerf Empty Config > Statistics

Overview **Statistics**

START TEST Test gracefully stopped. RESULT

Detailed views

Filter views by name

ApplicationAttackType Port All SNI All

Selected (1)

Application & Attack

Youtube

Yahoo Mail

Office365 Outlook

Modbus

Cisco Spark

Application & Attack Statistics (All)

	Bytes Sent	Bytes Received	Connections Received	Strikes Allowed	Strikes
Youtube	10,042,347,272	365,748,929	8,672	0	
Yahoo Mail	10,617,037,365	3,203,271,026	67,875	0	
Office365 Outlook	9,067,199,169	4,169,470,240	67,081	0	
Modbus	41,405,496	41,405,496	291,588	0	
Cisco Spark	837,713	2,691,514	16	4	

Summary

Client Application Profile

Client Attack Profile

Client Aggregated HTTP Statistics

Client Statistics per Action

Server Statistics

Server HTTP Statistics

TCP Statistics

Agent Traffic Statistics

Agent Resource Metrics

Cisco Spark

Modbus

Office365 Outlook

Yahoo Mail

Youtube

Statistics are organized into tabs displayed on the left side of the **Statistics** page. These tabs include but are not limited to the following:

- **Summary**
- **Streaming Statistics**
- **Client Application Profile**
- **Client Aggregated HTTP Statistics**
- **Server Statistics**
- **Server HTTP Statistics**
- **TCP Statistics**
- **UDP Statistics**
- **Agent Traffic Statistics**
- **Agent Resource Metrics**

About Clock Offset

The Agent Time Synchronization or Clock Offset feature allows the application to synchronize the time between the controller and the available agents.

NOTE: If the agents are not synchronized, an error is displayed on the **Events** page instructing you on how to synchronize the agents. You will also be notified about this clock offset through a sticky notification, which appears during a new test run and remains persistent in the session until you dismiss it. The error is triggered only when the time difference between the middleware and the agents is more than 1 second or less than 2 seconds.

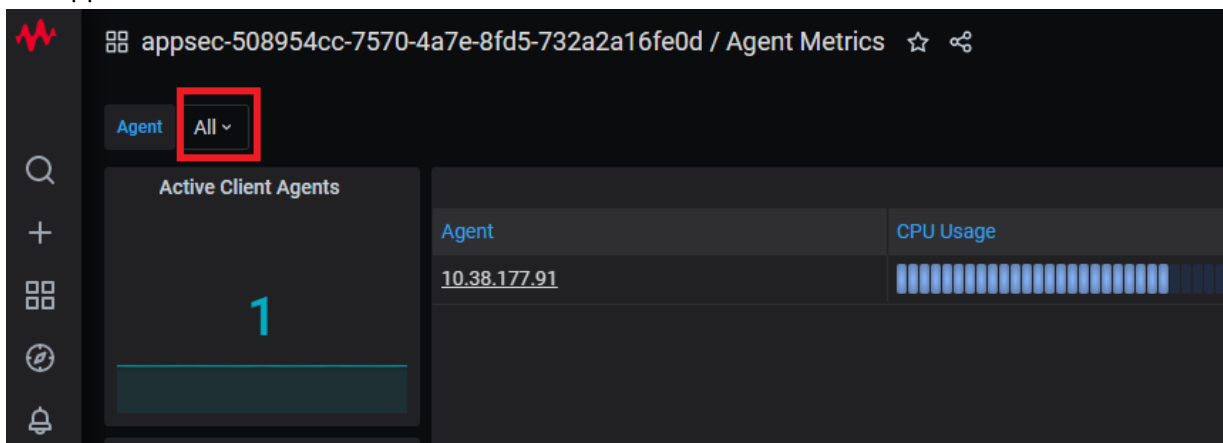
To set an NTP server on an *agent*, open a command line window and type '`cyperfagent ntpserver set`' followed by the IP address of the NTP server. The CyPerf controller itself or the CyPerf controller proxy can act as an NTP server. UDP port **123** needs to be opened for this.

For NTP settings on the *controller*, use the following commands:

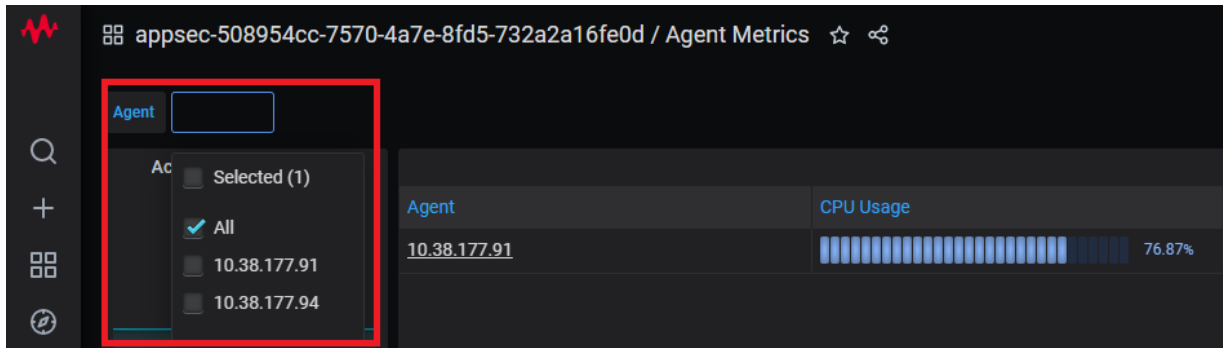
- `kcoss date-time ntp enable`: Enables NTP settings on the node.
- `kcoss date-time ntp disable`: Disables NTP settings on the node.
- `kcoss date-time ntp-servers set <NTP server addresses>`: Sets the list of NTP servers on the node. Use a space-separated list of one or more NTP server IP addresses.
- `kcoss date-time ntp-servers show`: Displays the list of known NTP servers and their status (enabled or disabled).
- `kcoss date-time ntp-servers delete`: Deletes the complete list of NTP servers from the node.

To view the clock offset of one or multiple agents:

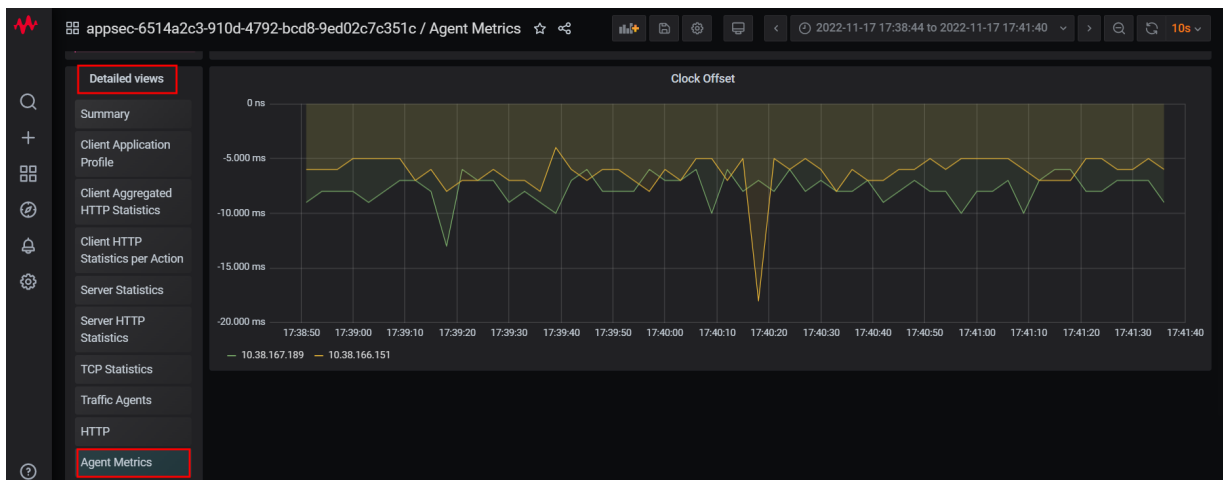
1. Under the **Statistics** page of a previously run test session, expand the **All** drop-down list on the left upper side of the user interface.



2. Select the **All** check box to view all the available agents, or select individual agents to view their statistics. You can also enter the IP address of the agent you wish to view in the search bar.



- Under **Detailed Views**, select the **Agent Metrics** tab. The **Clock Offset** graph is displayed at the bottom of the page plotting the clock offset values collected during the test run for the selected agent(s).



NOTE: The clock offset values for each agent per timestamp are also available in CSV format in the *agent-clock-offset.csv* file.

Artifacts

This chapter contains useful information and details on Playlist and Macro features.

Rules and Grammar for Playlists

Rules to support comma or double-quotes as a part of a playlist:

1. Each playlist item with comma or double-quote in the content **must** be enclosed within double-quotes.
2. Every double-quote used as a part of the content must be escaped with another double-quote.

Each record is located on a separate line, delimited by a line break (CRLF). For example: `record = value *(COMMA value):`

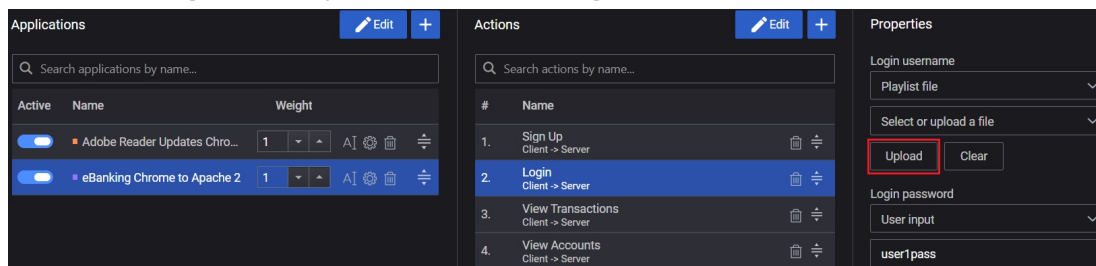
Record	value 1	value 2
abcd	abcd	
abcd,xyz	abcd	xyz
"abcd,pqr","xyz"	abcd,pqr	xyz
"abcd,pq""r","xyz"	abcd,pq"r	xyz

For all applications and attacks that have a **Sign In** or **Sign Up** action, the following parameters allow you to upload a playlist file:

- Login Username
- Login password
- SignUp Username
- SignUp password
- SignUp confirm password

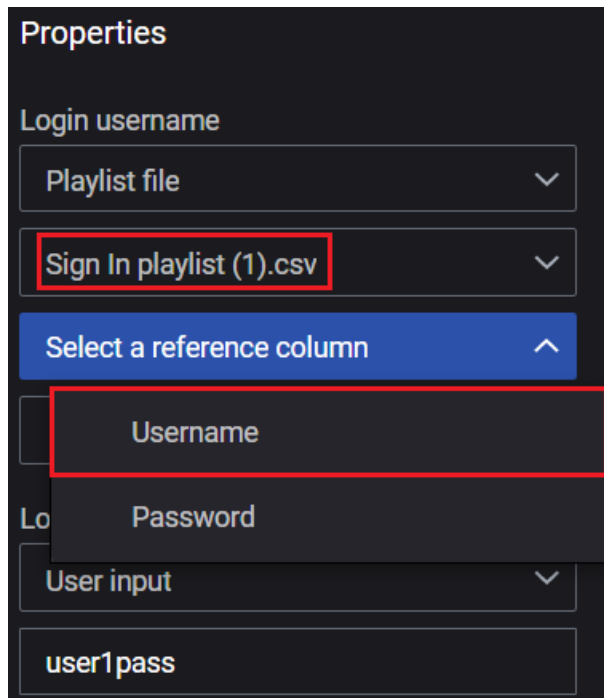
To upload a playlist file for **Sign In** or **Sign Up** actions:

1. Select the **Playlist file** option, and click the **Upload** button.



2. After the upload is complete, select one of the available Reference Column options from the

drop down. These options correspond to the uploaded CSV column names.



Properties

Login username

Playlist file

Sign In playlist (1).csv

Select a reference column

Username

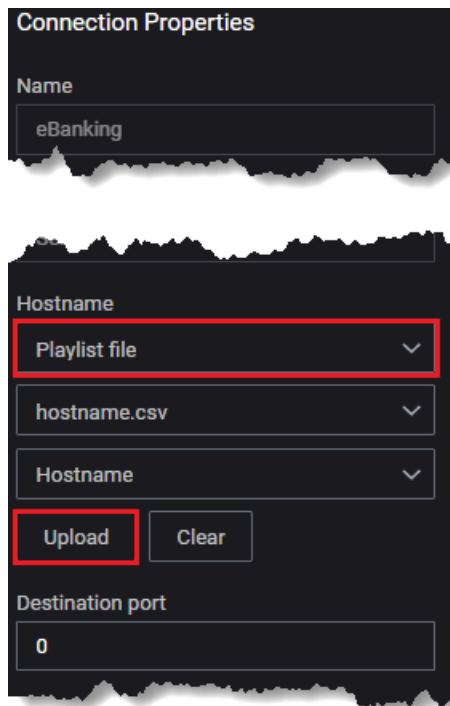
Password

User input

user1pass

For all attacks and applications (except *O365*, *Facebook*, and *Google Drive*), the **Hostname** option (under **Connection Properties**) allows you to upload a playlist file, as follows:

1. Select the **Playlist file** option, and click the **Upload** button.
2. After the upload is complete, select one of the available Reference Column options from the drop down. These options correspond to the uploaded CSV column names.



Connection Properties

Name

eBanking

Hostname

Playlist file

hostname.csv

Hostname

Upload Clear

Destination port

0

❗ **NOTE:** CyPerf currently does not support validation of empty **Hostname** values if they are fetched from a playlist file.

Example of a Hostname playlist file:

	A	B
1	Hostname	
2	server1.com	
3	server2.com	
4	server3.com	
5	server4.com	
6	server5.com	
7	server6.com	
8	server7.com	
9	server8.com	
10	server9.com	
11	server10.com	

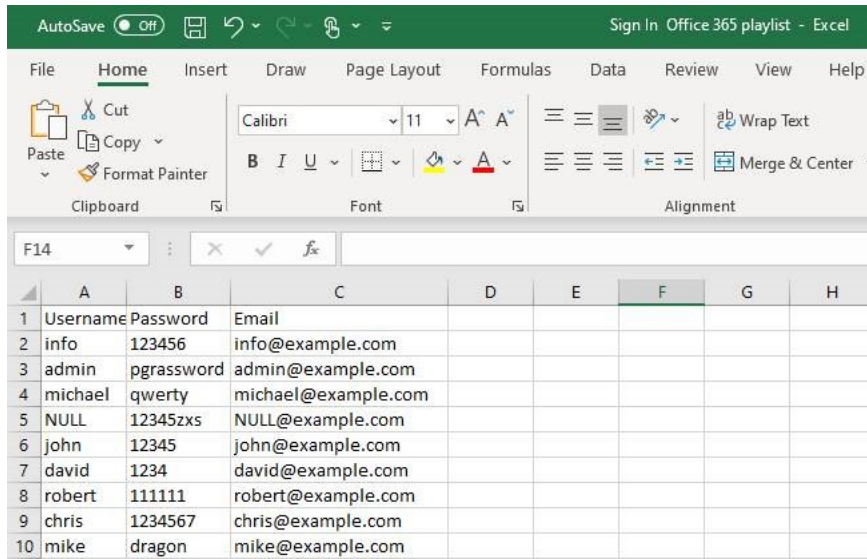
17	hostname(4107)	
----	----------------	--

About Playlists

For the **Sign In** action in eBanking, eShop, Social Network, Portal or the **Sign Up** action in eBanking applications or attacks, download this [Sign In playlist](#) file.

AutoSave ☐ Off									
Sign In playlist - Excel									
File	Home	Insert	Draw	Page Layout	Formulas	Data	Review	View	
Cut Copy Paste Format Painter Clipboard Calibri 11 A⁺ A⁻ B I U Font Color Background Color Font Align Center Wrap Text Merge & Center Alignment									
P10									
	A	B	C	D	E	F	G	H	I
1	Username	Password							
2	admin	pgrassword							
3	michael	qwerty							
4	NULL	123456789							
5	john	12345							

For the Sign In action in Office 365 (Outlook, Excel, OneDrive) applications or attacks, download this [Sign In Office 365 playlist](#) file.



For the Attack Cookie name and Attack cookie value parameters under Strike SQL Injection and XSS Vector, download this [Cookies SQL Injection and XSS playlist](#) file.

NOTE: To download the above-mentioned files, you have to be logged in on the Downloads & Updates Support page.

About Macros

A macro is a method or function which allows you to customize the payload text data with the following parameters. The `maxLength` limit is set to 1024:

Macros	Description
<code>\$(RandomIPAddress, 'IPv4')</code>	The RandomIPAddress macro randomly generates IPv4 address. IPv6 is not yet supported.
<code>\$(Rand, minValue, maxValue)</code>	The Rand macro generates one random number within the range [minValue, maxValue]. It takes one or two parameters. Range is 0 – N or N1 – N2.
<code>\$(RandomAscii, minLength, maxLength)</code>	The RandomAscii macro generates a sequence of random Ascii characters with values in the range: 0-127 minLength and maxLength are numbers. The sequence length is randomly chosen between the 2 values. If only one argument is specified, it is used as a fixed length.
<code>\$(RandomAlpha, minLength, maxLength)</code>	The RandomAlpha macro generates a sequence of random letters [A-Za-z]. minLength and maxLength are numbers. The sequence length is randomly chosen between the 2 values. If only one argument is specified, it is used as a fixed length.
<code>\$(RandomNum, minLength,</code>	The RandomNum macro generates a sequence of random digits minLength and maxLength are numbers. The sequence length is

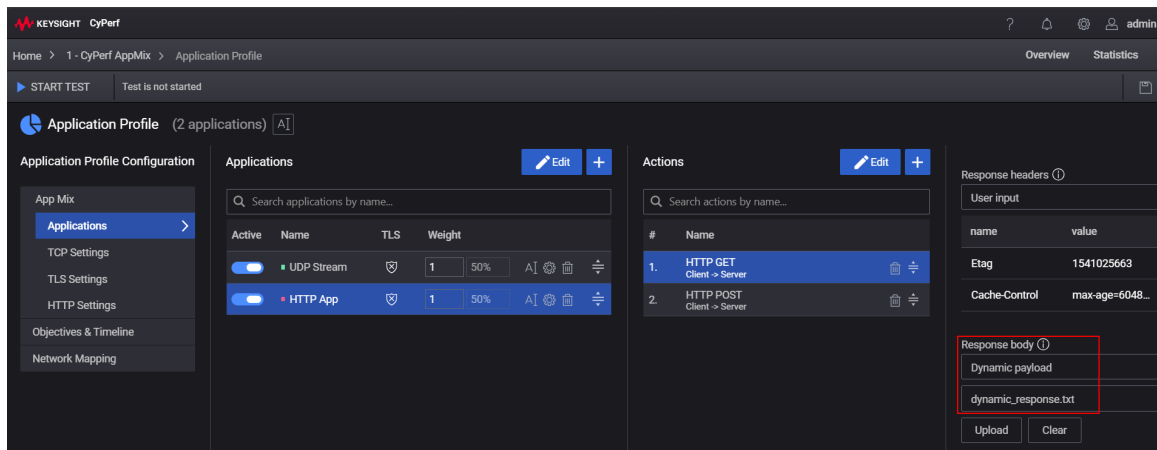
maxLength)	randomly chosen between the 2 values. If only one argument is specified, it is used as a fixed length.
\$(RandomAlphaNum, minLength, maxLength)	The RandomAlphaNum macro generates a sequence of random letters or digits [A-Za-z0-9]. minLength and maxLength are numbers. The sequence length is randomly chosen between the 2 values. If only one argument is specified, it is used as a fixed length.
\$(RandomByte, minLength, maxLength)	The RandomByte macro generates a sequence of random bytes. minLength and maxLength are numbers. The sequence length is randomly chosen between the 2 values. If only one argument is specified, it is used as a fixed length.
\$(PatternRepeat, pattern, minLength, maxLength)	The PatternRepeat macro generates a sequence of characters by repeating the <pattern> pattern. minLength and maxLength are numbers. The sequence length is randomly chosen between the 2 values. If only one argument is specified, it is used as a fixed length. If the chosen length is not an exact multiple of the length of <pattern>, the last repetition of <pattern> is truncated.

Download this [macros](#) file for the correct file format.

Example of a macro file structure:

```
dynamic_payload - Notepad
File Edit Format View Help
Random IPv4 $(RandomIPAddress, 'IPv4')
Random number between 200 and 400: $(Rand, 200, 400)
Random ascii $(RandomAscii, 10, 30)
Random alpha $(RandomAlpha, 5, 20)
RandomNum $(RandomNum, 10, 20)
Random Alpha Num $(RandomAlphaNum, 10)
Random Byte $(RandomByte, 2, 4)
Repeat pattern $(PatternRepeat, '!@#4QwEr', 5)
```

This feature is also available for the HTTP application, on both HTTP GET and HTTP POST actions, under the following parameters: Response body/Response body. Switch to the dynamic payload and upload the `dynamic_payload` file.



Assign the agents, enable the Traffic capture option, and start the test. After the test is finished, download the captured information and you can see the payload, as set in the macro file.

No.	Time	Source	Destination	Protocol	Length	Info
16	0.099346	192.168.10.91	192.168.10.90	TCP	66	[TCP Window Update] 48737 → 80 [ACK] Seq=1 Ack=1 Win=2896 Len=0 TSval=764983045 TSecr=807789185
17	0.099359	192.168.10.91	192.168.10.90	TCP	66	[TCP Window Update] 37775 → 80 [ACK] Seq=1 Ack=1 Win=2896 Len=0 TSval=764983119 TSecr=807784065
18	0.099366	192.168.10.91	192.168.10.90	TCP	66	[TCP Window Update] 58394 → 80 [ACK] Seq=1 Ack=1 Win=2896 Len=0 TSval=764983026 TSecr=807847657
19	0.099374	192.168.10.91	192.168.10.90	HTTP	371	GET /file.txt?name=val1 HTTP/1.1
20	0.099378	192.168.10.91	192.168.10.90	HTTP	371	GET /file.txt?name=val1 HTTP/1.1
21	0.099378	192.168.10.91	192.168.10.90	HTTP	371	GET /file.txt?name=val1 HTTP/1.1
22	0.099623	192.168.10.90	192.168.10.91	HTTP	590	HTTP/1.1 200 OK (text/plain)
23	0.099624	192.168.10.90	192.168.10.91	HTTP	602	HTTP/1.1 200 OK (text/plain)
24	0.099646	192.168.10.91	192.168.10.90	TCP	66	[TCP Window Update] 36116 → 80 [ACK] Seq=1 Ack=1 Win=2896 Len=0 TSval=764983171 TSecr=807846794
25	0.099651	192.168.10.91	192.168.10.90	HTTP	371	GET /file.txt?name=val1 HTTP/1.1
26	0.099685	192.168.10.90	192.168.10.91	HTTP	582	HTTP/1.1 200 OK (text/plain)

>	Frame 22: 590 bytes on wire (4720 bits), 590 bytes captured (4720 bits)
>	Ethernet II, Src: Vmware_a6:29:b9 (00:0c:29:a6:29:b9), Dst: Vmware_99:5a:02 (00:0c:29:99:5a:02)
>	Internet Protocol Version 4, Src: 192.168.10.90, Dst: 192.168.10.91
>	Transmission Control Protocol, Src Port: 80, Dst Port: 58394, Seq: 1, Ack: 306, Len: 524
>	Hypertext Transfer Protocol
>	Line-based text data: text/plain (12 lines)
	Random IPv4 150.243.103.228\r\n
	Random number between 200 and 400: 266\r\n
	Random ascii \016\035E\023TeI.'!]:\034+/?\031p*? \026qI\030\024<\r\n
	Random alpha VEXiKn\r\n
	RandomNum 2274992536640434\r\n
	Random Alpha Num R0tTu0nLmL\r\n
	Random Byte Y\r\n
	Repeat pattern !@#4QwEr!@#4QwEr!@#4QwEr!@#4QwEr!\r\n
	\r\n
	\r\n
	\r\n
	\r\n

CHAPTER 4

Troubleshooting

The following tables present the most common errors and their associated resolution if available:

General Errors	
TLS Handshake Failure	• Check if the proper cipher and its compatible certificate has been properly configured. • Check the agent SYSLOG to find if any error is reported.
Stopping a test during its execution (performing a test abort)	Doing a stop during a test does not close the tunnels gracefully on a DUT. In this case, the tunnels need to be manually cleaned from the DUT or you must wait until the VPN sessions time out before starting the next test in order to avoid tunnel creation failures.
Tunnel Failures in high scale tests	When running a test with the <i>default max pending (10%)</i> setting, tunnel failures might appear in a high scale test. To avoid this, reduce the <i>max pending simulated users</i> setting.

Failed to initialize test error messages	
Failed to initialize test: Application Profile has invalid objective combination. Throughput primary objective permits only the following secondary objectives: Simulated users. OR Failed to initialize test: Application Profile has invalid objective combination. Simulated users primary objective permits only the following secondary objectives: Throughput.	The configuration cannot run with the same primary and secondary objective flavor. Go to Edit Objective > Timeline and select a different primary or secondary value.
Failed to initialize test: Network Segment "IP Network Segment 1" is assigned as responder in	This error appears when an IP Network Segment is used at the same time as the initiator and responder in the Traffic/Attack Profile.

"Application Profile", which conflicts with its assignment as initiator in "Application Profile". OR Failed to initialize test: Network Segment "IP Network Segment 1" is assigned as initiator in "Attack Profile", which conflicts with its assignment as responder in "Application Profile".	Ensure that the Network Segment has only one function: responder or initiator, in Advanced traffic mapping.
Failed to initialize test: Network IP Network 1 has no agents assigned.	All the Network Segments added in the session must have agents assigned.
Failed to initialize test: There is no Application or Attack profile defined.	Make sure there is at least one profile (Application or Attack) defined to run.
Failed to initialize test: Application Profile 'Application Profile' must have at least 1 application.	The configuration needs at least 1 Application in the Application Profile. Go to Edit Appmix > Add Application and select one from the list.
Failed to initialize test: You can't run the action Sign Up of eBanking to nginx on the same network segment as eBanking to IIS. The eBanking server can't know when to send nginx and when to send IIS headers from IP Network 2.	When adding multiple applications of the same type on the same Network Segment, the HTTP Server Profile needs to be the same for all instances of the application. This is also valid for application based attacks.
Failed to initialize test: Unable to check out the requested 1 CYPERF-PERF licenses: features are not installed/valid: CYPERF-PERF	Check which license server is configured from the wheel menu. Then go to the License Management UI from the license server and make sure that the specified number of CYPERF-PERF licenses is available.
Failed to initialize test: Unable to reach license server.	Check if the license server IP can be reached from the UI. If the license server is reachable, make sure it is configured to allow incoming connections on port 443.
Failed to initialize test: Application Profile 'Application Profile' must have at least 1 application.	The configuration needs at least 1 Application to be added in the Application Profile. Go to Edit AppMix > Add Application and select one from the list.
Failed to initialize test: Trying to set multiple server HTTP profiles	When adding multiple applications of the same type, the HTTP Server Profile needs to be the same for all applications. This is

on same application type eBanking	also valid for application-based attacks.
--------------------------------------	---

Test stopped error messages	
Test stopped: Orphaned test detected and automatically aborted	You can see the error when the controller is restarted while a test run is in progress. The component that monitors test execution needs to be continuously running while a test runs. Any interruptions lead to this error.
Test stopped: Sending CONFIGURATION messages generated errors: configuration on IP Network Segment 2 generated errors: could not find IP on test interface ens192 for agent X.X.X.X	You can solve this error by: - configuring an IP on the test interface for the agent with the IP X.X.X.X, the test runs with the IP configured on the test interface from the agent CLI - unchecking the box from Network Segment > IP Range > Automatic IP → test runs with the configured IP.
Test stopped: Sending CONFIGURATION messages generated errors: configuration on IP Network Segment 2 generated errors: could not find NETMASK on test interface ens192 for agent X.X.X.X	You can solve this error by: - configuring an IP on the test interface for the agent with the IP X.X.X.X - unchecking the box from Network Segment > IP Range > Automatic IP prefix length.
Test stopped: FAIL: Failed to set objective. Core count = 4, Attacks per second = 2, Max concurrent attacks = 5. Configure attack rate to be greater than or equal to concurrent attacks when desired concurrency is lower than core count. Test has been cleaned and deconfigured	This error indicates that the Attack Profile can not run with attack rate or attacks per second lower than the max concurrent attacks value.
Test stopped: [Agent X.X.X.X is already a server and cannot be a client as well!]	One agent can not be assigned to more Network Segments.
Test stopped: Unable to verify that licenses are still checked out by this test.	Licenses used in this test are no longer present. Check if the licenses were deactivated while the test was running or if the connection to the external License Server is ok.
Test stopped: Unable to resolve agents: Insufficient online	Open the Assign Traffic Agents dialog for the specified

agents assigned to IP Network Segment 2. Detected 0 online agents assigned and the test requires at least 1.	network segment. If the agents are assigned by ID, you can see the assigned agent as offline or an error message about invalid ports selected. Make sure that at least one online agent is selected and try to start the test again. If the agents appear online, check the session events for additional messages regarding the failure and find the oldest error message that belongs to the current test. You can proceed based on the contents of that message.
--	---

Events messages	
Keysight Software Manager failed to process the request with: ERROR-License is expired.	This error is generated when an expired activation/entitlement code is used for activation. Use a valid code for this operation to be successful.
Keysight Software Manager failed to process the request with: ERROR-AC/Qty not available.	This error is generated when an incorrect quantity of licenses is entered. Wait for the information regarding the licenses to be completely loaded in the Activate Licenses dialog, and then click the Activate button.
Lost connection to server. Please check your network connectivity.	This may mean that the CyPerf application is not running or that there is no network connectivity between the browser and CyPerf. You need to check the status of the CyPerf machine. Sometimes a browser may fail to accept self-signed certificate even if it was previously accepted. This may cause errors to appear in the UI. Refresh the page and accept the certificate when prompted.
Unable to delete session 1 Generic web applications: Removing a running session is not allowed	Make sure the session you are trying to delete is listed as Test is stopped . If there is a test running on the session, stop the test and wait for it to reach the <i>stopped</i> state. You can then attempt to delete the session again. If the test is either starting or stopping and seems to have been left in this state for a long time (i.e. more than 10 minutes), you can manually PATCH the session and set its state to STOPPED . You should be able to delete it after this.

How to access agent captures and diagnostics

To access agent diagnostics, connect to the CyPerf agent, and go to the `/var/log/syslog` folder where you can find the diagnostics file. See also [Collect Diagnostics](#).

To access agent captures, connect to the CyPerf agent, and go to the `/var/log/cyperf/capture` folder where you can find the capture file. See also [Decrypt IPsec Packets](#).

NOTE: The **Collect Diagnostics** option from the CyPerf controller web UI generates both controller and agent diagnostics.



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